
Georgetown McDonough School of Business Consulting Case Interview Book



Fall 2015

Contents

Section	Page #
☐ Introduction	3
☐ Consulting Industry Guide Industry Overview Firm Overviews (by type)	5
☐ Interview Preparation Interview Overview Theoretical Frameworks / Helpful Accounting Information Sample Case Frameworks	9
☐ Practice Cases Case Certification Schedule 34 Practice Cases	27
☐ Additional Resources	149

Dear Consulting Club Member,

This casebook is meant to provide you with a brief overview of the consulting industry and more specifically, to prepare you for the case interview process. This casebook is not meant to be your sole resource. You will get the most from this casebook if you also attend the case certification lunch & learns, consulting club workshops, consult outside resources (suggestions in a later section), schedule regular career center appointments, and practice firm-specific cases prior to your interviews.

Every company looks for slightly different qualities from its candidates. This casebook is meant to give you a strong foundation in how to do a case interview, not to prepare you for a specific firm. For best results, wait until you've finished this book before practicing firm-specific cases. This will allow you to spend those firm-specific case prep sessions focusing on what that company wants to see, instead of trying to learn how to do a case while also worrying about what an individual company is looking for.

Good luck!

-2015 GSCG Co-Directors of Career Development

Contents

Section	Page #
☐ Introduction	3
☐ Consulting Industry Guide Industry Overview Firm Overviews (by type)	5
☐ Interview Preparation Interview Overview Theoretical Frameworks / Helpful Accounting Information Sample Case Frameworks	9
☐ Practice Cases Case Certification Schedule 34 Practice Cases	27
☐ Additional Resources	149

Industry Overview

What is management consulting?

Consultants provide advice or expertise with regard to a business problem. Consultants may engage with multiple industries, clients, and functions or take on a specific expertise within a given industry.

What will I do as a management consultant?

➤ *Client service*

- Interviewing clients and staff
- Conducting analyses, both quantitative and qualitative
- Producing client deliverables
- Making sound, actionable recommendations
- Managing client relationships

➤ *Business development*

- Identifying add-on work
- Developing white papers
- Presenting engagement pitches to clients
- Responding to requests for proposals/information (RFP/I)

➤ *Other*

- Training/learning & development
- Recruiting
- Periodic feedback
- Affinity groups
- CSR/volunteer work

Three Main Firm Categories

Big 3 Strategy

- Boston Consulting Group (BCG)
- McKinsey & Co.
- Bain & Co.

Big 4 Accounting

- Deloitte
- PricewaterhouseCoopers (PwC) + Strategy& (formerly Booz & Co.)
- KPMG
- Ernst & Young (EY)

Other Strategy & Boutique

- Accenture
- Oliver Wyman
- A.T. Kearney
- L.E.K.
- Cognizant
- Capgemini
- Censeo
- Charles River Associates + Marakon
- Gallup
- IBM (IT)
- Booz Allen Hamilton (public sector)
- Towers Watson (human capital)
- Cornerstone Research (law)
- BDO (finance & accounting)

Consulting Functions

Each consultant aligns with one or more functions (or horizontals) within the firm.

Upon joining the firm, consultants are aligned with a function. Strategy firms tend to interact with one or more of these functions. Boutique firms may specialize in a particular function. Gauging your interest and ability to jump between functions and industries for each case/client or embarking on a deep expertise in a given function is a great way to determine which firm(s) are best for you.

What is the difference between a horizontal & a vertical?

A horizontal is a function that can be applied across industries (e.g. finance and accounting consultants can serve banking, CPG, and public sector clients seeking this type of service). Verticals are the industries served (e.g. alignment with the technology vertical will allow you to serve IT clients but provide consulting across multiple functions, like strategy, human capital, and organizational effectiveness).



Contents

Section	Page #
☐ Introduction	3
☐ Consulting Industry Guide Industry Overview Firm Overviews (by type)	5
☐ Interview Preparation Interview Overview Theoretical Frameworks / Helpful Accounting Information Sample Case Frameworks	9
☐ Practice Cases Case Certification Schedule 34 Practice Cases	27
☐ Additional Resources	149

The Interview



The Process

- | | | | |
|---|--|--|---|
| <ul style="list-style-type: none"> • Wait in lobby with other candidates and recruiters/practitioners • Walk to interview area and small talk with your interviewer | <ul style="list-style-type: none"> • Opening discussion with interviewer • Questions about resume / background • Interviewer may give personal background | <ul style="list-style-type: none"> • Interviewer will start case • Keep track of time so that you can get to your risks and recommendation | <ul style="list-style-type: none"> • Your opportunity to ask questions • Walk back to lobby with interviewer or recruiter |
|---|--|--|---|

Your Job

- | | | | |
|--|---|---|---|
| <ul style="list-style-type: none"> • Try to relax and get comfortable • Appear friendly and professional | <ul style="list-style-type: none"> • Show that you're a good fit for the firm • Pass "airport test" | <ul style="list-style-type: none"> • Remain confident no matter what, display structure logic and creativity | <ul style="list-style-type: none"> • Ask good questions, learn more about interviewer's personal experiences |
|--|---|---|---|

Most Frequent Fit Questions

1. Why consulting?
2. Why our firm?
3. Tell me about yourself
4. What is your industry / function preference?
5. Tell me a time where you used strategic thinking to solve a problem / tell me about a time when you set a strategic direction for a team, project or organization
6. Tell me about a time when you designed or updated an existing system to improve efficiencies
7. Provide an example of a time when you demonstrated innovation
8. Tell me a time when you managed a decision, for which there were no previous guidelines
9. Tell me a time when you led a diverse team
10. Tell me a time when you demonstrated initiative by taking the lead without being told to do so
11. Tell me a time when you motivated a group of people in a changing environment
12. Tell me about a time when you successfully reprioritized the tasks of a team project
13. Tell me about a time when you anticipated potential problems and developed a proactive solution
14. Give me an example of a time when you convinced your manager about an idea or concept
15. Describe a time when you got people who didn't like each other to work well together
16. Tell me about a time when you recommended a change and were able to successfully implement that change
17. Tell me about a time when you successfully initiated a new program that effected many people
18. Tell me about a time when you had to collect and/or analyze a large amount of information and present it in an easy to understand manner to a key stakeholder
19. Tell me about a time when you demonstrated expertise in a core business function and were recognized positively for that knowledge
20. Tell me about a time when you saw potential in a colleague and helped them to develop a skill/competency

The Case: How It Will Flow

Get the question

- Take Notes
- Ask clarifying questions
- Ask if client has additional objectives

Structure your thoughts

- Ask for a few minutes to write down your thoughts
- Write out your initial framework with areas to explore in each
- Write down your initial hypothesis

Share your thoughts

- Turn your page to the interviewer
- State your initial hypothesis
- Explain how your buckets will prove/disprove your hypothesis
- Systematically guide the discussion through these buckets

Discussion

- As you talk through the case the interviewer will give you additional information or exhibits
- Do whatever calculations are necessary and continue to update your hypothesis with new information

Recommendations

- After you've exhausted the dialogue of the case move into brainstorming recommendations
- Try to aim for 3-5 possibilities
- Also, consider possible adjacencies

Risks

- After you've brainstormed recommendations / adjacencies discuss any risks associated with them → Porter's five forces is useful here
- Use risks to rule out some of your recommendations

Conclusion

- Answer the client's question and propose your solution, e.g. the client's issue was caused by X, so I recommend the client do Y for these 3 reasons.
- Briefly mention associated risks and ways to mitigate

Interview Do's

Do:

- 1) Know that as soon as you walk in the lobby your interview has begun. Every interaction counts.
- 2) Always ensure your stories somehow illustrate some quality you want the interviewer to remember you for.
- 3) Ask good questions.
- 4) Follow interviewer cues. Have fun on the outside even if you feel miserable on the inside.
- 5) Use common sense if you don't have direct industry experience (i.e. draw on personal experience as a consumer).
- 6) Maintain a dialog with the interviewer throughout the entire case except for when you are structuring at the beginning. Minimize long, silent pauses while you are doing math calculations.
- 7) Spend adequate time understanding the objectives and build a structure around meeting them. Provide the answer first.
- 8) Develop a hypothesis and use your MECE structure to either prove or disprove it. If it is wrong that is ok as long as your structure proves it.
- 9) Consider Porter's Five Forces, the market's attractiveness, and the company's position. Are they attractive?
- 10) Always follow the value and consider the impact on the bottom line.
- 11) Be organized in doing math. Use a separate sheet of paper and carry key numbers or findings over to the main sheet.
- 12) Highlight crucial numbers or findings with a different color pen.
- 13) Consider adjacencies when making recommendations on how to drive additional value
- 14) Balance being coachable with standing your ground when probed or challenged by the interviewer.
- 15) Highlight risks associated with a recommendation but do not equivocate based on a desire for more data or information.

Interview Don'ts

Don't:

- 1) Be late.
- 2) Try to interview with two different firms on the same day.
- 3) Ask how the interview is going.
- 4) Get flustered. If you make a mistake just keep doing your best and remember, no one has a perfect interview, you might still get it.
- 5) Speak about previous experiences or organizations in a negative light.
- 6) List accomplishments when describing yourself. Instead of regurgitating your resume, use your behavioral answers to illustrate qualities.
- 7) Ask about compensation.
- 8) Ask personal / political / religious questions of your interviewer.
- 9) Show up to an interview without having done significant research about that firm.

Theoretical Frameworks: Porter's Five Forces

Barriers to Entry: *How difficult is it to enter a market? Consider:*

- Economies of scale
- Differentiation
- Capital requirements
- Switching costs (barriers to exit)
- Access to distribution channels
- Necessity for proprietary technology / knowledge (learning curve)
- Favorable access to raw materials, locations, distribution channels
- Government policies / subsidies

Substitute Products: *How easily can other products be substituted for this industry's? Consider:*

- Does industry compete with substitutes based on price?
- Are substitutes a high-margin industry?
- How powerful are the companies producing substitutes?

Rivalry: *How intense is competition? Consider:*

- Structure of industry (closer to oligopoly or perfect competition)?
- High fixed costs?
- Lack of differentiation or switching costs?
- Economies of scale?
- Growth in industry?

Supplier Bargaining Power: *What is the balance of power between the industry and its suppliers? Consider:*

- Substitutability of supplies
- Importance of industry to supplier
- Importance of supplier to industry
- Differentiation amongst suppliers
- Can suppliers forward integrate?
- Can industry backwards integrate?

Buyer Bargaining Power: *What is the balance of power between the industry and its customers? Consider:*

- Do buyers purchase large percentage of seller's sales?
- Are the buyers purchasing differentiated or undifferentiated products?
- Can buyers backwards integrate?
- Can industry forward integrate?
- Does buyer have full information?
- How important is seller's product to buyer's products / services?
- Do buyer's face switching costs?

Theoretical Frameworks: Porter's Diamond

Note: this model attempts to explain the competitive advantage that nations may have as a result of the listed factors. This is useful for analyzing the competitive atmosphere in those locations, and why a company may wish to operate there. For instance, a company may enter an unprofitable market if doing so will help the company innovate or improve its competitiveness elsewhere. These are called "Strategic Markets." (Think high-end fashion designers co-located in Milan).

Chance: Events beyond the firm's control.

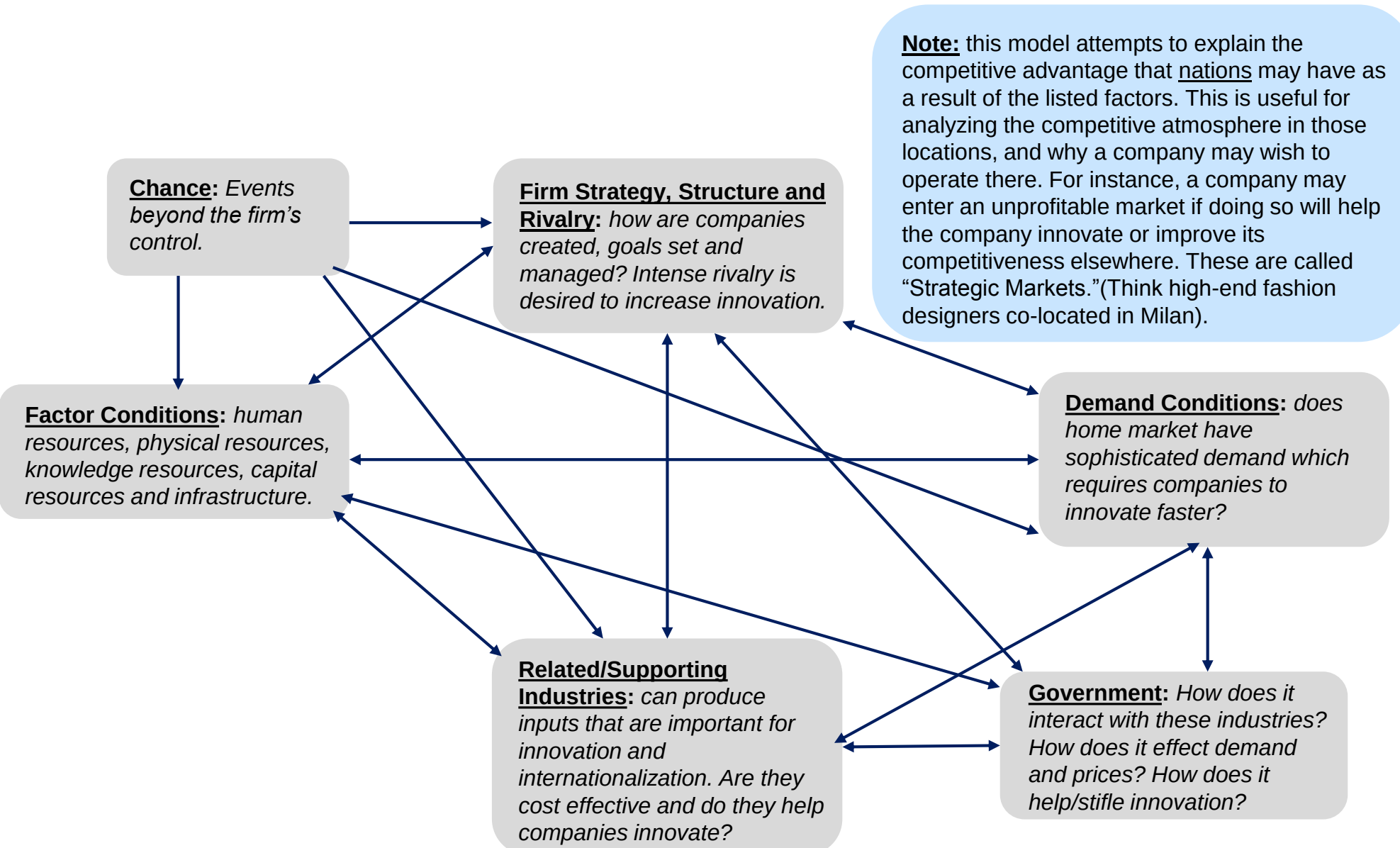
Firm Strategy, Structure and Rivalry: how are companies created, goals set and managed? Intense rivalry is desired to increase innovation.

Factor Conditions: human resources, physical resources, knowledge resources, capital resources and infrastructure.

Demand Conditions: does home market have sophisticated demand which requires companies to innovate faster?

Related/Supporting Industries: can produce inputs that are important for innovation and internationalization. Are they cost effective and do they help companies innovate?

Government: How does it interact with these industries? How does it effect demand and prices? How does it help/stifle innovation?



Theoretical Frameworks: 4 C's

Customers:

- Define the market: what needs are we aiming to serve?
- What is the situation of purchase and how is product used? How frequent are purchases and how much do they purchase?
- Who uses the product (consumer) and who purchases the product (customer)? How are their needs different? What influences them to buy?
- What is the company's relationship with its customers?
- What could get the customer to consider purchasing a substitute good?

Company:

- How is their value chain organized? What is the company's level of integration?
- *Operation Factors*: product mix, inventory turnover, sales force, cost structure
- *Competitive Standing*: size of company, brand equity, customer loyalty, trend-setter? Economies of scale/scope?
- *Organizational Structure*: presence of skilled labor / tacit knowledge? Properly aligned incentives?
- Financial situation?
- Are the company's resources/competencies aligned with its goals?

Competitors:

- How is their value chain organized? What is the company's level of integration?
- What are the company's goals?
- How does the company view itself?
- What is the competitor doing and what can it do?
- Strengths and weaknesses?
- Recent actions? Have they sent any signals to the market?

Collaborators:

- Are you working with a buyer or a supplier?
- What is the buyer/supplier's bargaining power with the company?

Buyers

- *Quality determined by*: purchasing potential, growth potential, intrinsic bargaining power / propensity to use it, and cost of servicing.
- Has company built up switching costs and eliminated high-cost buyers?

Suppliers

- *Key issues*: stability and competitiveness of suppliers, optimal degree of vertical integration, allocation among suppliers, leverage with suppliers.

Theoretical Frameworks: 4 P's

Promotion:

- What is the goal of promotion?
- What are the obstacles to achieving this goal?
- How is this different than what competitors are doing?
- Is this a pull or push strategy?
- How much money is being allocated?
- What kind of ROI does the company want to see? What conversion rate and customer lifetime value is required to achieve that ROI?

Product:

- How is it positioned in the market?
- Is the product a: branded commodity? Premium product? Commodity? Or under-priced product?
- How is the market segmented and which segment does the product target?
- Is the product differentiated?
- What kind of brand equity does the product uphold?
- Is it consistent with the company's image and competencies?
- How does the product fit within the company's goals?
- How does the product look?

Place/Distribution:

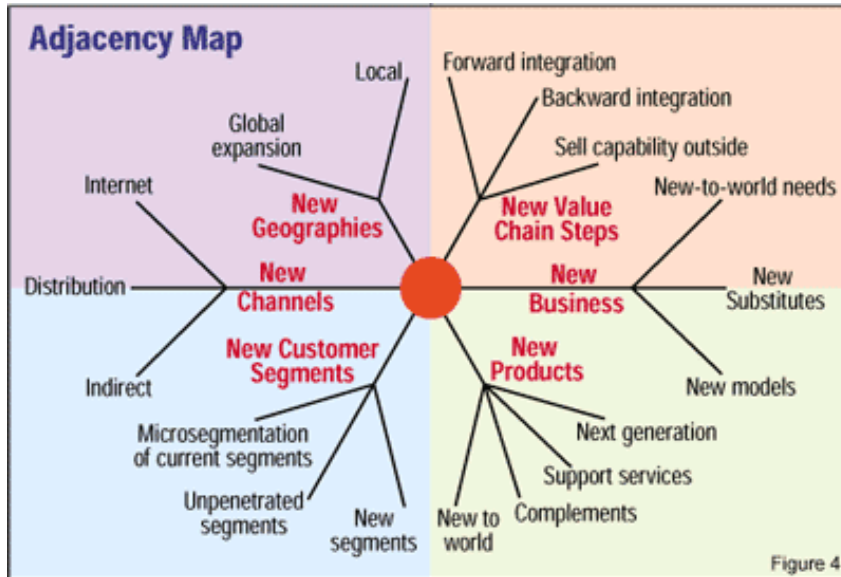
- Where is product being sold?
- Which channels are most closely aligned with company's strategy?
- What functions does company want channel to serve
- Can/should company go direct to consumer?
- What are economics of channel? Who is capturing what margin? How does this effect selling price?
- How much control does company want over distribution?
- How much volatility is company willing to accept in channel?

Price:

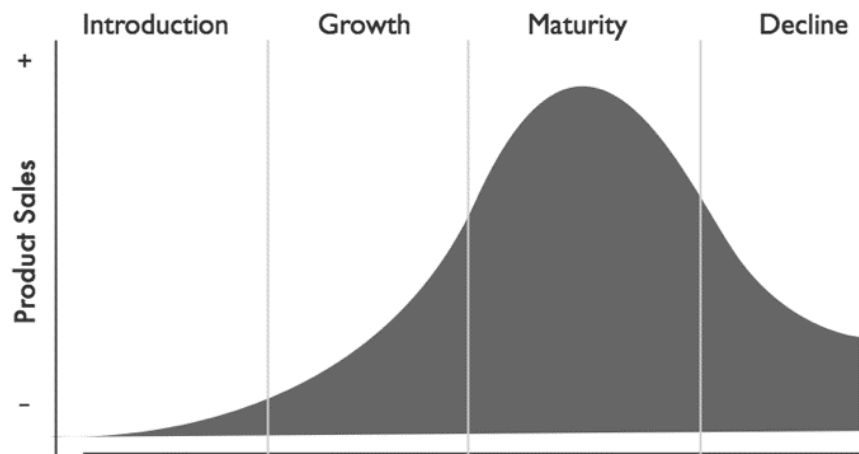
- What is perceived value of product?
- What is elasticity of demand?
- How much does it cost to produce?
- How much do substitute goods cost?
- How much did previous versions of this product cost?
- What is the expected profit margin and expected consumer surplus?

Theoretical Frameworks

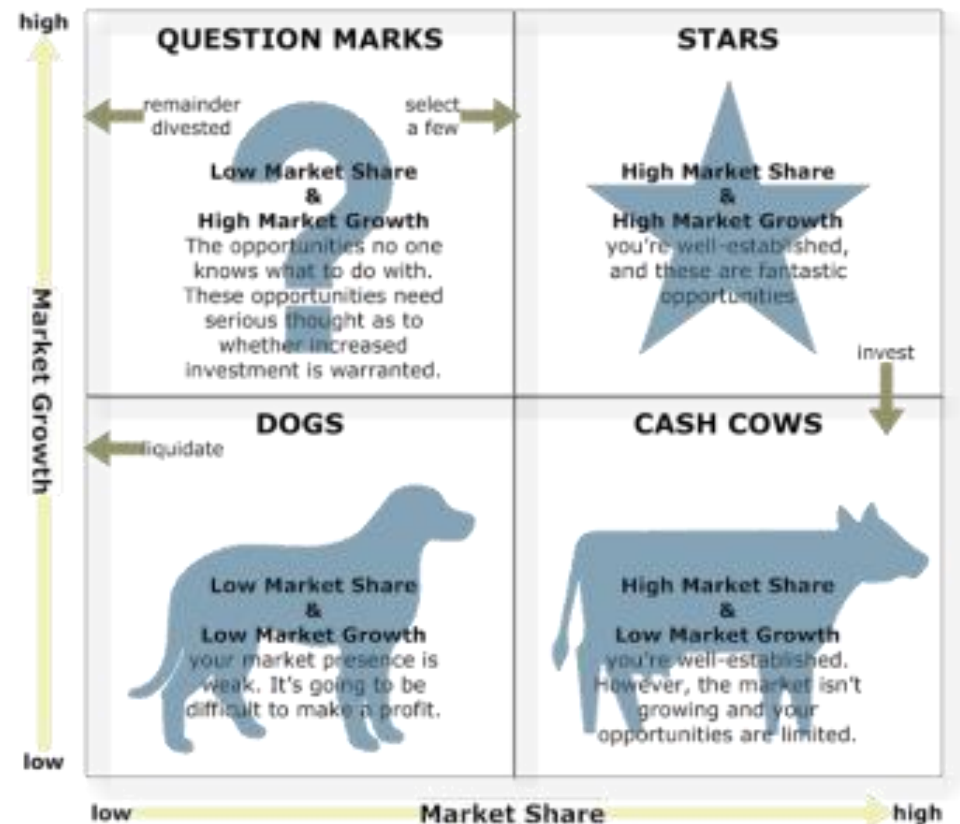
Adjacency Map



Product Life Cycle



BCG Matrix



Helpful Accounting Info: Definitions

Definitions:

Cost: sacrifice of economic resources

Expense: cost charged against revenue in an accounting period

Opportunity cost: foregone benefit from taking an alternative action

Product cost: traced or assigned to products ["Cost to Provide" in CLV]

Direct costs: costs that can be traced to products (i.e., direct materials, direct labor)

Indirect costs: product costs that are assigned to products (i.e., indirect materials, indirect labor)

Period costs: non-product costs, expensed in period incurred

Marketing expenses: advertising, sales commissions, direct sales activities, proposal/bid development

Admin expenses: shared services (accounting, finance, legal, IT, HRM, customer service, billing...)

Period costs include the "Cost to Acquire", "Cost to Serve", and "Cost to Retain" customers.

Traditional Income Statement

Revenues	
Less: Cost of goods sold	
(including variable and fixed	
<i>manufacturing costs</i>)	
Gross margin	
Less: Marketing and administrative costs	
(including variable and fixed <i>marketing</i>	
<i>and administrative costs</i>)	
Operating profit	

Contribution margin Income Statement

Revenues	
Less: Variable costs	
(including <i>variable</i> manufacturing,	
marketing and administrative costs)	
Contribution margin	
Less: Fixed costs	
(including <i>fixed</i> manufacturing,	
marketing and administrative costs)	
Operating profit	

Helpful Accounting Info: Contribution Analysis

<u>Step</u>	<u>Measure</u>	<u>Example</u>
1. Calculate the contribution that each unit provides to cover fixed/overhead costs	Unit Contribution: Unit Selling Price - <u>Variable Cost</u> =Unit Contribution	\$100 - <u>\$30</u> \$70
2. Determine the number of units that need to be sold to break-even	Break-even Volume: <u>Fixed Costs</u> Unit Contribution	<u>\$70,000</u> \$70 =1,000 units
3. Assess the percentage of market share that needs to be captured	Break-even Market Share: <u>Break-even Volume</u> Total Market Share	<u>1,000 units</u> 10,000 units =10% Share
4. Estimate the total contribution margin	Total Contribution: Unit Contribution <u>x # of units sold for period</u> =Total Contribution to Fixed Costs & Profit	\$70 <u>x 1,500 units</u> =\$105,000
5. Calculate net profit	Net Profit: Total Contribution to Fixed Costs & Profit - <u>Total Fixed Costs</u> =Net Profit	\$105,000 - <u>\$70,000</u> =\$35,000

Sample Case Frameworks: Profitability

How Should the Client Handle An Issue About Profitability?

Revenues

Costs

Market:

- What are trends in the market?
- What are other firms experiencing?
- Has there been a recent shock to the market?
- How fierce is rivalry?
- What is the power relationship between firms, suppliers, and buyers?
- Have there been any new entrants?

Price:

- What is the price?
- How is it determined?
- Has this price changed?
- Are there any other pricing options?
- What contribution margin does each product have?

Quantity:

- What is our product mix?
- How has that mix changed?
- Are quantities increasing or decreasing?
- How are products segmented? Regionally? By customer type? By distribution channel?

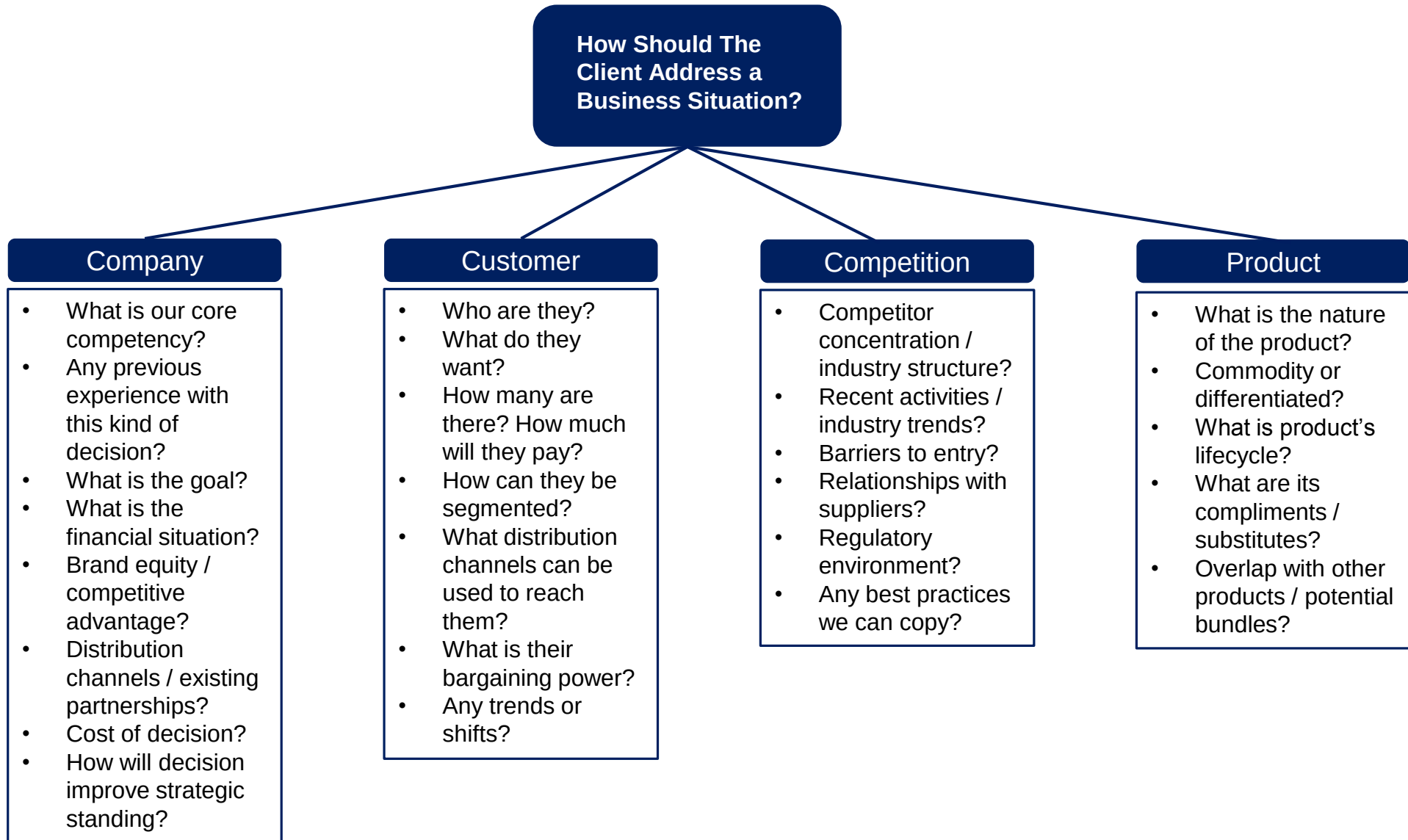
Fixed Costs:

- Has anything changed?
- What are the largest costs?
- Are there any ways to reduce these costs?
- How are these costs trending?
- What does the value chain look like?

Variable Costs:

- What are the main cost drivers?
- How have these changed?
- Are these the same as others in the industry?
- Are there any ways to reduce these changes?
- What is client's relationship with suppliers?

Sample Case Frameworks: Business Decision



Sample Case Frameworks: Market Entry

Should The Client Enter a New Market?

Company

- What is our core competency?
- Any previous market entry experience?
- What is the goal?
- What is the financial situation?
- Brand equity / competitive advantage?
- Distribution channels / existing partnerships?
- Cost of entry?
- How will entry improve strategic standing?

Customer

- Who are they?
- What do they want?
- How many are there? How much will they pay?
- How can they be segmented?
- What distribution channels can be used to reach them?
- What is their bargaining power?
- Any trends or shifts?

Competition

- Competitor concentration / industry structure?
- Recent activities / industry trends?
- Barriers to entry?
- Relationships with suppliers?
- Regulatory environment?
- Any best practices we can copy?

Product

- What is the nature of the product?
- Commodity or differentiated?
- What is product's lifecycle?
- What are its compliments / substitutes?
- Overlap with other products / potential bundles?

Method

- **Joint Venture:**
 - Licensing, non-equity alliance, equity alliance?
 - Least control, best suited for more stable and less competitive environments
- **Acquisition:**
 - Costly, but fast. Must consider integration costs/difficulty
- **Organic:**
 - Highest level of control, longest process, requires knowledge to grow capability

Sample Case Frameworks: M&A

Should the
Client Conduct
a Merger or
Acquisition?

Strategic Fit

Should we do it?:

- How attractive is the market?
- How attractive is the partner?
- What are the industry trends?
- What is the goal?
- Is this deal strategically important enough to justify a negative NPV? If so, why?

Method:

- Vertical integration
- Horizontal integration
- Equity Alliance
- Non-Equity Alliance

Economics

Value of deal:

- Revenues & Costs
- Required CAPEX
- Tax Rate?
- Cost of capital?
- Value = {[Revenue – (Cost + CAPEX + Working Capital)]*(1 – Tax)} / Cost of Capital

Synergies:

- How much are they worth?
- What kind?

Deal Price:

- If the deal price is less than the combination of the value of the deal + synergies then it makes economic sense.

Risks

Capability:

- Has the company done this before? More experienced company = higher chance of success.

Is it a good fit?

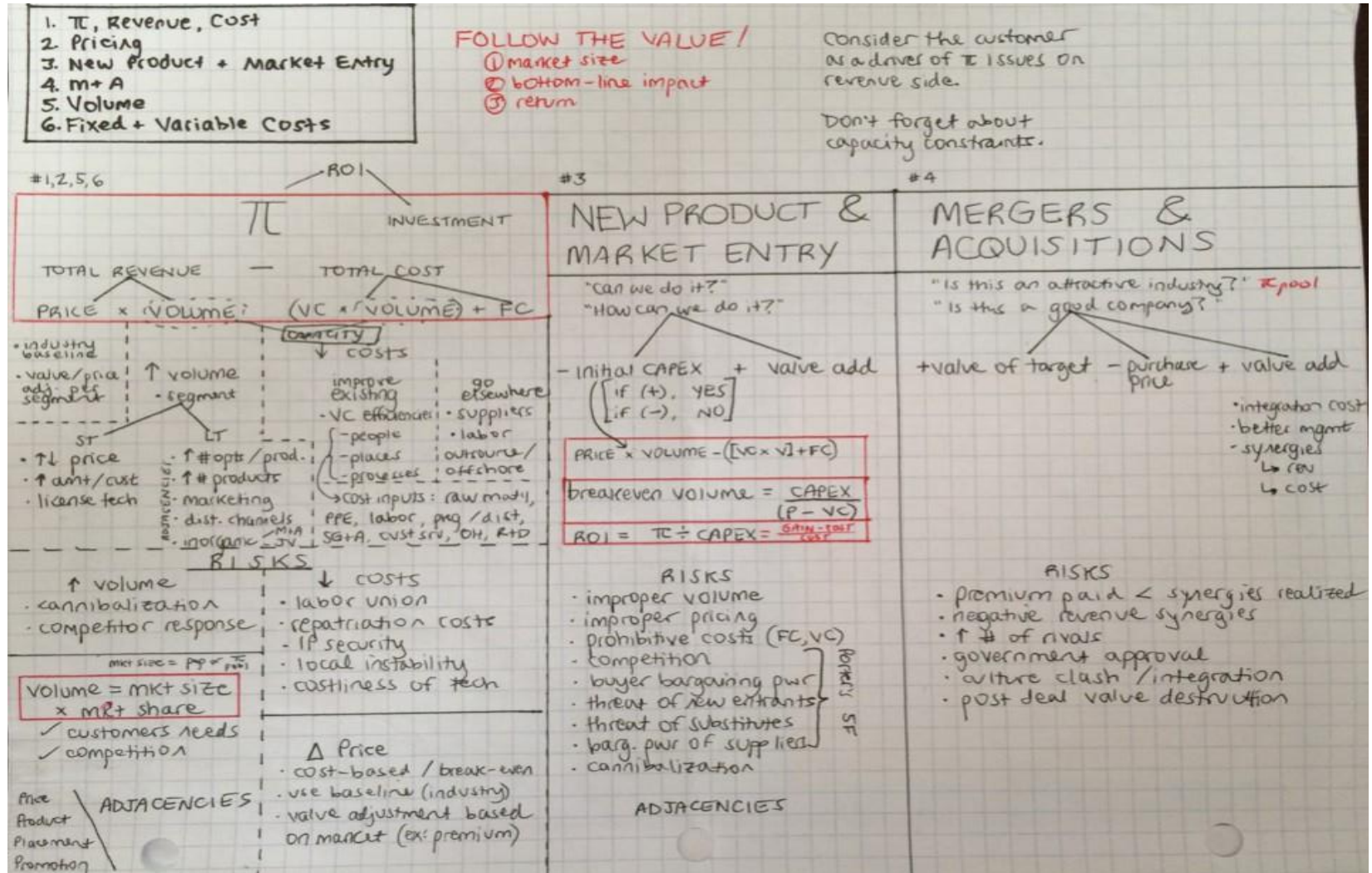
- Cultural distance?
- Administrative distance?
- Geographic distance?
- Economic distance?
- Will overlaps in value chain allow for savings?

Post-Merger Integration:

- How difficult will it be to integrate the two companies?

Developing an Overarching Framework

Some people find it helpful to take multiple frameworks and boil them down to 6-8 common buckets, then use 3-4 of those buckets for any given case. Below is an example of how one student developed his overarching framework:



Contents

Section	Page #
☐ Introduction	3
☐ Consulting Industry Guide Industry Overview Firm Overviews (by type)	5
☐ Interview Preparation Interview Overview Theoretical Frameworks / Helpful Accounting Information Sample Case Frameworks	9
☐ Practice Cases Case Certification Schedule 34 Practice Cases	27
☐ Additional Resources	149

2015 Case Certification Schedule

Week	General Type	Date Range	Cases
1	Profitability	October 6 th – October 13 th	1 - 6
2	Profitability / Business Decision	October 14 th - October 27 th	7 - 10
3	Business Decision	October 28 th – November 1 st	11 - 15
4	Market Entry	November 2 nd – 8 th *Super Weekend 1*	16 - 21
5	M&A / Interviewer Led	November 9 th – 15 th	22 - 28
6	Hard Cases	November 16 th – 22 nd *Super Weekend 2*	29-34

Case 1: Flying High

Prompt: In the United States, how many people are in the air right now?

Notes for Interviewer: This is about the candidate's thought process, not getting the right number. At the onset, this case involves careful organization of ideas; and, at the end, offers an opportunity for creativity. First, the interviewee should create categories/buckets of people "in the air". The majority of people in the air are in airplanes, so consider both cargo and passenger flights, private planes, military/emergency planes/helicopters, etc. Notice that the prompt asks "in the air," so the interviewee can include any other reasons why people are not on the ground, such as paragliding, bungee jumping, jumping on a trampoline, or other recreational activities.

Data	Assumption (Example)	Calculation
# of Commercial (Passenger and Cargo) Planes in the Air	- On average, 3,500 passenger planes and 5,000 cargo planes fly each day. - Flights are evenly distributed across 24 hours (i.e. a flight takes off every hour)	3,500 planes / 24 hrs = 145 → can estimate/round up to 150 5,000 planes / 24 hrs = 208 → can estimate/round to 200 - Total planes in air at a given time = 150 + 200 = 350
# of People on Planes in the Air	- Average number of seats in a passenger plane = 300 - Average occupancy rate = 75% - Average number of staff members on a passenger plane = 6 - Average number of staff members on a cargo plane = 10	300 passengers x 75% = 225 x 150 = 33,750 6 staff x 150 = 900 10 staff x 200 = 2000 - Total # of people = 33,750 + 900 + 2000 = 36,650
Other People in the Air	Approx. 500 people in the air doing "other activities"	Total = 36,650 + 500 = 37,150

Case 2: Count the Fries?

Prompt: How many individual French fries does McDonald's sell in the US each year?

Notes for interviewer: This is about the candidate's thought process, not getting the right number. The main thing is ensuring that the interviewee has a well laid out thought structure based off reasonable assumptions. Also, if the interviewee is going to round his/her numbers it's a good practice to ask if that is ok before doing so. Below is one way of thinking through this problem. After the interviewee arrives at his/her answer always ask the interviewee "If I told you that your estimate is too high/low, which assumption(s) would you look at first?" This question is a typical way to show flexibility and a realistic perspective about which assumptions may be off.

Data	Assumption	Calculation
# of McDonald's Locations	320M people in US. Each store serves 32,000 people.	10,000 stores
Avg fries per pack	3 sizes: small (15 fries), medium (25 fries), large (50 fries). Each sells w/ same frequency.	$(15+25+50)/3 = 30$ fries per pack
Customer frequency	1 customer per minute	60 customers per hour
Fries sold per hour per store	50% of customers buy fries	$30 \times 60 \times 50\% = 900$ fries per hour
Fries sold per day per store	Store open 6 AM to 11 PM, sells fries 11 AM to 11 PM	$12 \text{ hours} \times 900 \text{ fries per hour} = 10,800$ fries per day
Fries sold per day	10,000 stores / 10,800 fries per day	$10,000 \times 10,800 = 108\text{M}$ fries per day
Fries sold per year	Fries sold 365 days per year	$108\text{M} \times 365 = 39.42\text{B}$ fries per year

Case 3: Firearms

Prompt: Our customer, Houston Firearms, is the 300 year-old national market leader of firearms manufacturing and sales. Recently, they have seen a decrease in sales. What are the reasons of this drop and what can we do to proliferate sales?

Interviewer Information: provide this additional information ONLY if asked by the interviewee

- *Market share:* In 2009 was 45%; in 2007 was 55%
- *Industry growth:* 3% annually in the last two years
- *Product mix:* The most popular “Houston Rocker” constitutes 70% of our sales.
- *Costs:* Costs of production, distribution, and marketing have not changed
- *Price:* Prices have increased at the rate of inflation
- *Competition:*
 - Austin based “Wal-Arms” was established in 2005 and has since been growing rapidly. They outsource their production to China and thus offer a cheaper alternative: the “FireEasy”
 - Besides a few small local companies, no new company has entered the industry

Notes for Interviewer: The point of this case is to get the interviewee to explore possible reasons for a decrease in sales and then brainstorm solutions and risks. Sales have dropped because Houston Firearms generates most of its revenue from the “Houston Rocker” and a new Austin-based gun manufacturer has entered the market and gained market share rapidly by offering a cheaper version. A good interviewee will be able to identify that Houston firearms has several opportunities. Houston firearms could position itself as a premium product against “Wal-Arms.” Going through the 6 adjacencies (new geographies, new channels, new customer segments, new products, new businesses, changes in value chain steps) should also result in numerous options for Houston Firearms. Finally, a good candidate should identify the risks associated with whichever recommendations he/she makes and explain any necessary mitigation. After the interviewee has identified several solutions and risks ask the interviewee to summarize with a 30-second recommendation to the CEO of Houston Firearms.

Case 3: Firearms (Example Framework)

Main questions	Why are sales declining? How can we reverse this trend?
Key areas to explore	<u>Client</u> - Product Mix? - Quality - Marketing? - Operational issues? <u>Competition</u> - Change in landscape? - Differentiation? - Price competition? - New Entrant? <u>Customers</u> - Changes in Preferences? <u>Other</u> - Government regulation? - Macro-economic issues?
Analysis	- Market share in 2009 was 45%; in 2007 was 55% - The industry has grown by 3% annually “Houston Rocker” constitutes 70% of sales - COGS, distribution, and marketing unchanged - Industry grown 3% annually - Prices increased at the rate of inflation “Wal-Arms” new entrant growing rapidly, outsource → competes on price
Recommendation	Current Position: Strengthen by focusing on offering a premium product, “buy American” campaign, if brand equity is strong enough increase price Adjacencies: Offer a new product (like a cheaper gun), sell in new geographies, explore new distribution channels, add/reduce/outsource value chain steps, target new customer segments, look at complimentary gun products
Other factors (risks)	- Break-even point for any new products - New government regulations - Possible price war

Case 4: Drug Store Profitability

Prompt: Our client is a drug store chain, similar to CVS. The client has been losing profits in the last few years. Can you help the client identify the reasons for this loss and means to improve profits?

Interviewer Information: Stores are typical to CVS, located in several areas. They have 3 sections pharmacy, health and beauty, and general merchandise. 60% of the stores are located around hospitals, with heavy competition and in some crime infested areas– these stores make 10% loss, the remaining stores make 25% profit.

Notes for Interviewer: There is no 100% correct answer to this case – the client has several potential problems, with several potential solutions. A good candidate will identify those problems and recommend solutions, i.e. “solve the case.”

- *Issues/causes:* 1) store locations – 60% make 10% loss, remaining 40% make 25% profit. 2) product mix – client’s main business is its’ pharmacy, pharmacies occupy the most space and provide the smallest margin.
- *Potential solutions:* 1) relocate away from hospitals – lots of competition, some dangerous neighborhoods, and lower margins. 2) Find a way to adjust product mix – maybe move pharmacy to back of store so consumers have to walk through beauty section to get there, or rearrange the store to devote more space to beauty products.

A good candidate’s response will look something like what is above. However, a great candidate will turn this into a real conversation. Consider this from the client’s perspective. If the client is similar to CVS, then they are best known for being a pharmacy. As a pharmacy, they need to be located near hospitals since hospitals are where people get prescriptions. Additionally, as a drug store, they need to have a good pharmacy section. It is tempting to look at this case and just say: “move away from hospitals, decrease pharmacy size, and increase health and beauty section,” but the issue you then run into is that you’re essentially telling a CVS that their best strategy is to become Sephora. Instead you have to look at ways to increase profitability without harming their core business. One idea would be expanding the beauty section and marketing that aspect of the store further. Another possibility would be reorganizing the store to encourage pharmacy customers to make impulse purchases in other sections. These solutions are by no means exhaustive, but a great candidate will turn the case into a discussion like this and generate many solutions.

Case 4: Drug Store Profitability (Example Framework)

Main questions	Why is profitability declining? How can we reverse this trend?
Key areas to explore	<u>Market</u> - Increase in competition? - Industry trends? <u>Revenue</u> <u>Price</u> - How have prices changed? <u>Quantity</u> - How have sales quantities changed? <u>Product Mix</u> - Changes in product mix? - Selling less profitable products? <u>Costs</u> <u>Fixed</u> - Changed to more expensive locations? <u>Variable</u> - Utilities costs changes? - Product cost changes?
Analysis	60% of stores are in locations with a 10% loss. Other 40% generate a 25% profit. - Client is a drug store, and their pharmacy generates the smallest margin despite occupying the largest portion of the store. Other sections of the store generate higher margins.
Recommendation	Decision: Expand stores sizes to increase Health & Beauty and General Merchandise sections. Open more stores away from hospitals. Rearrange stores to encourage pharmacy customers to make impulse purchases in other sections. Adjacency Examples: Offer new, higher-margin products in stores; move more of business online to reduce costs, expand into new geographies.
Other factors (risks)	- Client needs to maintain core business - Dangers of expanding store sizes in competitive areas

Case 4: Drug Store Profitability (Cont'd)

Exhibit 1

Product Type	Sales/Sq. Ft	Contribution Margin
Pharmacy	\$20,000/ sq. ft.	5%
Health & Beauty	\$10,000/ sq. ft.	20%
General Merchandise	\$5,000/ sq. ft.	10%

Case 5: McNews

Prompt: McNews is a global newspaper firm, producing and distributing newspapers globally. Recently executives realized a decrease in profits and have hired you to investigate the issue. What kind of suggestions can you provide?

Interviewer Information:

- Assume costs have remained constant
- Similar/major competitors have seen a similar decline
- The decline is mainly in 4 cities: New York, London, Chicago and Frankfurt
- Prices have not changed

Notes for Interviewer:

This is a very typical profit and loss case. When examining the exhibits the candidate should notice a several things. First, print is more profitable than electronic. Second, non-daily is more profitable than daily. Third, the largest segment (print daily) has seen a sharp decline, but was the only segment to decline. Fourth, overall electronic is increasing while print is decreasing. Fifth, print non-daily increased. Finally, North America and Europe are largest markets, but Asia is also substantial. At this point the candidate should look at adjacencies and ways for the client to shift away from the rest of the declining market. Some potentials include: moving to a new geography (Asia is very appealing), selling a new product mix, finding new products (maybe TV), acquiring new businesses, divesting less profitable/promising businesses, merging with key competitors, finding ways to cut costs, etc. The candidate should then discuss the risks associated with whatever recommendations he/she made. The biggest risk in a scenario involving product mix is that the products are complimentary in some way and thus, a shift may not work. For instance, if the non-daily print business was successful because of the popularity of the daily print business, then shifting away from daily print would hurt non-daily print. A great candidate will be able to list multiple adjacencies, discuss the risks associated with each, and make a clear recommendation without too much guidance.

Case 5: McNews (Example Framework)

Main questions	Why is profitability declining? How can we reverse this trend?
Key areas to explore	<u>Market</u> • Increase in competition? • Industry trends? <u>Revenue</u> <u>Price</u> • How have prices changed? <u>Quantity</u> • How have sales quantities changed? <u>Product Mix</u> • Changes in product mix? • Selling less profitable products? <u>Costs</u> <u>Fixed</u> • Changed to more expensive locations? • New Investments? <u>Variable</u> • Utilities costs changes? • Product cost changes?
Analysis	Electronic daily - \$.10, electronic non-daily - \$1.40, print daily - \$.40, print non-daily - \$9.20 → print is more profitable than electronic and non-daily is more profitable than daily. Non-daily segment is increasing, while print segment is decreasing and electronic segment is increasing. Client's largest geographies are North America and Europe, where profit is declining. Asia is also substantial market with no decline.
Recommendation	Decision: This is an industry-wide problem so McNews needs to find a way to make itself different than its' competitors. That could mean just focusing on non-daily publications, focusing on its' online presence, divesting its print business, an increased marketing spend, looking to expand into Asia since that's a growing market, look for ways to cut costs, etc.
Other factors (risks)	Staying in a declining industry is always risky. What is the connection between daily/non-daily, what is the connection between print and electronic? If they're complimentary then cutting one hurts the other. What are competitors doing?

Case 5: McNews (Cont'd)

Exhibit 1: Revenue by Geography

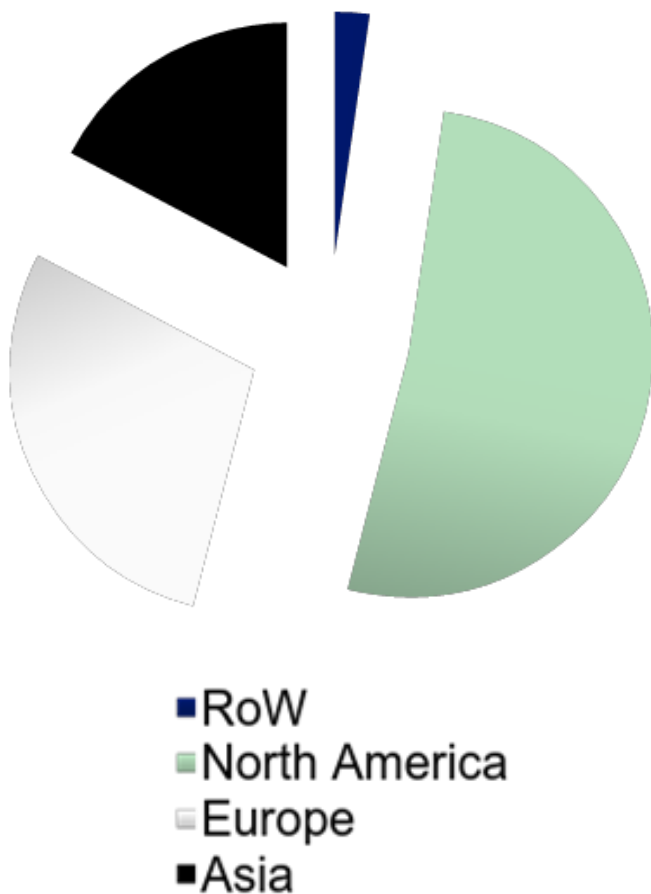


Exhibit 2: Revenue by Product Line

	Sales in \$mm		Avg. selling price per unit (\$)
	2008	2009	
Electronic daily	41	41	0.4
Electronic non-daily	33	39	2.5
Print daily	223	188	2.5
Print non daily	98	103	12.0

Exhibit 3: Cost Per Unit by Product Line

	Cost in \$	
	Production	Distribution
Electronic daily	0.20	0.10
Electronic non-daily	0.30	0.80
Print daily	2.00	0.10
Print non daily	2.00	0.80

Case 6: Playing with the Law

Prompt: A reputed law firm has retained you to inspect the company's finances. The company sells to several industries mainly in its local region: New York City. The CEO just reviewed the company's performance and projections and is not happy—he wants your ideas in helping boost profits. What do you recommend?

Interviewer Information:

- Business Model: Law firms earn revenue by paying employees a fixed wage per hour (plus a discretionary year-end bonus), then billing clients at a higher rate.
- Revenue = \$46 million. The company's lawyers bill \$300 per hour on average
- Profits = \$4.5 million
- Number of employees: 78
- Industry focuses: healthcare, energy, financial services, and telecommunication
- Headquartered in New York City and sells almost entirely to that region

Notes for Interviewer:

This case seems challenging because it is not a traditional business case, but like others it is really just about profit and loss. Since billable hours are the main driver of revenue and profit, the interviewee should focus on increasing billable hours in order to increase profitability. After examining the exhibits the interviewee should note that things are stable across all industries, except financial services which has seen a decline in billable hours in recent years. This means that the decline in this firm's financial services practice is the cause of its overall profitability decrease. Following this conclusion the candidate should identify several ways to increase the number of billable hours or to decrease the company's cost structure.

Case 6: Playing with the Law (Exhibit Explanation)

Exhibit 1: New York is a hub of Financial Services, yet the client has one of the lowest share of revenue from that industry. The interviewee should indicate that she/he wants to know why this is such.

Exhibit 2: Operating costs are rising dramatically: a 25% increase this year, and a projected 43% increase next year

- Main cost increase from personnel (30% increase); sales and marketing (10% increase), and SG&A (5%)

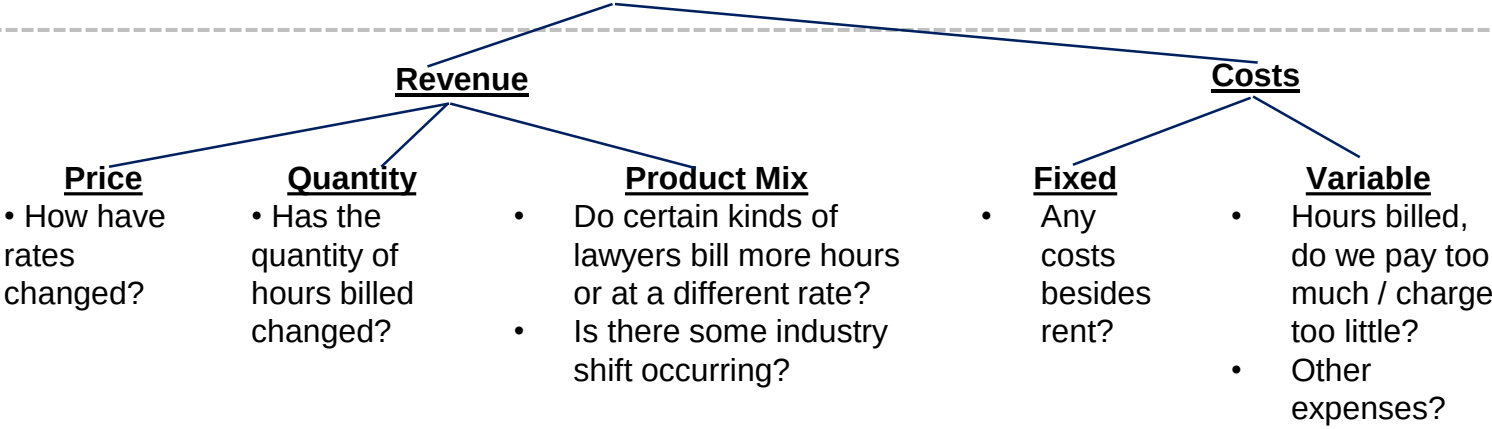
Exhibit 3: Does not provide too much insight, but does highlight that most of the employees are getting their year end bonuses. This could be the reason why personnel cost has been increasing.

- Total revenue/total employees: \$46 Million/78 = ~ 600K is the revenue per employee
- 600K/ hourly rate = ~ 2,000 hrs per year
- Spread evenly over 52 weeks, this results in 38 hrs/week → Lawyers are being underutilized.

Exhibit 4:

- Financial services hours dropping dramatically
- Industry mix not changing, except financial services

Case 6: Playing with the Law (Example Framework)

Main questions	Why is profitability declining? How can we reverse this trend?
Key areas to explore	 <pre> graph TD Root[Why is profitability declining? How can we reverse this trend?] --> Revenue[Revenue] Root --> Costs[Costs] Revenue --> Price[Price] Revenue --> Quantity[Quantity] Revenue --> ProductMix[Product Mix] Costs --> Fixed[Fixed] Costs --> Variable[Variable] Price --> PriceQ[How have rates changed?] Quantity --> QuantityQ[Has the quantity of hours billed changed?] ProductMix --> ProductMixQ[Do certain kinds of lawyers bill more hours or at a different rate? Is there some industry shift occurring?] Fixed --> FixedQ[Any costs besides rent?] Variable --> VariableQ[Hours billed, do we pay too much / charge too little? Other expenses?] </pre>
Analysis	Billable hours are stagnant across all segments, except financial services. Financial services billable hours are actually declining. Increasing personnel costs mean that client is paying employees more, despite a lack of increasing revenue.
Recommendation	Decision: In order to boost profits the client either needs to increase the number of hours billed or cut costs. There are several ways the client can accomplish this: For boosting revenue the client could market its' financial services better – being in NYC this should be the client's largest sector. Alternately the client could divest of its' financial services division and try to become a leader in healthcare or energy. With respect to costs, the client could look at its' bonus structure to ensure that it is not giving out unwarranted bonuses. Also, the client could assess whether its employees working less than 40 hours/week actually qualify for healthcare or other ancillary benefits.
Other factors (risks)	There may be risks with trying to change what the law firm is known for and risks with changing employee benefits if employees have signed certain contracts.

Case 6: Playing with the Law (Cont'd)

Exhibit 1: Revenue Breakdown (This Year)

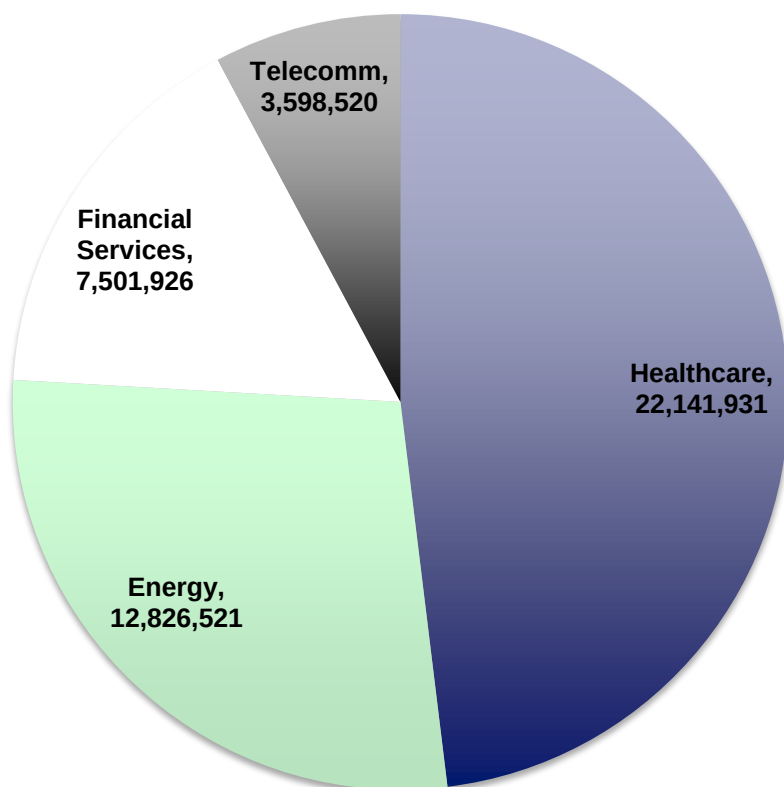
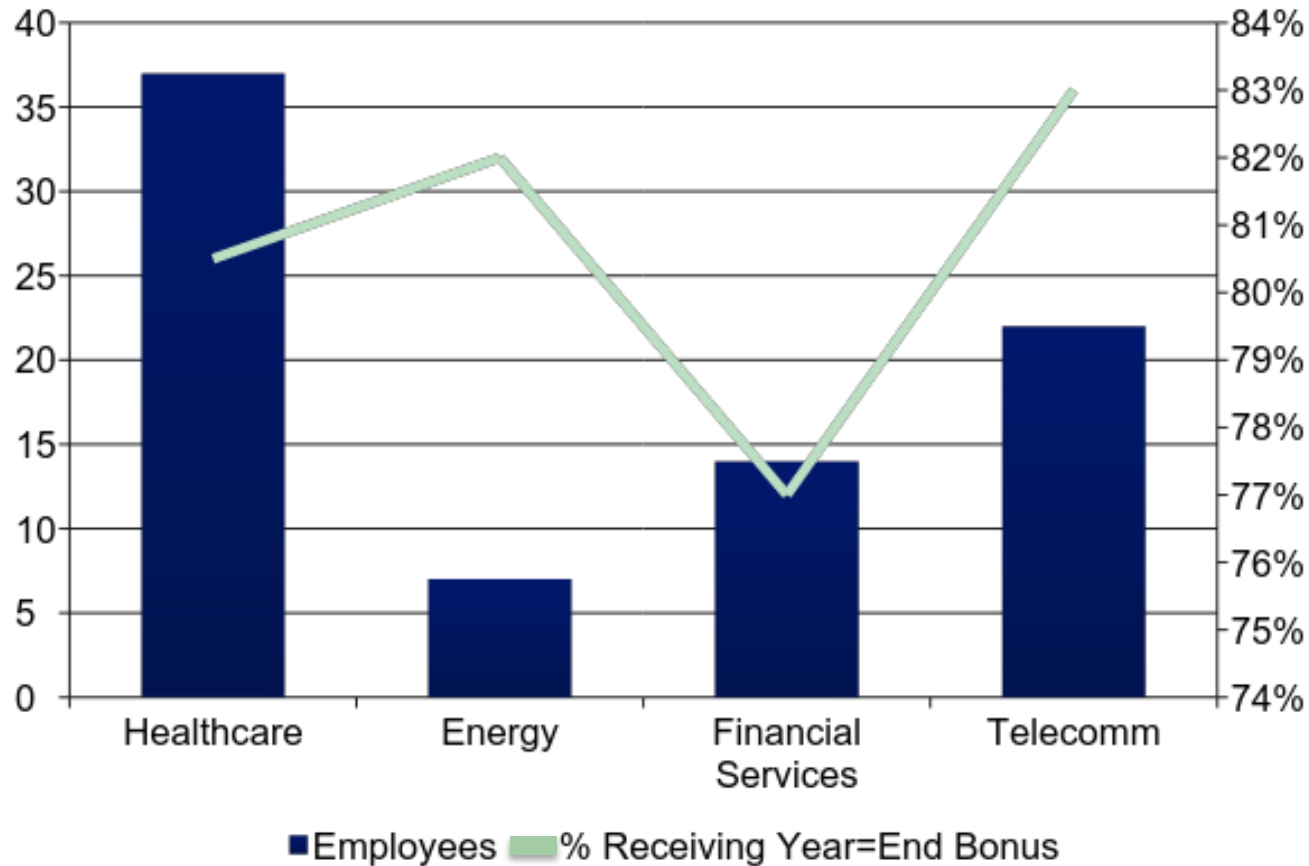


Exhibit 2: Operating Costs

	Last	This Year	Next Year
Sales and Marketing	439,311	461,277	484,341
Rent	153,509	153,509	153,509
General & Administrative	68,672	72,106	75,711
Research & Development	3,600	3,600	3,600
Personnel	3,151,385	4,096,801	6,145,202
Depreciation & Amortization	58,182	58,182	58,182
Interest	22,500	22,500	22,500
Taxes	269,697	336,881	480,483
Total Operating Costs	\$4,166,856	\$5,204,856	\$7,423,528

Case 6: Playing with the Law (Cont'd)

Exhibit 3: Employees by Industry



Case 6: Playing with the Law (Cont'd)

Exhibit 4: Billable Hours per Employee

	Last	This Year	Next Year
Billable Hours	Year	End	Projected
Healthcare	2,101	2,150	2,123
Energy	2,077	2,032	2,049
Financial Services	1,800	1,676	1,372
Telecommunications	2,140	2,142	2,147

Case 7: Gardening Retailer

Prompt: You have been approached by the COO of a \$2B Midwest Industrial Company that makes fertilizers, pesticides, etc. for lawns and gardens. They have recently acquired a \$100M retail chain selling patio furniture, garden tools, clothing, décor, ornaments, etc. The retail chain has both retail stores and direct order business. The retail chain has a gross margin of 50%, but is still losing money. Why are they losing money? What should they do?

Interviewer Information:

- *Locations:* 50 retail stores in the U.S, stores are upscale and located in expensive places, goods sold by salaried sales people
- *Customers:* Most of the customers are in warm locations (CA, AZ, FL), 40% of customers in California.
- *Procurement:* Purchased from China
- *Transportation:* Shipped in large containers to Long Beach, CA. Shipping costs included in COGS (50%)
- *Warehousing:* Goods transported by trucks to the warehouse in Kentucky, stored in warehouse and redistributed according to demand
- *Distribution:* Transported by trucks to different retail stores

Notes for Interviewer:

The important part of this case is to not get stuck down in answering “why are they losing money?” and to instead quickly shift to ways to increase profitability. It should be obvious that the client’s method of shipping to Kentucky and distributing from there is causing a significant portion of its’ losses. The client also has numerous options to expand to grow revenue and profit. One method of depicting those options is a 2x2 matrix:

	Same Products	New Products
Same Customers	No Change	Sell flowers, plants, etc.
New Customers	Expand in NV, TX, GA, etc.	Combination

Case 7: Gardening Retailer (Example Framework)

Main questions	Why is profitability declining? How can we reverse this trend?
Key areas to explore	<u>Revenue</u> <u>Price</u> - Have prices changed? <u>Quantity</u> - How have sales quantity changed? <u>Product Mix</u> - Any changes in product mix sold/offered? - Selling less profitable products? <u>Costs</u> <u>Fixed</u> - Ways to reduce distribution costs? <u>Variable</u> - How do COGS compare to industry? - Ways to lower COGS?
Analysis	- Distribution: China → CA → KY → CA/AZ/FL. 40% of sales in CA. - COGS = 50%. - Segmentation: High end customers in high end, warm-weather locations
Recommendation	Decision: Move to distribution centers on East and West Coast – should save 40% on costs of shipping CA – KY – CA. Then look at ways to expand sales. Adjacency Examples: Offer new, higher-margin products in stores. Sell online or explore other distribution channels. Explore selling in new locations or begin targeting lower-end customers like Walmart and Target.
Other factors (risks)	Moving distribution center: CAPEX of building new distribution center could outweigh savings. Expansion: CAPEX could outweigh new revenue if margins still too low. Expansion could tarnish brand if not done properly.

Case 7: Gardening Retailer (Cont'd)

Exhibit 1: Income Statement

Income Statement	
Revenue	\$100M
COGS	\$50M
Transportation/Distribution	\$15M
Marketing	\$11M
Fulfillment	\$11M
Labor + Store Rent	\$11M
SG&A	\$6M



Direct Order Costs
Designing
Printing
Distribution/Shipping

Case 8: Choco Bars

Prompt: You are consulting for a company that produces and distributes chocolate bars (ChocoBar). They are the second largest of two major global players and compete largely in the United States. Last year, they had net revenue of \$800 million. They have noticed a profit margin gap relative to their competitors and so they have asked you to identify what the cause of the margin gap is and how to improve it, if possible.

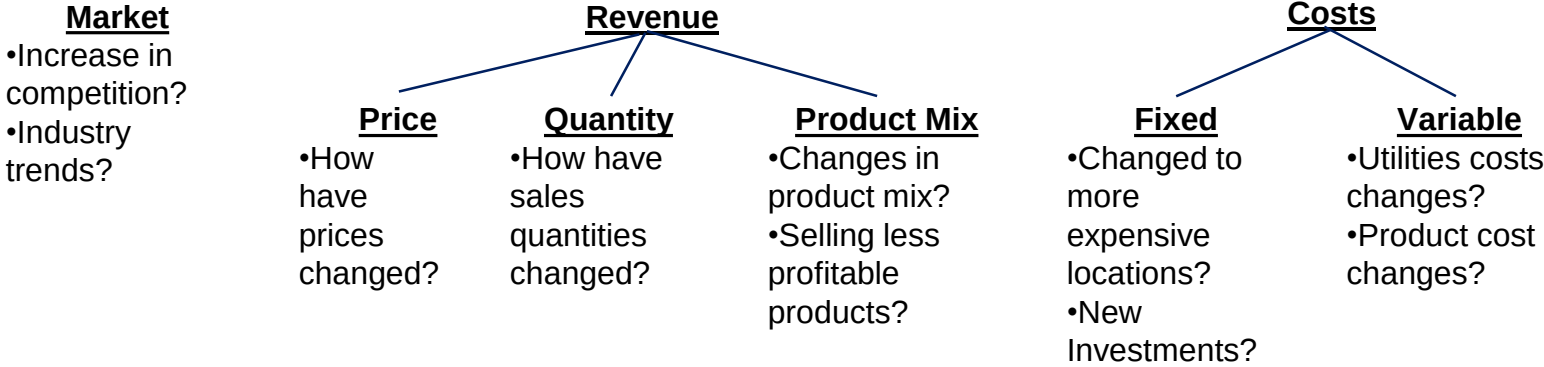
Interviewer Information:

- *Market*
 - Major competitor, CompoBar, has \$1 billion in revenue, and the market is growing at 3%.
- *Product*
 - CompoBar sells its chocolate bars just about everywhere that ChocoBar does, both have a similar U.S./International mix and differentiation is minimal. They sell each bar for \$2.00, but ChocoBar's sales people have been adding a \$0.50 discount. CompoBar has not been giving this discount. ChocoBar sell 5 different types of bar, but CompoBar also sells products 3, 4, and 5, and NOT 1 or 2.
- *Company*
 - Big, branded company that sells only confectionary products

Notes for Interviewer:

- After the interviewee does his/her initial framework give exhibits one at a time and ask the candidate what it means and how it affects his/her initial hypothesis. From exhibit 1 the candidate should notice similar cost structures, but \$200M of discounts for the client. From exhibit 2 the candidate should notice that the client will not see a tremendous drop in sales from removing the discount. Over 60% of the company's sales come from bars 1 and 2, which will only see a 5-10% volume drop when price is increased by 33%. Exhibit 3 should confirm the candidates predictions from exhibit 2. Have the candidate fill out his/her copy of exhibit 3. After exhibit 3 ask the candidate for some alternate ideas and any risks/challenges associated with those suggestions.

Case 8: Choco Bars (Example Framework)

Main questions	Why is profitability declining? How can we reverse this trend?
Key areas to explore	 Market • Increase in competition? • Industry trends? Revenue Price • How have prices changed? Quantity • How have sales quantities changed? Product Mix • Changes in product mix? • Selling less profitable products? Costs Fixed • Changed to more expensive locations? • New Investments? Variable • Utilities costs changes? • Product cost changes?
Analysis	Cost structure is similar to competitor, but \$200M of discounts is hurting company. ChocoBar can remove discounts (effectively increasing price by 33%), and see a 57% increase in gross profit. All five lines of chocolate bars are profitable and profit will not decrease for any of the five with removal of discount.
Recommendation	Decision: Stop allowing sales people to offer discount on products. This should increase gross margin by 57% Adjacency Examples: Offer a new/higher margin bar, try to differentiate products, explore new distribution channels or new locations. Diversify beyond chocolate bars. Do some kind of M&A to enter a new market.
Other factors (risks)	Biggest risk with recommendation is alienating sales people. Right now they're probably incentivized based off volume sold, so they lower prices to sell more. Realigning their incentives may be the best way forward, possibly changing incentives to be based off the margins of their sales instead of volume.

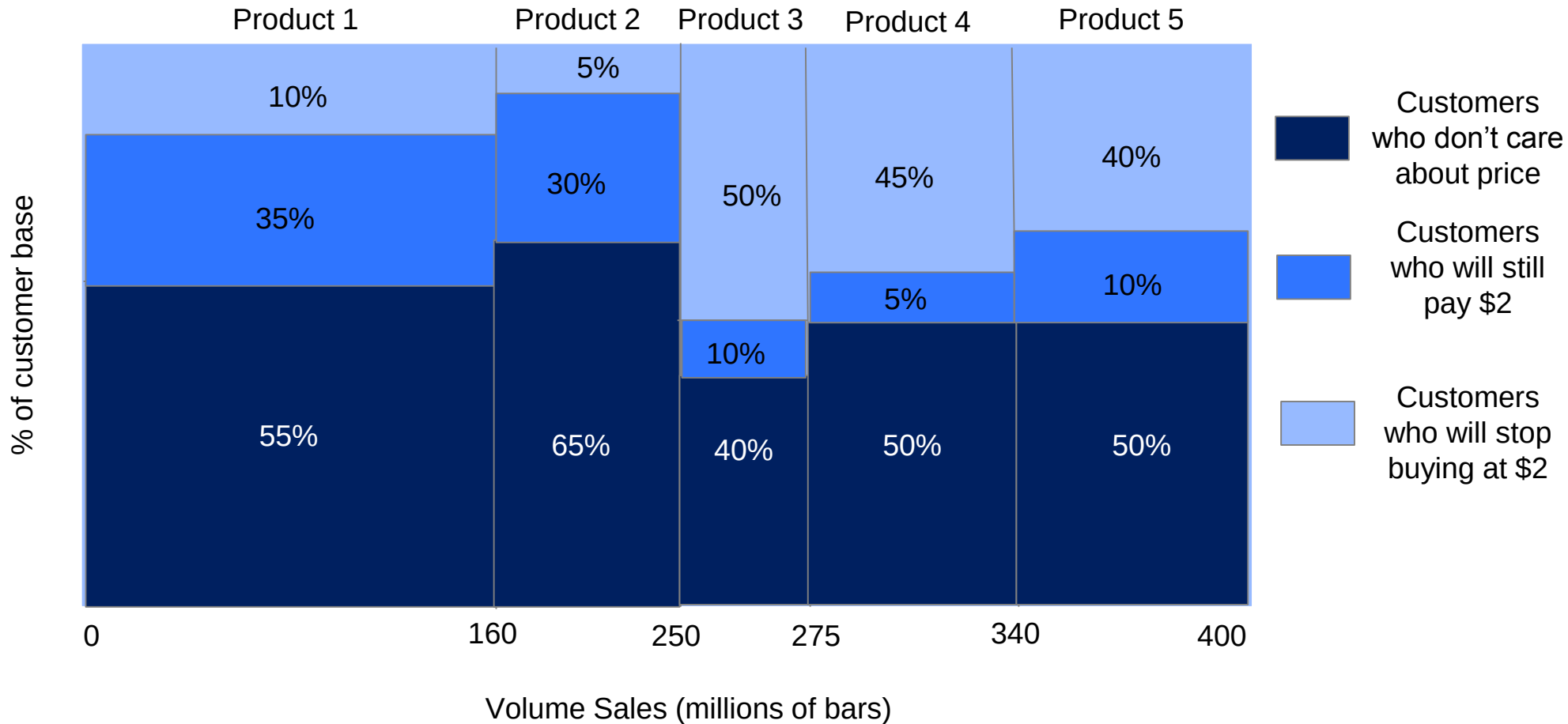
Case 8: Choco Bars (Cont'd)

Exhibit 1: Comparable Income Statement

	ChocoBar	% of Total Rev	CompoBar	% of Total Rev
Total Revenue	\$800		\$1,000	
Less: Discounts	200		0	
Net Revenue	\$600	75%	\$1,000	100%
COGS	400	50%	500	50%
Gross Profit	\$200	25%	\$500	50%
S&M	51	6%	60	6%
G&A	40	5%	45	5%
R&D	31	4%	43	4%
Total Op. Exp.	122	15%	148	15%
EBITDA	\$78	10%	\$352	35%
D&A + Int. + Taxes	10	1%	14	1%
Net Income	\$68	9%	\$338	34%

Case 8: Choco Bars (Cont'd)

Exhibit 2: Pricing by Product



Case 8: Choco Bars (Interviewer Copy)

Exhibit 3: Gross Profit

Existing Gross Profit

	Product 1	Product 2	Product 3	Product 4	Product 5	Total
Existing Volume	160	90	25	65	60	400
Price per Bar	2	2	2	2	2	
Total Revenue	\$320	\$180	\$50	\$130	\$120	\$800
(Less): \$0.50 Discount per Bar	\$(80)	\$(45)	\$(12.5)	\$(32.5)	\$(30)	\$(200)
Net Revenue	\$240	\$135	\$37.5	\$97.5	\$90	\$600
GOGS per Bar	\$1	\$1	\$1	\$1	\$1	\$1
Gross Profit	\$80	\$45	\$12.5	\$32.5	\$30	\$200

New Gross Profit

	Product 1	Product 2	Product 3	Product 4	Product 5	Total
New Volume	144	85.5	12.5	36	36	314
Price per Bar	\$2	\$2	\$2	\$2	\$2	
Total Revenue	\$288	\$171	\$25	\$72	\$72	\$628
GOGS per Bar	\$1	\$1	\$1	\$1	\$1	\$1
Gross Profit	\$144	\$85.5	\$12.5	\$36	\$36	\$314

Case 8: Choco Bars (Interviewee Copy)

Exhibit 3: Gross Profit

Existing Gross Profit

	Product 1	Product 2	Product 3	Product 4	Product 5	Total
Existing Volume	160	90	25	65	60	400
Price per Bar	2	2	2	2	2	
Total Revenue	\$320	\$180	\$50	\$130	\$120	\$800
(Less): \$0.50 Discount per Bar	\$(80)	\$(45)	\$(12.5)	\$(32.5)	\$(30)	\$(200)
Net Revenue	\$240	\$135	\$37.5	\$97.5	\$90	\$600
GOGS per Bar	\$1	\$1	\$1	\$1	\$1	\$1
Gross Profit	\$80	\$45	\$12.5	\$32.5	\$30	\$200

New Gross Profit

	Product 1	Product 2	Product 3	Product 4	Product 5	Total
New Volume						314
Price per Bar	\$2	\$2	\$2	\$2	\$2	
Total Revenue						
GOGS per Bar	\$1	\$1	\$1	\$1	\$1	\$1
Gross Profit						

Interviewee should fill out blank spots on New Gross Profit table

Case 9: Energy Drinks

Prompt: Our client is a large, well-known beverage company that produces a popular energy drink. Last month they ran a distinct trade promotion in three separate stores, and they have hired you to tell them whether or not the promotions were successful. How would you go about helping them figure this out?

Interviewer Information: The interviewee must think through what constitutes a “successful” promotion, and then synthesize data to provide a recommendation to the client. *For this case, a “successful” promotion is one that generates more profit than would be attained without running any promotion at all.* However, the interviewer should ask the interviewee to define how success might be measured, and encourage the interviewee to brainstorm several other barometers of success, besides profits. This information should be provided ONLY if asked for by the interviewee.

- *Promotions run:* Store 1 was given a 10% discount; Store 2 was given a “buy 4, get 1 free” deal; Store 3 was given \$10,000 to promote the product however they wanted
- *Volume purchased by each store during promotion:* Store 1 purchased 12,000 cans; Store 2 purchased 15,000; Store 3 purchased 14,000
- *Volume base case (non-promotion environment):* In a non-promotion month, each store typically buys 10,000 cans
- *Price:* Each can is sold for \$5
- *COGS:* The cost to produce a can is \$2.50. This figure includes all relevant costs (i.e. raw materials, manufacturing, distribution, etc.)

Notes for Interviewer: Interviewee should identify there is a base case (status quo) profitability and promotion-based profitability for each store. They should calculate profits for the non-promotion environment (applies to all 3 stores) and the promotion environment for each of the 3 stores (will be different for each store) to compare which promotions, if any, were successful.

Case 9: Energy Drinks (Example Framework)

Main questions	Are the promotions successful? Specifically, are they generating higher profits than the status quo?
Key areas to explore	<u>Profit</u> - More revenue brought in via increased volume? - Marketing? - Do the discounts, bundles, and marketing money adequately offset status quo/base case? <u>Other Factors</u> - Repeat customers (stores and individuals)? - Market share?
Analysis	<u>Base Case Profit Calculation</u> - Normal, or baseline sales: (10,000 cans * \$5.0 revenue per can) – (10,000 cans * \$2.5 cost per can) = \$25,000 profit <u>Profit Calculations for Promotions</u> - Store 1: (90% * 12,000 cans * \$5.0 revenue per can) – (12,000 cans * \$2.5 per can) = \$24,000 profit - Store 2: (80% * 15,000 cans * \$5.0 revenue per can) – (15,000 cans * \$2.5 per can) = \$22,500 profit - Store 3: (14,000 cans * \$5.0 revenue per can) – (14,000 cans * \$2.5 per can) - \$10,000 = \$25,000 profit
Recommendation	- Promotions 1 and 2 lead to lower profits, so not useful - Promotion 3 yields the same profits as non-promotion operations - Does this mean that none of the strategies are worthwhile? Not necessarily. See 'Other Factors (Risks)' below
Other factors (risks)	- Moving forward with the 3rd strategy should be based on things such as: - Market trends: perhaps the competitors are heavily promoting their products - Long term effects of promotion: market share, brand loyalty, affecting customer buying habits

Case 10: Cloudy Days Ahead

Prompt: Our client is an Atlanta base healthcare company that specializes in children's diseases. They have been among the highest revenue-earning players in the 10 southern states. Their IT department realized that their data storage capacity is not large enough to meet the increasing needs. They are interested in using cloud computing for additional storage need. Should they do so? If so, how long until our client take to break even (compare to the current system)? If not, what strategic alternatives can you offer?

Interviewer Information.

- Our client needs 1 server for each \$300 million in new revenue in each location. For example, a location with \$200 million in new revenue requires 1 server, \$400 million in new revenue requires 2 servers, and \$700 million in new revenue requires 3 servers.
- Servers run 24 hours per day, 365 days per year
- The company is also considering “self implementation” of cloud computing: purchase and implement the equipment itself, rather than pay a service provider.
- Vendors being considered: O’Rak Corp and Mell Computers

Notes for Interviewer: This is a fairly straight forward case. The interviewee should talk about comparing the current costs and the cost in the future with the various cloud computing options before making the final decision. Also, the interviewee should talk about the cost driver, which in this case is new revenue. Finally, the interviewee should discuss the dangers of working with a new partner.

- Annual costs:
 - Self implementation: $((50\text{kW/hr} * 24 \text{ hrs}) * 12 \text{ servers}) * 365 * \$0.05 / \text{kW}) + (10 * \$100,000) = \$1,262,800$
 - O’Rak: $(\$2.5/\text{server} * 365 \text{ days} * 12 \text{ servers}) + (2 * \$100,000) + \$700,000 = \$910,950$
 - Mell: $(\$2.0/\text{server} * 365 \text{ days} * 12 \text{ servers}) + (3 * \$100,000) + \$600,000 = \$908,760$

Case 10: Cloudy Days Ahead (Example Framework)

Main questions	Which option should the company use? How much money will it save them?			
Key areas to explore	<u>Cost Savings</u> How much money will we save by switching?	<u>Cost of Change</u> - Implementation cost of each option? - Annual cost of each option?	<u>Options</u> - Are there other vendors? - Could client do this themselves?	<u>Strategic Fit</u> - Do any of these options fit better with our strategy? - Do we have any non-financial concerns about options?
Analysis	- Self Implementation: Annual savings are \$230,000. So the client will take $5M / \\$230,000 = 21$ years to break even. - O’rak Corp: Annual savings: \$580,000. So to break even, the client will take $2.1M / \\$580,000 = 3.6$ years. - Mell Computers: Annual Savings: \$590,000. So to break even, the clients will take $\\$2.7 / \\$590,000 = 4.5$ years			
Recommendation	Decision: Use O’rak Corp → fastest payback period and there seems to be no quality difference between options. Other Ideas: Try to get companies to bid lower (compete bids with each other). Look for ways to cut costs in contracts (like finding cheaper engineers). Search for other storage options besides current system and cloud.			
Other factors (risks)	Working with a new partner always poses some risks, especially if there is not any history of working together			

Case 10: Cloudy Days Ahead (Cont'd)

Exhibit 1: Revenue and Local Market Share

	2009 Expected	Local Market
	Revenue Growth \$mm	Share (%)
Florida	300	30
Kentucky	80	33
Georgia	550	66
Alabama	400	55
Texas	180	21
North Carolina	120	27
South Carolina	40	29
Louisiana	50	35
Oklahoma	60	42
Virginia	40	22

Case 10: Cloudy Days Ahead (Cont'd)

Exhibit 2: Vendors

	Cloud Computing Market Share (%)	Public Stock: Year-over-Year Change	Relations with Us	Recent News
O'rak Corp.	12	2.1%	Medium	Acquired Southern U.S. Sunny Systems, a major data storage company
Mell Computers	8	(12.3)%	Medium	Entered 4 different emerging markets in last 3 months

Exhibit 3: Costs

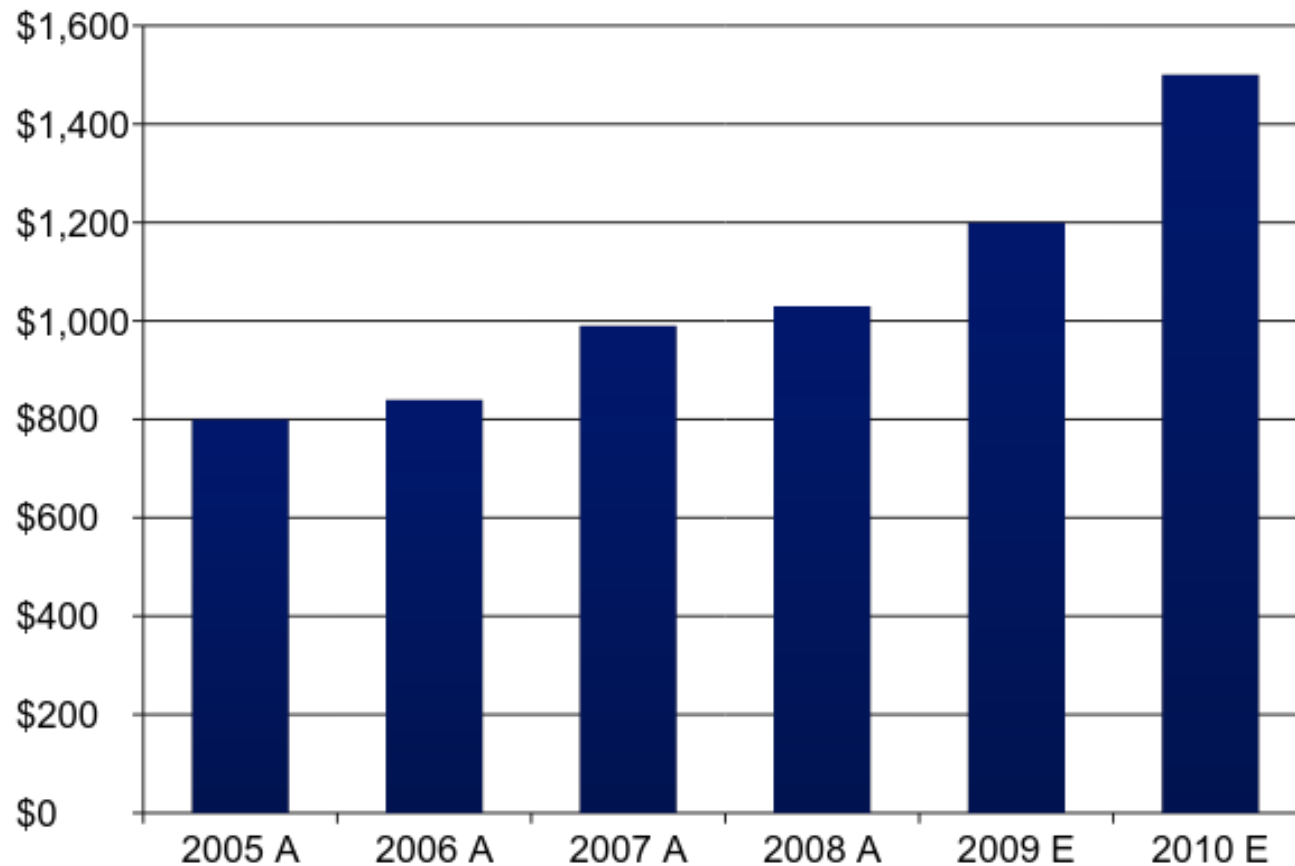
Cloud Options	Installation Cost*	Daily Fees (\$)	Energy Consumption**	Full-Time Engineers	Annual Fee (\$)
Self Implementation	\$5.0 million	0	50 kW per hour per server	10	0
O'rak Corp.	\$2.1 million	\$2.50 / day per server	0	2	700,000
Mell Computers	\$2.7 million	\$2.00 / day per server	0	3	600,000

- This is the cost to install all 12 servers

** The cost per kW is \$0.05

Case 10: Cloudy Days Ahead (Cont'd)

Exhibit 4: IT Costs for Continuing with Current Storage System (\$ Ten Thousands)



Case 11: Shipping

Prompt: Our client is in the business of delivering time sensitive shipments all over the United States. Currently, when a customer sends a package, he or she must call the client's call center and provide a tracking ID in order to find out where the shipment is and when it is expected to be delivered. Our client has asked you to advise whether or not to implement an online tracking system for their customers to be able to track shipments.

Interviewer Information:

Data (given to candidate when asked):	Calculation
1 Million customers / yr	None - given
10 shipments / cust.	10 Million shipments per year
80% call, w/ 1 call per shipment,	8 Million calls per year
Cost per call: \$1.50 (Assume this is 100% variable)	Annual VC of call center: \$12M
Cost of implementing online system: \$1.5M	None - given
Break-even # of calls saved:	$\$1.5M / \$1.5 = 1M$ calls saved
Required customer conversion rate:	$1M / 8M = 12.5\%$
Estimated conversion rate: 90%	$90\% > 12.5\% \rightarrow \text{GO}$

Notes for Interviewer: This is a straight forward cost-benefit analysis. Candidates should be able to setup a framework (almost like market sizing) to determine the break-even # of phone calls saved. However, a great candidate will also inquire about payback period and the company's WACC. A great candidate will also lay out a basic framework like on the next slide to put this decision into a greater context. Finally a great candidate will attack the numbers and give the payback period and annual savings with 90% conversion rate.

Case 11: Shipping (Example Framework)

Main questions	Should client implement online tracking system?
Key areas to explore	<u>Client</u> - Cost of implementation? - Potential cost savings? - Impact on organization - Financial expectations (WACC / payback) <u>Competition</u> - Do competitors offer similar service? <u>Customers</u> - Would they want this? - Can they use new system?
Analysis	- Economically sound decision, only 12.5% conversion required to break-even in year 1. - If company's 90% estimate is correct → 90% x 8M calls = 7.2M calls saved → \$10.8M annual savings, \$9.3M savings in year one, payback occurs within 2 months. - Potential competitive advantage if competitors do not offer a similar service.
Recommendation	Decision: Make the investment, required conversion rate is far below company's projections and the decision could give the client a competitive advantage. How to Raise Awareness / Increase Conversion: Automated phone call to customers, direct mail to current customers, sales people at shipping points, labels on boxes, etc.
Other factors (risks)	- If the system doesn't work - If clients don't convert - Downside of reducing call center employees - Maybe this is a company in a developing location and many customers don't have access to internet

Case 12: Groceries

Prompt: Our client is a local organic/natural foods grocery store. They have seen the success national chains have had with prepared food offerings and are looking to duplicate that success. They are looking to us to help them decide which of the three options to pursue:

- Option 1: Pre-made Food – Our client will buy a custom cooler and have employees produce and wrap a number of sandwiches every morning before the store opens
- Option 2: Full Service Counter – Our client will build a full-size deli counter and staff it throughout the day with employees who will make sandwiches to order
- Option 3: Partnership – Our client will partner with an outside chain (e.g. Subway, Quiznos, etc.) and install a mini-restaurant in front of the store

Interviewer Information:

- Company has a payback period of 2 years
- Company stats:
 - Customers served per store = 2000/day
 - Average purchase per customer = \$100
 - Groceries have a 50% margin

Notes for Interviewer: This is a typical “entering a new market” case: the $P = R - C$ framework will be a useful element. The challenge is to maintain a clear and consistent framework throughout multiple calculations. Once the interviewee lays out his/her framework ask what kind of information the candidate would need to calculate the attractiveness of each of the options. Then give the candidate exhibit 1 and encourage him/her to attack the numbers to come up with a daily profit. After the candidate identifies an option (either 1 or 2 can be argued for), encourage the candidate to think of the ramifications. For instance, how will the competition respond, what kind of marketing can you do, can you save money operationally by using safe, but unsightly store produce, are there other ways to partner with outside stores?

Case 12: Groceries (Example Framework)

Main questions

- Which option should the company go with?

Key areas to explore

	Option 1	Option 2	Option 3
Expected Revenue	# of customers? Avg \$ per customer?		
Expected Costs	Variable costs? Fixed Costs? Installation Costs?		
Other considerations	Increase in # of customers overall? Effect on customer grocery purchases?		

Analysis

- Option 1:** \$307 profit per day, but not additional grocery revenue. Payback period 33 days.
- Option 2:** \$170 profit per day, but drives an additional \$5,000 of grocery profits. Payback period 97 days.
- Option 3:** -\$2,683 profit per day plus \$10,000 of grocery profits. Payback period of 35 days.

Recommendation

- Decision:** Option 3. While the option on its own is a loss, the additional grocery profits from bringing in more customers significantly outweighs the total profitability of Options 1 and 2. Additionally, the payback period is modest at 35 days.
- Other options:** Do options 1 and 3, use soon-to-expire groceries in foods to reduce spoilage, payback period is approx. the same time period for both options.

Other factors (risks)

- How will the competition respond?
- If demand increases or decreases rapidly how will this effect options?
- What if margin from groceries declines to 10%, does that change which option?

Case 12: Groceries (Cont'd)

Exhibit 1: Revenue and Cost Breakdown

Revenues	Options 1: Pre-Made/Cooler	Option 2: Full Service Counter	Option 3: Full Service Partnership
Customer effect	0% change	5% increase	10% increase
Meals bought	5% of customers	10% of customers	10% of customers
Client share	100%	100%	20%
Price of meal	\$5	\$10	\$8
Costs			
Initial outlay	\$10,000	\$500,000	\$250,000
Overhead (power/maintenance/etc)	\$3/day	10% of meal revenue	10% of meal revenue
Materials (food/utensils/etc)	30% of meal	20% of meal	\$0
Labor	\$40	\$300	\$0
Opportunity cost (lost grocery revenue from space reduction)	0 (small cooler)	\$1000	\$3000

Case 12: Groceries (Solution)

Revenue	Options 1: Pre-Made/Cooler	Option 2: Full Service Counter	Option 3: Full Service Partnership
Meal revenue (% buying meals * total customers * meal price * client share)	$5\% * 2,000 * \$5 * 100\% = \500	$10\% * 2,100 * \$10 * 100\% = \$2,100$	$10\% * 2,200 * \$8 * 20\% = \352
Cost			
Variable – overhead	\$3	$\$2,100 * 10\% = \210	$\$352 * 10\% = \35.20
Variable – materials/food	$30\% * \$500 = \150	$20\% * \$2,100 = \420	\$0
Fixed – opportunity	\$0	\$1,000	\$3,000
Fixed – labor	\$40	\$300	\$0
Total Costs/Day	\$193	\$1,930	\$3,035
Profit/Day	\$307	\$170	-\$2,683
Consumer effect revenue (consumer effect * avg. total purchase)	\$0	$5\% * 2,000 * \$100 = \$10,000$	$10\% * 2,000 * \$100 = \$20,000$
Daily Margin (50%)	\$0	\$5,000	\$10,000
Total Daily Profit from Option	\$307	\$5,170	\$7,317
Payback Period	$10,000/307 = 33$ days	$500,000/5,170 = 97$ days	$250,000/7,317 = 35$ days

Case 13: Pet Food Co.

Prompt: Our client, Pet Food Co., competes in the \$16 billion United States pet food market. They have four segments: dry food for dogs, wet food for dogs, dry food for cats, and wet food for cats. They are considering a \$10 million marketing campaign for one of their segments. Which segment do you recommend?

Interviewer Information:

- *Market:* There are more dog owners, but cat owners tend to own many more cats. So, the total number of each animal is approximately the same. On average, a dog weighs twice as much as a cat.
- *Product:* Foods have the same inputs: general processed meats. Each can sells for the same price.

Notes for Interviewer:

The goal is to have the candidates lay out a good framework with minimal information and then have to adjust based off more refined data from the exhibit. Candidates should, at minimum, identify # of pets, # of meals per day, and size of each meal as drivers of demand. A good candidate will also ask about the effectiveness of a \$10M spend in each segment and the client's contribution margin from each type of food. Tell the candidate to assume they are the same at this point.

Once the candidate is given the exhibit they should do the calculations to determine the profitability of each option. Probe the candidate about possibilities after they adjust their running hypothesis. For instance, what happens if the market share increase lasts for two years? What happens if the Dog Food market increases by 10%? What other things could the client do with \$10M to increase profitability?

		Dry Dog	Wet Dog	Dry Cat	Wet Cat
Calculations For Interviewers Only:	New market share	31%	42%	14%	10%
	New total revenue	\$2,325M	\$1,050M	\$560M	\$200M
	Marginal revenue	\$75M	\$50M	\$160M	\$100M
	Marginal profit	\$9M	\$9M	\$8M	\$10M

Case 13: Pet Food Co. (Example Framework)

Main questions

- Which segment should the client market to?

Key areas to explore

Dogs		Cats	
Wet Food	Dry Food	Wet Food	Dry Food
Demand? Client market share? Expected Increase in Market Share? COGS?			
Profitability of marketing to Wet Dog Food	Profitability of marketing to Dry Dog Food	Profitability of marketing to Wet Cat Food	Profitability of marketing to Dry Cat Food

Analysis

- Client much stronger in dog food than cat food. More room to expand in cat food.
- Increases in profit: Wet/Dry dog food - \$9M, dry cat food - \$8M, wet cat food - \$10M. Wet cat food is only option with a 1 year payback period.

Recommendation

- Decision:** Invest in marketing for wet cat food. Smallest overall market, but produces highest expected return in year one, and it's the market where our client has the most potential market share to gain.
- Alternative Option Examples:** New "organic pet food" product line, start selling in other countries, investigate new distribution channels, segment customers differently (what about people who buy multiple kinds of products?), could spending \$5M in two segments have greater impact? Spending \$10M on decreasing costs?

Other factors (risks)

- How reliable is anticipated market share gain data? What if marketing spend costs more? How long do gains last for? What is client's payback period? How will competitors respond? Is it better to try to completely dominate dog food market or is it better to increase presence in cat food market? What about the growth rates for each market segment?

Case 13: Pet Food Co. (Cont'd)

Exhibit 1: Campaign Estimates by Segment

	Dry Dog	Wet Dog	Dry Cat	Wet Cat
Market size	\$7.5B	\$2.5B	\$4.0B	\$2.0B
Pet Food Co. market share	30%	40%	10%	5%
Competitor's market share	30%	10%	30%	40%
Pet Food Co. profit margin	12%	18%	5%	10%
Market share increase from \$10M campaign	+1%	+2%	+4%	+5%

Case 14: Airplane De-Icing

Problem Statement Narrative: Your client is AirCo, a U.S. airline that has significant operations at one of the Chicago airports. Due to the cold weather, the client's planes often have to be de-iced. The need to de-ice planes is very unpredictable, so the client decided to outsource the de-icing to IceCo last year. However, IceCo's performance has not been satisfactory. The client is considering in-sourcing airplane de-icing, but currently does not have enough resources to do the de-icing in-house. The client requires a 4-year payback on investments and wants to know if they should in-source or outsource the de-icing.

Interviewer Information:

- If the client in-sources the de-icing, they will need to hire an average of 30 workers per month.
- Each worker costs \$4,000/month
- There are 5 months in the icy season
- The performance problems result from IceCo taking too long to de-ice the planes, leading to delays. We cannot quantify the impact of this.

Notes for Interviewer:

This case has a good case of quantitative and qualitative aspects. When you give the client their exhibit and the information above, instruct them to complete the exhibit. This should lead the candidate to see that the investment does not meet the payback period that the client requires. However, there is a lot more to this case than the numbers. The candidate should notice that the cost of delays is not included in any of the calculations. The candidate should also talk about some of the risks (labor costs or weather changing) and generate some alternative suggestions.

Case 14: Airplane De-Icing (Example Framework)

Main questions	Should the client invest?		
Key areas to explore	<u>Strategic Fit</u> - Does this fit with client's other activities? - Are there any potential synergies? - Are there any benefits on service?	<u>Economics</u> - Valuation of deal? - Costs of new deal? - Is valuation > costs? - Will it meet the payback period?	<u>Customers</u> - Cost overruns? - What if winters stop being so bad? - Learning curve?
Analysis	Benefits: Will save \$140 per snow storm, will prevent delays resulting from current contract. Considerations: Demand is unpredictable, which means airline could invest in capacity that goes unused for long periods of time. Payback period is 7 years, but company's goal is 4. Cost of delays uncertain.		
Recommendation	Decision: Payback period is 3 years too long, do not invest. Other Options: Re-negotiate contract with IceCo to have strict performance standards, tie IceCo's fees to meeting those standards. Re-examine payback period – add in cost of delays, explore more flexible labor options, explore leasing equipment instead of purchasing		
Other factors (risks)	Labor costs can change, weather is unpredictable – a year with a lot of snow decreases payback period, while a winter with no snow would increase it. What if there are some tacit knowledge requirements the client lacks.		

Case 14: Airplane De-Icing (Interviewer Exhibit)

Exhibit 1: Cost Comparison

	IceCo	Client
Number of Events	3,000	3,000
Fee per Event	\$300	N/A
Labor Costs	N/A	\$200
Cost per Gallon of Chemicals	\$5	\$4
# of Gallons of Chemical per Event	40	40
Cost per Event	\$500	\$360

Savings per Event	\$140
Annual Savings	\$420,000
Investment Cost	\$3,000,000
Payback Period	~7 years

Case 14: Airplane De-Icing (Interviewee Exhibit)

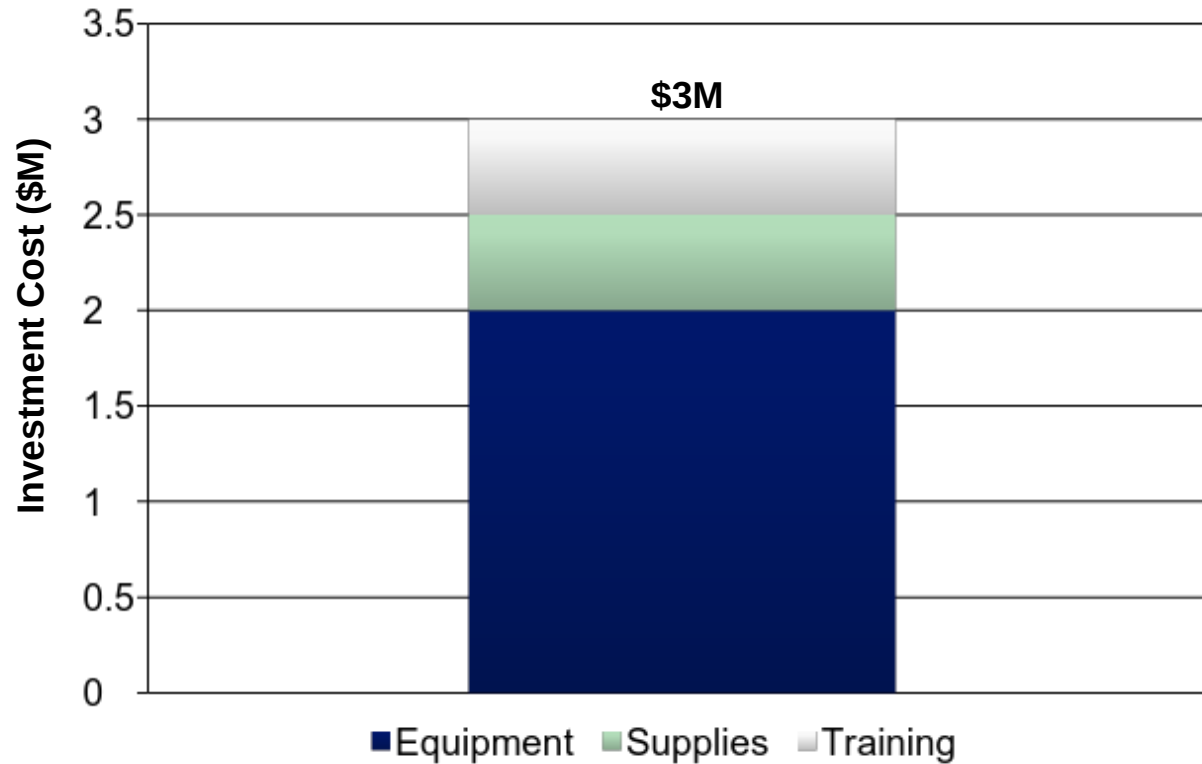
Exhibit 1: Cost Comparison

	IceCo	Client
Number of Events	3,000	3,000
Fee per Event	\$300	N/A
Labor Costs	N/A	_____
Cost per Gallon of Chemicals	\$5	\$4
# of Gallons of Chemical per Event	40	40
Cost per Event	_____	_____

Savings per Event	
Annual Savings	
Investment Cost	
Payback Period	

Case 14: Airplane De-Icing (Cont'd)

Exhibit 2: De-Icing Investment Costs



Case 15: Gas Branding

Prompt: You have been approached by the CEO of a convenience store chain. His convenience stores all have gas stations where he sells branded gas. He has been using the same supplier of gas for the past 20 years. The contract with this supplier is coming up for renewal and the CEO is considering his options to either renew the contract, sign a contract with a different gas supplier, or start his own new brand of gas. What do you recommend and why?

Interviewer Information:

Competition: Assume there are 2 levels of branded competitors - high and low
Product Mix: Assume there are 2 types of gas in the market - premium and regular. For high branded competitors, 10% of their volume sales are premium, 90% are regular. For low branded competitors, 10% of their volume sales are premium, 90% are regular

- **Sales:** High branded competitors sell 90,000 gallons per month on average, while low branded ones sell 110,000 gallons per month on average
- **Gas Type:** Used Currently using mid-premium supplier (right in the middle of premium and regular) - implies that supplier's price / profits are in the middle of high and low brands
- **Gas Quality:** Gas quality is the same as different suppliers. There is a perceived difference, but for our purposes, consider it negligible. For example, premium unleaded from Shell is of the same quality as premium unleaded from Exxon Mobil
- **Customer Behavior/Preferences:** 70% prioritize convenience (location), 20% prioritize price, 8% prioritize quality (brand), and 2% prioritize other services (car wash, convenience store, etc.)
- **Price:**
 - There is a \$0.30 difference between premium and regular gas prices at high branded gas stations; while this difference is \$0.20 at low branded gas stations
 - Low branded competitors sell regular gas for \$1.67 and high branded sell it for \$1.70
- **Profit:** The profit margin on premium and regular gas at a low brand gas station is \$0.15/gallon and \$0.10/gallon, respectively
- **COGS:** Relative to low branded gas, for high branded gas, the cost per gallon of regular gas is \$0.30 higher and cost of premium gas is \$0.05 higher

Case 15: Gas Branding (Example Framework)

Main questions	Should the client renew the contract, sign a contract with a different gas supplier, or start his own new brand of gas?
Key areas to explore	<u>Product</u> - Is one type of product more profitable than the other? - Is this product a commodity or differentiable? <u>Consumer</u> - How do consumers buy gas? Preferences? Patterns? - What would be required to change behavior? <u>Company</u> - What risk would the company face if change suppliers? - What are the set up costs assoc. w/ changing suppliers? - What marketing/ advertising would need to occur to maintain/ grow sales? <u>Competition</u> - Competitive response? - Does anyone offer something similar? - Is overall market growing? - Industry trends?
Analysis	- Calculate premium and low quality gas profitability (see next slide); approx. same profit - Gas consumers are generally sticky (based on 70% preferring convenience) so changing suppliers would not impact market share unless targeting other preference-based customers - High set up cost associated with new suppliers
Recommendation	- Maintain current supplier/renew contract <i>Potential Alternatives:</i> Interviewee could argue that new customers or changing market dynamics lend themselves to changing suppliers or offering their own brand of gas (e.g. customer loyalty, more control over value chain, customer indifference to low and premium quality)
Other factors (risks)	- Price volatility could change the profitability dynamic; less control over value chain - Consumer/industry trends for less dependence on traditional energy sources - Economic conditions decreasing the use/need for gasoline

Case 15: Gas Branding (Cont'd)

Breakdown of Gas Volume (A)

	Premium	Regular
High-brand	20%	80%
Low-brand	10%	90%

Total Volume of Gas (B)

High-brand	90K gallons
Low-brand	110K gallons

Profit Margin of Gas per Gallon (C)

	Premium	Regular
High-brand	Price: \$2.00 Cost: <u>\$1.77</u> Profit: \$0.23	Price: \$1.70 Cost: <u>\$1.60</u> Profit: \$0.10
Low-brand	Price: \$1.87 Cost: <u>\$1.72</u> Profit: \$0.15	Price: \$1.67 Cost: <u>\$1.57</u> Profit: \$0.10

**Tell interviewee: assume
difference is negligible:
#s are close enough to
be exactly the same**

Total Profits (A*B*C)

	Premium	Regular	Total
High-brand	Rev: \$36.00K <u>COGS: \$31.86K</u> Profit: \$4.14K (18K gallons * \$0.23)	Rev: \$122.40K <u>COGS: \$115.20K</u> Profit: \$7.20K (72K gallons * \$0.10)	\$4.14 + \$7.20 = <u>\$11.34K</u>
Low-brand	Rev: \$20.57K <u>COGS: \$18.92K</u> Profit: \$1.65K (11K gallons * \$0.15)	Rev: \$165.33K <u>COGS: \$155.43K</u> Profit: \$9.90K (99K gallons * \$0.10)	\$1.65 + \$9.90 = <u>\$11.55K</u>

Case 16: German Luxury Car Maker

Prompt: A German luxury car maker wants to grow its business and is looking to sell cars in Bangladesh. The GDP growth in Bangladesh is 5% per year. Currently, the only luxury car sold in Bangladesh is Mercedes-Benz and has been in the market for the past 10 years. The CEO wants to find out if the company enters the market, can they break even in three years?

Interviewer Information:

- Mercedes-Benz imported and sold 10,000 cars in this market over the past 10 years, and has its own dealership in Bangladesh. Existing owners replace their car every 10 years.
- There are 1000 new buyers each year and this is expected to continue for at least 5 more years.
- The client already entered the Vietnamese and other similar markets and gained on an average 30% of the market share. We can, therefore, safely assume that the client will gain a 30% market share. Assume the market share will be gained in the very first year and will remain at the same level; no increase or decrease expected.
- The price Mercedes-Benz charges is \$100,000 per car
- Cost structure:
 - Initial investment: \$7 million
 - Variable Cost (per car):
 - Manufacturing: \$20,000
 - Transportation: 120% of manufacturing cost = \$24,000
 - Customs/taxes: 95% of (manufacturing + transportation costs) = \$41,800
 - SG&A: 12% of all the above costs combined = \$10,296

Notes for Interviewer: This is a straight-forward market entry case. The candidate should do the calculation to determine that the investment meets the client's criteria for proceeding and should be able to speak at length about the various risks associated with the decision.

Case 16: German Luxury Car Maker (Example Framework)

Main questions	Should the company enter the Bangladeshi market?			
Key areas to explore	<u>Company</u> - Point of differentiation? - Financial Resources? - Experience entering new markets?	<u>Competition</u> - Mercedes-Benz - How have they fared in Bangladesh? - How difficult to displace?	<u>Customer</u> - How much demand? - Does 5% GDP growth mean 5% demand growth? - What do they value?	<u>Economics</u> - Initial investment required? - Estimated volume and margin? - Payback period?
Analysis	Company expects to get 30% market share. Market is 2000 new cars sold per year in year 1, that means 600 cars. Margin per car is \$4000 if client charges same as Mercedes, meaning \$2.4M margin and payback achieved within 3 years.			
Recommendation	Decision: Invest. Enter at same price as Mercedes. Payback within three years. Other considerations: look at other markets, explore three methods of entry: organic, JV, or organic. Company is probably considering only organic, but a JV with a local player could yield higher margins. Is this a strategic market (ie, are there for entry besides profit, like innovation or access to resources)?			
Other factors (risks)	Political instability in Bangladesh, new entrants into the market, Mercedes-Benz responds aggressively, what if the client doesn't achieve 30% market share (what is the break-even market share), what if the economy reverses?			

Case 17: Small Flying

Prompt: The CEO of an airline has approached you and said that he would like to expand into a small California town called Scottsdale. You have 5 days to do preliminary research and get back to him. In your short time, what should you research? What options does the airline have? What will you recommend?

Interviewer Information:

- **Company:**
 - Major airline, operates out of many airports of various sizes.
 - Part of larger effort to increase presence in smaller airports
 - Wants to be profitable in first year
 - Can devote either 0, 1, or 2 planes to airport.
- **Competition:**
 - Most flights operated by smaller local airline, other major airlines already offer 1 or 2 flights per day.
- **Customer:**
 - Typical small airport clientele. Travelling for a variety of reasons.
- **Product:**
 - Regular small airport flights. Small planes going to major hubs or other small local airports.

Notes for Interviewer:

This is a typical market entry case. A good candidate should go through the process of determining whether this is an attractive airport to expand into. The profitability in year 1 with the two plane option, as well as the growing customer base make this an attractive option. A good candidate will bring up a variety of issues, including: airline may be looking to expand service to please airline's frequent fliers, risks of costs being incorrect or economy slowing down, airline's typical razor thin margins, risk of competitive response, etc. After conquering the math, a good candidate will be able to turn this into a conversation about factors airlines should consider when expanding into smaller airports.

Case 17: Small Flying (Example Framework)

Main questions	Should the company operate at this airport?
Key areas to explore	<u>Company</u> - Typical clientele? - Why here? - Expansion options? - Size of presence? <u>Competition</u> - How many other airlines? - How will they respond? <u>Customer</u> - Who are they? - What are their preferences? - What matters to them? - Trends/growth? <u>Product</u> - What kinds of flights are we offering? - Is that normal for this airport?
Analysis	Company: We're a big airline that operates out of all size airports. Can enter with 1 or 2 planes. Only profitable to enter with 2 planes. Competition: AVG of 50 flights per day from airport. Mostly one local airline, a few of the other big airlines offer 1 to 2 flights per day Customers: Customers per year growing at strong rate. Product: Local flights.
Recommendation	Decision: Use 2 planes: provides estimated 180k profit in year 1. Other considerations: Airlines make most of their profit from business travelers and frequent fliers. A key to their success with those segments is ensuring that they can provide a flight to customers to/from anywhere. Expanding into local markets accomplishes this objective. Other options: Contract with local airline, 3 planes, nearby airports.
Other factors (risks)	What if airport fees increase, or airline unions request pay raise? How stable are the variable costs? How will other big airlines respond? Will they cut prices to drive you out? How easy will it be to get a gate at the airport? What if there are no available gates?

Case 17: Small Flying (Interviewer Exhibit)

Exhibit 1 (interviewer copy)

	1 Plane	2 Planes
Revenue	1,800,000	3,060,000
Variable Costs	720,000	1,440,000
Crew	360,000	720,000
Airport Fee	720,000	720,000
Profit	0	180,000

- (1) Revenue would be double that of 1 plane option, but assume a 15% drop in capacity utilization for both planes
- (2) 40% of revenue
- (3) Multiply VC of 1 plane by two
- (4) Crew fees double

Case 17: Small Flying (Interviewee Exhibit)

Exhibit 1: Option Comparison

	1 Plane	2 Planes
Revenue	1,800,000	(1)
Variable Costs	(2)	(3)
Crew	360,000	(4)
Airport Fee	720,000	720,000
Profit	(5)	(6)

- (1) Revenue would be double that of 1 plane option, but assume a 15% drop in capacity utilization for both planes
- (2) 40% of revenue
- (3) Multiply VC of 1 plane by two
- (4) Crew fees double
- (5) Calculate profit for 1 plane option
- (6) Calculate profit for 2 plane option

Case 18: Home Garden Chemicals

Prompt: Our client is a home garden chemicals manufacturer with annual revenues of \$1B. Its product lines include herbicides and manures. The client primarily targets the consumer market. Product are sold through “big-box” retailers, such as Home Depot and Lowe's. The client has recently come up with a new technology that enables it to manufacture a new type of genetically modified grass. The client is considering selling this genetically modified grass directly to golf courses. Is this a good strategy?

Interviewer Information:

- Herbicides used to kill weeds damage traditional grass, but new grass will not be affected by herbicides
- Water and herbicides can be delivered together through pipes/sprinkler systems
- Assume the total grass market is \$100 million, and the golf course grass market is 10% of that with no growth
- Replanting the grass on a golf course costs \$20,000
- Cost of developing the technology: \$3 million per year for the first three years with no sales; 50% profit margin from the 4th year onwards
- Revenue projection: 30% of the golf course market for the first 3 years (i.e. starting from 4th year); 70% market share after that each year
- Client's policy presses on a 5 year payback period
- **Only if asked:** Client is hoping to use this market as a testing ground before eventually marketing it to the rest of grass market.

Notes for Interviewer: An important question to ask when a client wants to enter a market is why? There are many reasons to enter an unprofitable market (to create innovation, gain access to resources or customers, potential growth, etc.). Some of the reasons can make an unprofitable market strategic for a client. This case seems fairly straight forward – it's a drop in the bucket amount of revenue that doesn't meet the client's payback period. However, by asking the right questions the candidate can find out a much more important thing – the client is actually looking to target the entire grass market, which represents 10% of their current revenue.

Case 18: Home Garden Chemicals (Example Framework)

Main questions	Should the client sell genetically modified grass to golf courses?		
Key areas to explore	<u>Company</u> - Why are they doing this? - Do they have the capability? - Do they have the capacity? - How will it fit in with their other businesses?	<u>Market</u> - How big is the opportunity? - What are the margins? - Are there any competitors? - Customer KBFs? - Is the market growing? - Is this a strategic market for the client?	<u>Product</u> - Investment required? - Potential Market Share? - Expected revenues and costs?
Analysis	Company: \$1B company looking trying to target customers worth \$10M of revenue. Real reason to enter market is to test product before targeting entire grass market. Based off information given client will lose \$3M annually in yrs 1-3, profit \$1.5M annually in yrs 4-6, and profit \$3.5M annually in year 7+ and beyond. Payback occurs in yr 8 with 0% discount rate.		
Recommendation	Decision: Enter market. Client’s payback period is 5 yrs and this project has an 8 yr payback, but there are strategic reasons for entering this market. Client’s ultimate goal is to enter \$100M grass market, so a short-term loss is worth perfecting product prior to entry. Other Options: Consider just going straight to consumer market, find ways to speed up R&D to bring product to market sooner.		
Other factors (risks)	- How long will the patent on this technology last? - Will consumers want the same thing as golf courses? - Will they be willing to pay the same price? - Are golf courses really a good test market? - Will the client still care about its’ payback period?		

Case 19: Paper Labels

Prompt: Our client is a \$100 million manufacturer of public signs. These consist mostly of public and traffic street signs, but are also used in some businesses, construction sites, and universities. The company is considering entering the paper label industry next year (2016). Paper labels have a wide variety of uses and so our client wants to know if there is any particular segment to enter, how to enter, and whether you recommend entry at all.

Interviewer Information: Industry growth is expected to be approximately 2% in next few years. Management believes it can capture 10% of current market.

- *Customer Segments:* healthcare (pharmaceutical labels), business (cards), manufacturing, food, and apparel. Major players exist equally in all these markets. However, some markets are more quality/premium driven, others are more cost driven
- *Product:* Can use a razor-blade model where the printer only works with same company's labels. Premium labels are mostly razor-blade model. Cheap segment is never razor-blade.
- *Competition:* A few dominant players may make entry difficult: perhaps they are already employing razor-blade model. They currently focus on premium; it's the small players who are focused on the cheap segment

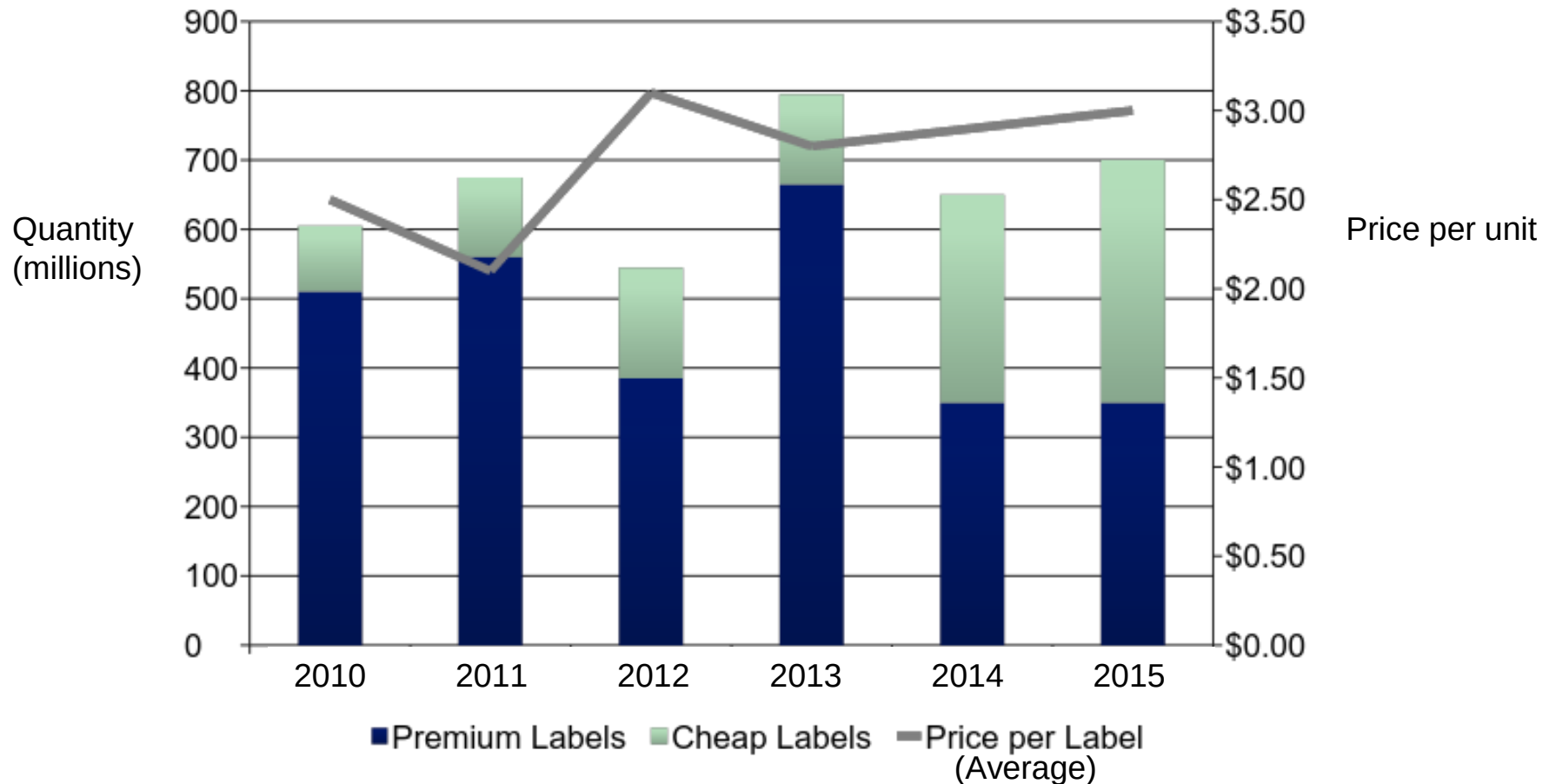
Notes for Interviewer: The market opportunity is very large, but growth is slow and there are several large players who dominate the industry and make entry difficult. High market capture may be difficult; however, management's estimate of 10% means revenue of \$210 million, over twice the size of the existing company! Also, the different industry segments present interesting options for entering the market as a premium or cheap player. Importantly, with printers and labels, the client can use a razor-blade model where the printer only works with same company's labels, thus retaining customers (offering both labels and printers is wise). The candidate should identify which market he/she thinks is more appealing and be able to explain selected entry method accordingly. The goal of this case is to create a good discussion about the pros/cons of various entry methods.

Case 19: Paper Labels (Example Framework)

Main questions	Should the client enter the paper label industry?			
Key areas to explore	<u>Company</u> - Current customers? - Size of Co.? - Entry options? - Related competencies?	<u>Competition</u> - How will they respond? - Barriers to entry? - How do they segment?	<u>Customer</u> - Different segments? - What are their preferences? - What matters to them? - Trends/growth?	<u>Product</u> - What kinds of paper labels? - Adjacent products?
Analysis	Company: Good reputation, some in-roads with current clients, entry into \$2.1B market represents a big opportunity. Competition: Very strong, but even getting 1/3 of “other” market share would double size of client. Customers: Significant overlap with current customers. Product: Premium and cheap segments. Cheap segment growing quickly.			
Recommendation	Decision: Enter the market via a Joint-Venture and produce cheap paper labels. The cheap paper label market is growing quickly so the client should try to enter immediately. With no industry knowledge the client should look to partner with another firm. Premium label market is shrinking so client may be able to use its current customer base to make in-roads. Other options: Use an acquisition or organic entry to target the premium market. It’s a shrinking market, but should provide higher margins.			
Other factors (risks)	What will be the effect on the client’s core business? What if they make a big investment and the big firms decide to force them out of the market? If they choose to make cheap paper labels will it hurt their reputation?			

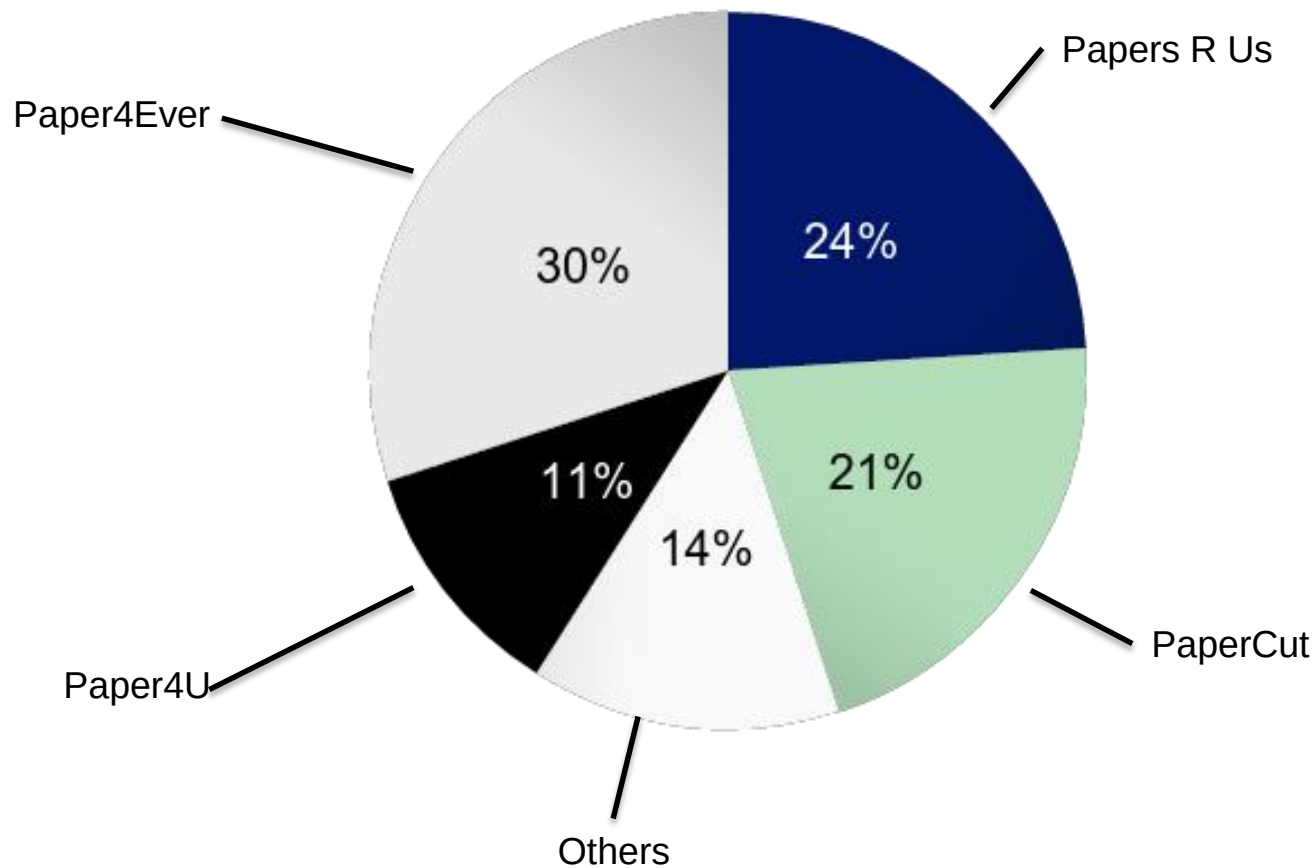
Case 19: Paper Labels (Cont'd)

Exhibit 1: Sales



Case 19: Paper Labels (Cont'd)

Exhibit 2: Competitors



Case 20: Fore Golf

Prompt: Our client, Fore Golf, is a high-end golf club and apparel manufacturer. Fore Golf is recognized as a technology leader in the golf industry. Fore Golf is considering whether or not to launch a golf ball division in the United States. Fore Golf has hired you to determine the following: is there an opportunity in the market, and if so, should the company invest in the new division? Fore Golf requires a 3 year payback on investments and estimates that the cost of entering will be \$16 million.

Interviewer Information:

- *Market dynamics:* Market is broken into 3 segments – Premium, Mid, and Low (see Exhibit 1)
- *Sales (in dozens) per year:* 35,000,000
- *Average industry sale price (per dozen):* \$30 → approx. \$1B in overall sales
- *Market share (in sales) / Competition:* See Exhibit 2 → Interviewee should note the 25% 'Other' which consists of 7 other companies
- *Reputation of Fore Golf:* Fore Golf brand is stronger than High Flite and comparable to Ballaway, Rike, and Title → Interviewee should conclude that Fore Golf brand is strong enough to enter this market and eventually take market share from all segment

Notes for Interviewer: The case focuses on both a qualitative market analysis and quantitative profitability analysis. Both suggest that the investment is a good idea, but a strong case will involve the strategic considerations that may or may not support Fore Golf's US golf ball division.

Case 20: Fore Golf (Example Framework)

Main questions	Should client implement online tracking system?						
Key areas to explore	<table><tr><th><u>Revenue</u></th><th><u>Costs</u></th><th><u>Other factors</u></th></tr><tr><td> - Price? - Volume? - Product mix between premium, mid, low? </td><td> - Investment of \$16M - Other costs? Relevant? </td><td> - Competition? Competitor response? - Marketing? - Seasonality? - Ties with overall economic performance? - Other opportunities for \$16M investment? - Number of rounds of golf played? - Distribution networks? - Promotions with clubs? - Outsource, partner, purchase other golf ball manufacturer? </td></tr></table>	<u>Revenue</u>	<u>Costs</u>	<u>Other factors</u>	- Price? - Volume? - Product mix between premium, mid, low?	- Investment of \$16M - Other costs? Relevant?	- Competition? Competitor response? - Marketing? - Seasonality? - Ties with overall economic performance? - Other opportunities for \$16M investment? - Number of rounds of golf played? - Distribution networks? - Promotions with clubs? - Outsource, partner, purchase other golf ball manufacturer?
<u>Revenue</u>	<u>Costs</u>	<u>Other factors</u>					
- Price? - Volume? - Product mix between premium, mid, low?	- Investment of \$16M - Other costs? Relevant?	- Competition? Competitor response? - Marketing? - Seasonality? - Ties with overall economic performance? - Other opportunities for \$16M investment? - Number of rounds of golf played? - Distribution networks? - Promotions with clubs? - Outsource, partner, purchase other golf ball manufacturer?					
Analysis	- Determine profitability based on price per product mix and project sales volume over the next 3 years. Does it exceed the investment made? (Use Exhibits 3 & 4) - Other alternatives for the investments and/or overall strategy decisions?						
Recommendation	- Yes, invest in US golf ball division because profitable and meets 3 year breakeven criteria (see solution on next slide) Potential Alternatives: Invest in outsource/partner/production already in place, adjust price, market/advertise						
Other factors (risks)	- Golf is closely tied to economic performance so should adjust price/strategic decision based on the state of the economy - Competition may lower price to gain market share or make other strategic positioning plays						

Case 20: Fore Golf (Solution)

Solution: The interviewer should understand that Fore Golf's profit is the wholesaler mark up because Fore Golf will not be selling directly to retail, but rather that next step in the distribution chain. First, calculate the profit per item from Exhibit 3. Answers shown below. Then multiply the profit per item by the forecast of sales volume (Exhibit 4) to arrive at profit per year and total profit in the first three years. Notice that the expected \$18 million earned is greater than the \$16 million investment. Therefore, it is within the 3 year payback period.

Profit per Item (Calculated from Exhibit 3)				
		Wholesale	Store	
	MCF Cost	Mark-Up	Mark-Up	Profit
Premium	\$25	\$15	\$5	\$15
Mid	\$15	\$10	\$5	\$10
Low	\$10	\$5	\$5	\$5

Forecast of Sales Volume (Exhibit 4) * Profit per Item = Total Profit				
	Year 1	Year 2	Year 3	Total
Premium	\$2,250,000	\$3,000,000	\$4,500,000	\$9,750,000
Mid	\$1,000,000	\$1,500,000	\$2,000,000	\$4,500,000
Low	\$1,000,000	\$1,250,000	\$1,500,000	\$3,750,000
	\$4,250,000	\$5,750,000	\$8,000,000	\$18,000,000

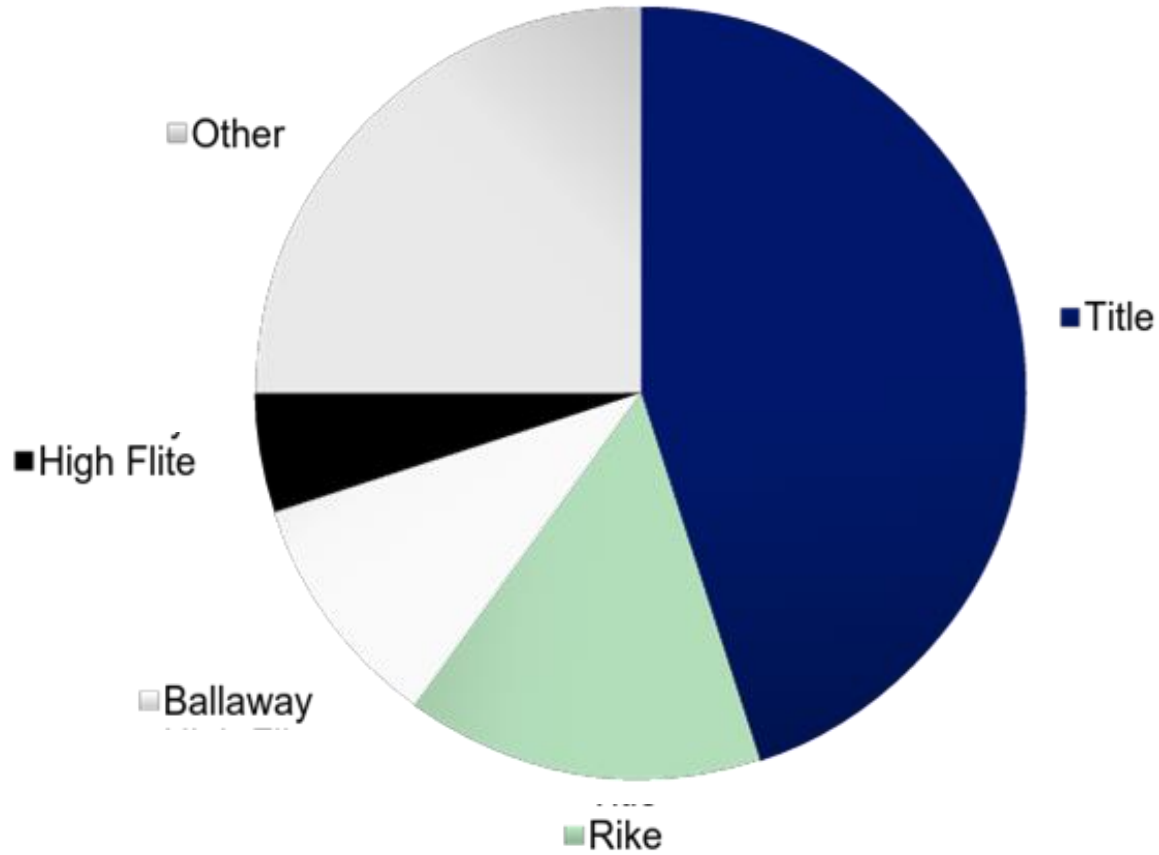
Case 20: Fore Golf (Cont'd)

Exhibit 1: Segment Information

Segments	Store Sale Price	Dozens Sold per Year
Premium	\$45	8,000,000
Mid	\$30	8,000,000
Low	\$20	19,000,000
		35,000,000

Case 20: Fore Golf (Cont'd)

Exhibit 2: Competitors



Case 20: Fore Golf (Cont'd)

Exhibit 3: Percentage Mark-Up Calculated from Final Price

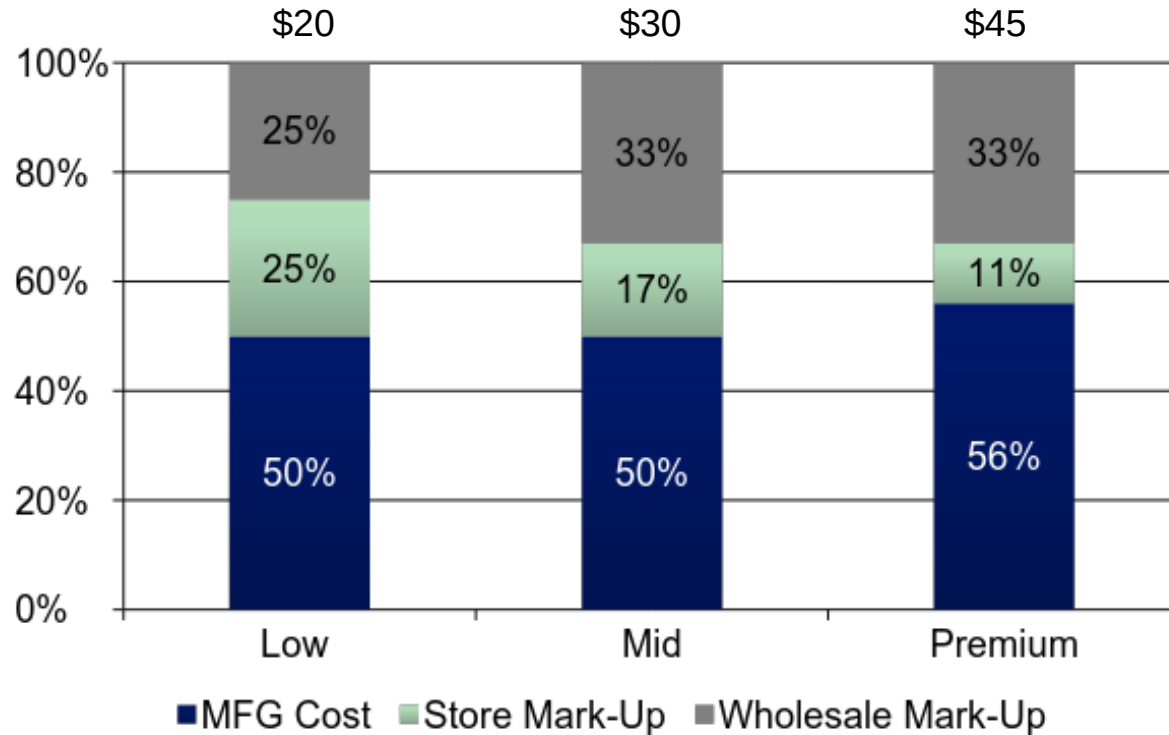


Exhibit 4: Forecast of Sales Volume

Segments	Year 1	Year 2	Year 3
Premium	150,000	200,000	300,000
Mid	100,000	150,000	200,000
Low	200,000	250,000	300,000

Case 21: Diabetes Testing Meter

Problem Statement Narrative: Our client is a laboratory that provides diabetes testing services to hospitals in England. They have developed a self-diagnosis meter that patients can use to do testing on their own. They have hired us to determine if we should take this product to the market.

Interviewer Information:

- *Demand Information:* England's population is ~50M, 3% of people younger than 65 have diabetes, 5% of people older than 65 have diabetes, 20% of population is over 65, assume no population growth.
- *Competition:* There are 4 other competitors, with market shares of 25%, 25%, 15%, 15%. Client has a 20% share. Growth was over 20% until 2 years ago and has been stagnant since.
- *Revenues and Costs:* Fixed cost of investment is \$2.5M. Per unit revenue is \$25 and per unit cost is \$20.
- *Additional points:* Patients could opt to use both methods (e.g. self test and also the hospital testing), Product could be promoted as a prevention device, assume 1 device per patient.

Notes for Interviewer:

Candidates should set out to determine the client's # of customers (potential demand), and use the financial information to determine if the client would break even. Candidates should determine that client would lose \$800k on investment. Getting to the investment decision is straight-forward, but this case allows a lot of follow on discussion. For instance, how much would this cannibalize the client's current business? Would consumers want this, and if so, could the client use this to steal market share? How much market share would the client need gain in order to break-even? What are some of the client's other options with this device? After the candidate has laid out his/her framework and developed a solution, ask these questions along with any others you think of before asking the candidate for his/her final recommendation.

# of people with diabetes:	# of customers:	Marginal Revenue:	Total Cost (Investment + VC):
1.7M	340K	\$8.5M	\$9.3M

Case 21: Diabetes Testing Meter (Example Framework)

Main questions	Should client take this product to market?
Key areas to explore	<u>Company</u> - Is the client in a position to market this device? - Per unit contribution margin? - Initial investment? - Cannibalization concerns? <u>Competition</u> - Is the client in a position to market this device? - Per unit contribution margin? - Initial investment? - Cannibalization concerns? <u>Consumer</u> - Would people want this? - Would hospitals want this? - What would the demand be? <u>Product</u> - What does it do? - What services does it replace?
Analysis	- Client has 340K customers, \$5 CM per device mean \$1.7M CM for device, \$2.5M investment means loss of \$800k → don't invest. - Stagnant market, high potential for cannibalization → don't invest. - Break-even volume = 500k new customers (~30% market share gain) → don't invest
Recommendation	- Don't invest. Break-even volume is 500k customers. If some of these are coming from cannibalization, then that number increases. The only way this makes sense is if the device will allow the client to gain significant market share, or if this is a precursor/test before launching in larger markets like the US. Potential Alternatives: Sell devices to hospitals instead of consumers, look at other markets which are larger or offer higher growth, find ways to market as a prevention device
Other factors (risks)	- What is the contribution margin from client's current customers? → determines how much cannibalization would hurt. - Is a competitor going to coming out with something similar and gain market share as a result?

Case 22: Deposit Slips

Prompt: Your client manufactures and sells deposit slips to banks at a price of \$1/slip. The client is the leading firm in this \$100M industry with 60% market share. There is one other competitor in the market. Federal regulations will cause the industry size to shrink by 10% next year. Your client wants to maintain profits to fund other projects. Should it go ahead with a price war?

Notes for Interviewer:

This is an interviewer led case so do not let the candidate start writing a framework. Instead follow the steps listed below and turn this into a conversation with the candidate.

1) Should the client go ahead with a price war?

An important stipulation here is that this is a commodity market with no differentiation. The candidate should talk about how in a price war competitors will drop their price to marginal cost and if price drops below a competitor's marginal cost, then that competitor will exit the market. After a theoretical discussion supply the candidate with the following info:

- Competitor outsources manufacturing for a total cost of \$0.90 per slip
 - Total cost for client is \$0.70 per slip
- Under these conditions the client should drop its price to \$0.89.

2) Ask the candidate to calculate profitability after the government regulation:

- $Q = 60\% \times 90\text{mm} = 54\text{mm}$
- $TR = Q \times \$1/\text{slip} = \54mm
- $TC = Q \times \$0.70/\text{slip} = \42mm
- $\text{Profit} = TR - TC = \12mm

Case 22: Deposit Slips (Cont'd)

3) Ask the candidate to determine profitability if there is a price war

- With the price war, price will be brought down to \$.89
- $Q = 90\text{mm}$ (client has 100% of market now, but market has shrunk by 10% due to federal regulation)
- $TR = Q \times \$0.89/\text{slip} = 80.1\text{mm}$
- $TC = Q \times \$0.70/\text{slip} = 63\text{mm}$
- Profit = \$16.1m → profits increase → Go ahead with price war

4) Ask the interviewee for potential risks

- Capacity constraints: Q is increasing by significant amount (50%), so plant expansion may be necessary or the client may have to make its employees work longer hours, start something like a shift system.
- Competitor may act irrationally and further lower the price.
- Federal regulation may be a sign of future trends → lower industry profitability in future years
- What about threat of new entrants? How much longer will companies continue to use paper slips?

5) We know that the competitor has another business division that manufactures specialty chemicals sold to banks for fraud detection. Does this change your analysis?

- The competitor's other division comprises 75% of its revenues
- This division is highly dependent on bank relationships build through the deposit slip business
- The competitor is more likely to sell deposit slips below cost because its other division is so dependent on it
- Price war may not be a good idea

Case 23: Gassy Trips

Prompt: Your client is a private equity group considering the acquisition of a company that produces ground transportation gas tankers. Imagine the sort of trailers that hold big containers of gas and deliver them to gas stations. How large is the market for these trailers in the United States? Should the client go ahead with the acquisition?

Interviewer Information:

If asked, provide the following information before candidate does market sizing.

- Market drivers
 - Volume: total gas needed for delivery/year
 - Distance: how far gas needs to be transported

Notes for Interviewer:

This is an interviewer-led market sizing case. Follow the format below:

- (1) Estimate the market size based on drivers that govern demand for gas
- (2) Once market size is estimated, challenge the final number and see the interviewee's reaction to pushing on different assumptions.
- (3) After there is a market size, say: "In due diligence, the private equity group has been told by the company that demand has been surging recently, but your client is suspicious about this sudden growth. What might cause a change in demand for gas trailers?" Let the interviewee brainstorm.
- (4) After brainstorming, provide this information. "A legal regulation 9 years ago required upgrades, and so at the time, there was a sudden spike in purchases. Now that we are approaching the 10-year average replacement time, purchases are spiking again. How does this affect your recommendation to the private equity group?"
- (5) Wrap up the case with a summary and recommendation about whether to acquire.

Case 23: Gassy Trips (Solution)

Step 1/2: Market Size: To calculate the market size for these gas trailers, we could start of by estimating the total demand for gas in the US:

- Demand: 200B gallons of gas/ year
- 75% of the demand needs transportation, 20% of that is transported in ways besides trailers
- Each trailer holds 7,500 gallons on average
- Replacement is every 10 years
- Total number of trailers required: $120B / 7,500 = 16M$

Step 3: What could cause change in demand for gas trailers

- Increased production in the U.S.
- Greater rate of gas consumption
- Gas is travelling further, say to nearby nations, Mexico
- Average size of tankers is declining
- Legal regulations that may affect the quality of trailers, such that the replacement time decreases
- Bad maintenance of roads, causing replacement time to decrease
- Shippers switch from other forms of transportation to trailers

Step 4: Provide the piece of information from the Notes section and ask for recommendations

- In the next 1 year, there will be a huge demand for these trailers, because of the regulation, however, the demand will drastically drop once the replacement cycle is complete
- The company stands a good chance to make a lot of money over the next 1 year, however, competitors, prices they charge and the cost of acquisition are factors that need to be looked into before making the final decision.

Case 24: ATVs

Prompt: We are doing private equity work for an ATV manufacturer. The manufacturer want to understand the cyclicity of the ATV business. He has some information about the industry wide sales of ATVs from 1980 until today (2007). Please see Exhibit 1. The client wants to know the reasons for the cyclicity of the business. So, tell me the following: Why did the sales fall in 1980? Why did sales then flatten out around the 2000s? Why did they rise in the 2000s? What will happen after 2007?

Interviewer Information:

- ATV (All Terrain Vehicle) is primarily an off-road vehicle used for recreational purposes. It has 4 wheels, is typically smaller than a motorcycle and people ride it like motorcycle
- Users: it is very popular among kids and adults from ages 15-50
- Demographics: the use population remained about the same
- ATV prices grew with inflation

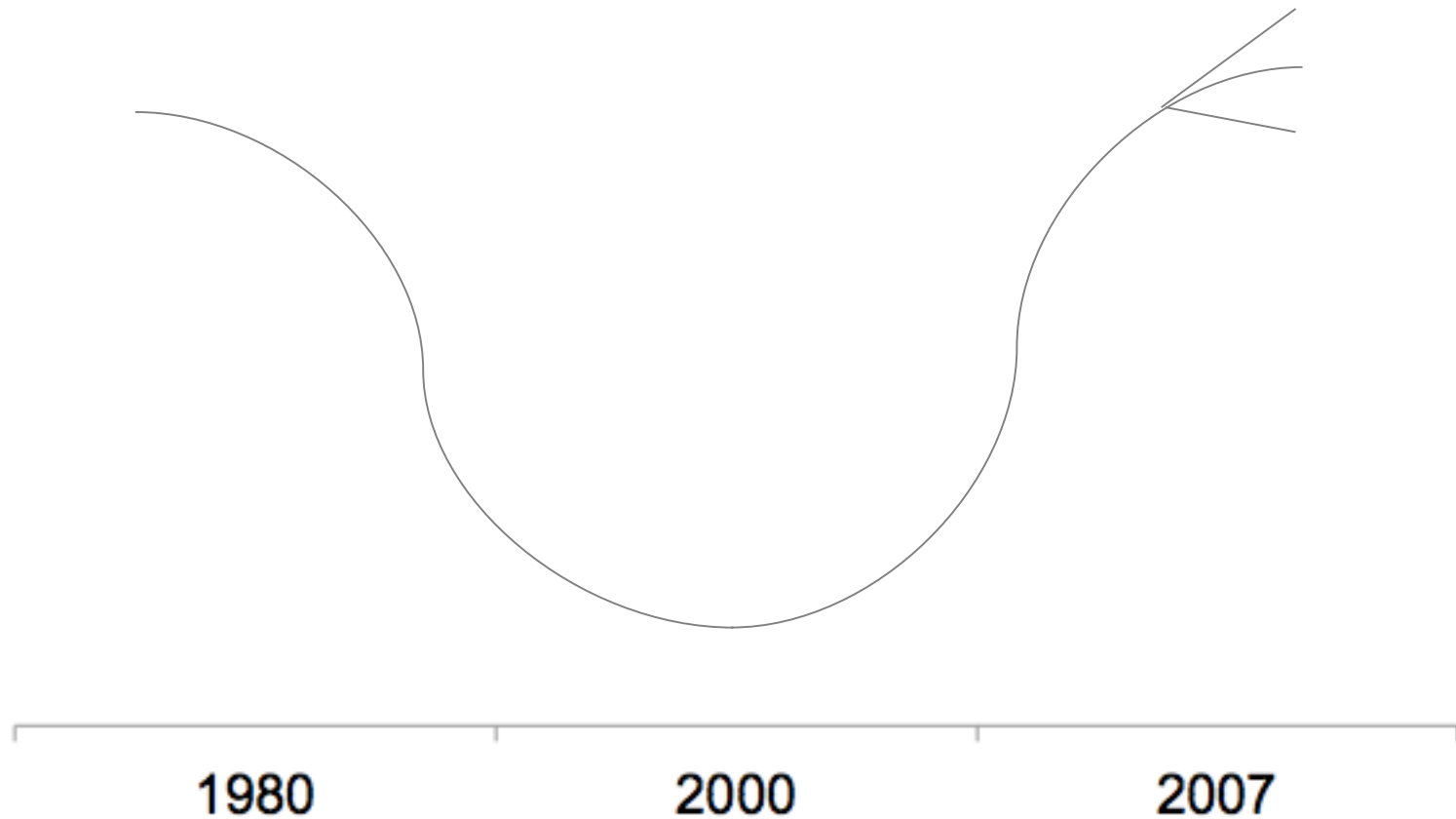
Notes for Interviewer: Don't allow the interviewee to open with a framework. Instead dive straight into the case. This presents a more uncommon challenge: having to think on one's feet constantly. There are hundreds of reasons why annual ATV sales have fluctuated so much. The interviewee should be able to brainstorm ideas and conduct conversation with the interviewer to identify which reasons make the most sense and which one do not. The interviewer should allow the conversation to float to different areas, challenging the interviewee as he/she analyzes different factors. This is largely a discussion case, but the interviewer should keep the discussion focused and should press the interviewee constantly.

Solution:

- 1980s: ATVs were classified as 3 wheeled vehicles and the public had many safety concerns after a number of accidents. In response, the government essentially banned them. That explains the drastic fall.
- 1990s: 4 wheeled ATVs became popular as they were safer. People wore helmets ATVs were restricted to off-road trails. This caused the stabilization.
- 2000-2007: The economy was going through a housing market bubble. People had a lot credit available. They used the money to buy ATVs (and other luxury/discretionary goods), explaining the rise.
- Future: sales will fall and stabilize somewhere in the half-way mark by 2011.

Case 24: ATVs (Cont'd)

Exhibit 1: Rough Representation of #ATV Sales/Year



Case 25: Cowboy Game Theory

Prompt: There are three cowboys– Cowboy A, Cowboy B, and Cowboy C. They are standing in an equilateral triangle. Each cowboy has one gun and one bullet and they all must shoot each other at the same time. Whoever Cowboy A shoots has a 100% chance of dying. Whoever Cowboy B shoots has a 60% chance of dying. Whoever Cowboy C shoots has a 40% chance of dying. Which cowboy is most likely to walk out alive?

Notes for Interviewer: This is not a traditional framework based case. This is a game theory conversation with interviewee. Don't allow the interviewee to make a framework for this one. Jump straight into the case and present the questions successively as the interviewee answers each one. Try to throw the interviewee off pace by not allowing a framework and changing the scenarios. It should feel different for the interviewee, but a well-practiced one will roll with the punches and take ahold and drive the case.

Interviewer Information:

Ask the candidate the following italicized questions and challenge their line of thinking so that the candidate must defend their solution – even if the candidate has the correct answer. Proper logic and answers are provided for the interviewee beneath the questions. If the candidate has sound logic, but does not come up with the given answer, just begin the next question with, “assume you had determined ____ for the previous question.. Then ask the question as written”

Question 1: Which cowboy is most likely to walk out alive?

- Probability of each cowboy dying:
 - Cowboy A: $(60\% + 40\%) / 2 = 50\%$
 - Cowboy B: $(100\% + 40\%) / 2 = 70\%$
 - Cowboy C: $(100\% + 60\%) / 2 = 80\%$
- **Answer:** Cowboy A is the most likely to walk out alive

Case 25: Cowboy Game Theory (Cont'd)

Question 2: In the previous situation, the cowboys didn't know each other's probabilities. Now assume that the cowboys know everyone's probabilities; then who is the most likely to walk out of there alive?

- Cowboy A will shoot Cowboy B
- Cowboy B will shoot Cowboy A
- Cowboy C will shoot Cowboy A
- **Answer: Cowboy C is the most likely to walk out alive** (each cowboy will shoot their most deadly foe)

Question 3: What is the relationship between parts 1 and 2? What exactly caused the change in the most likely to survive?

- **Answer: There is an inverse relationship between the probability of survival and the amount of information known. Being a better shooter makes the cowboy a more likely target, and thus hurts his probability of survival. The key driver is the availability of information**

Question 4: Can you provide a business example of when having asymmetric information can lead to different outcomes?

- **Answer: There are almost countless examples. The interviewee should brainstorm several and describe the relation to each. Some examples: mergers and acquisitions, insider trading, contract bargaining, government negotiations, new market entry, etc.**

Case 26: Tobacco Leaf Processor

Prompt:

Our client is a tobacco leaf processor. Tobacco is a commodity and revenues in the tobacco industry are flat. The #2 and #3 tobacco leaf processors are considering a merger. The new merged company will be 75% of the #1 company. Over the past five years, #2 has had flat revenues and slight profitability. #3 has had declining revenues and profitability. Both are global organizations with operations, plants, and distribution at a global level. They overlap in most, but not all, markets. Should the client go through with the merger?

Interviewer Information:

- **Market:** Only one product and segment. There are 4-5 end consumers that purchase processed tobacco leaf.
- **Competition:** #1 has a total market share of 35-40%. Besides the top three players, there are 5-6 smaller firms.
- **Innovation:** There has been no innovation in tobacco leaf processing for many years.
- **Organizations:** #2 is highly centralized with a bureaucratic structure. #3 is decentralized and entrepreneurial.
- **Value Chain:** Tobacco Farmers → Tobacco Leaf Processors (our client) → End buyer (e.g. Phillip Morris)

Notes for Interviewer:

Mergers & acquisitions are one of the most important types of cases. This case does not have a quantitative aspect, though M&A cases lend themselves to an NPV analysis of the deal coupled with a qualitative evaluation. This case is purely qualitative. This case is a good test of a candidate's M&A framework as much of this case is simply applying one's framework to a merger in the tobacco leaf processing industry. Beyond having a good framework a good candidate will be able to speak about the effects of: being in a commoditized industry, the competitive response of the #1 company, and merging companies with very different cultures in addition to other considerations.

Case 26: Tobacco Leaf Processor (Example Framework)

Main questions	Should the companies conduct merger?
Key areas to explore	<u>Strategic Fit</u> - Client's competitive strategy? - Reason for merger? - Areas of potential synergies? - Will it foster innovation? <u>Economics</u> - Valuation of deal? - Costs of new deal? - Is valuation > costs <u>Risks</u> - Culture differences? - Frequency of mergers in industry / company's history? - Challenges of post-merger integration?
Analysis	Benefits: market overlap means they'll have redundancies and can cut costs. Merged company will have greater market share → greater bargaining power. Possible innovation benefits. Considerations: #2 company is larger than #3 and will have greater standing in relationship. #3's culture is unique → won't want to change. With such different cultures retention could be issue.
Recommendation	Decision: Go through with merger. Merged company will be much stronger than individual companies could be individually. Implementation: Pre-merger forums to manage expectations. Develop ways to implement aspects of both company's cultures. Identify areas of overlap to eliminate redundancies.
Other factors (risks)	Regulatory issues in some countries where merged company will have greatest market share. #1 company may respond by conducting its' own mergers.

Case 27: Chemical Brothers

Prompt: Your client, Chemical Brothers International (CHEMBRO), is a major chemical producer, has retained your firm's services to evaluate the feasibility of acquiring another major player in the industry, Plastics of America (POA). Both companies are bulk commodity chemical producers. Your task is to analyze the future prospects of POA's major product line, a chemical used in the production of plastics. Should Chembro acquire POA?

Notes for Interviewer:

There are four main motives for a merger or an acquisition:

- 1) "The overcapacity M&As" – goal is to increase market share, conduct more efficient operations (economy of scale), or to take capacity out of the industry.
- 2) "The geographic roll-up M&As" – typical of early stage industry life cycle, acquirer lowers operating cost and brings value to the customer base (hotels and banking in some countries). This also includes acquiring another company in order to access a new geography.
- 3) "Product of market expansion M&As" – relative size matters and M&A helps new company to innovate or diversify its products.
- 4) "M&A as R&D" – useful to shorten product life-cycle. This is more typical in the tech/pharmaceutical industries. For instance, a small company will begin development of a drug and then be acquired by a larger company which wants access to that R&D.

The main point of this case is to eventually get into a discussion of why CHEMBRO would consider entering an unattractive market. The information in the case paints the industry in a very negative light, and a good will discuss why this should lead CHEMBRO away from this industry. However, a great interviewee will be able to discuss some of the reasons why CHEMBRO would consider entering anyways. Ultimately the case does not suggest this so the interviewee's recommendation should still be to not acquire POA, but the interviewee should discuss some of the other potential reasons anyways.

Case 27: Chemical Brothers (Cont'd)

Interviewer Information:

- Market Analysis:
 - End-users come primarily from the automotive industry, market size has been slowly declining over the last five years, within the last couple of years, prices have also declined rapidly
- Competition / Industry Analysis:
 - There are 10 major producers; the largest one with a 35% share, number two has 25%, and POA has 20%; the remaining share is divided among others, the two largest competitors earn a small return; POA is slightly above break-even; the rest are operating at break-even or at a loss
 - Relative capacity utilization in the industry is 60% to 70% and has been so for the last three years; POA is currently working at 75% capacity
 - The two largest competitors are highly diversified with this particular product line representing no more than 20% of their revenues
 - Highly regulated industry with expensive pollution control equipment
 - High barriers to entry because of the low profits and high investments required
- Product Value Proposition / Brand Portfolio:
 - The price has been driven by self-destructive cuts from the leaders to gain temporary share points
 - We do not foresee the development of any significant by-products, no other possible uses
 - Complementary assets: 50% of POA's sales are to the automotive industry
- Finance and Operations:
 - Cost is based on size/efficiency/age of plant, etc. POA is in an above average position
 - There are several operational improvements that could be implemented, and management has not been aggressive in its pursuit of quality and cost controls
 - Great economies of scale exist in marketing and transportation (not quantifiable)
 - Operational synergies could represent an additional \$30 million in profits

Case 27: Chemical Brothers (Example Framework)

Main questions	Should the client take this device to market?			
Key areas to explore	<u>Market</u> - Trends? - Price changes? - Quantity changes? - Margin changes?	<u>Competition</u> - Trends? - Main competitors? - Client's position? - Barriers to entry?	<u>Product</u> - Price changes? - Product mix? - Different uses? - Room for innovation?	<u>Economics</u> - NPV? - Value of synergies - Economies of scale - Economies of scope
Analysis	NPV is \$1B ($c/(r-g)$) = ($\\$90M/ (.12-.03)$). Industry is not attractive – margins are shrinking, price is dropping, no alternate uses (no economies of scope synergies), highly dependent upon auto industry → quickly becoming a commodity product. POA has room for additional cost cuts and cost is dependent upon size.			
Recommendation	Don't invest. Overall this does not look like an attractive industry to enter. This is an industry towards the end of its life cycle when profits decrease and the product becomes commoditized.			
Other factors (risks)	There are some circumstances when it is beneficial to enter an unattractive market. It is mentioned in the interviewer information that cost is related to size. If CHEMBRO's acquisition of POA allows CHEMBRO to improve its cost structure through scale that could be beneficial. Also, if this is a highly competitive industry, CHEMBRO may be able to learn from entering, so innovation could be a reason. CHEMBRO may want access to POA's customer, or the offer price for POA might be significantly undervalued.			

Case 27: Chemical Brothers (Cont'd)

Exhibit 1

Purchase Price	\$950M
Annual Operating Income Before Tax	\$90M
Cash	\$30M
No. of Employees	2,000
Return of Capital	12%
Market Risk Premium	7%
Growth Rate	3%
Tax Rate	40%

Case 28: Retirement Plan

Prompt: Our client is a full service asset management firm, Georgetown Securities Co. (“GSC”). GSC is looking to enter the market for Individual Retirement Accounts (“IRAs”) or Defined Contribution Plans (“DCPs”). In case you are not aware, an IRA is a retirement account with certain tax advantages while a DCP is a retirement account where the individual makes contributions on a regular basis. Today, both the IRA and DCP markets are the same size, which is around \$3 trillion, and both are expected to generate the same fees for GSC. GSC is interested in knowing the five year growth outlook for each of these markets. What information would we need to estimate this growth and how should we advise GSC to best position itself?

Interviewer Information: Provide this additional information ONLY if asked by the interviewee

- *Customers:* Provide Exhibit 1 and Exhibit 2.
 - Note that Exhibit 1 helps us understand our potential customer base.
 - Note that “population growth” in Exhibit 2 relates to total percent growth over the five year period and not the compounded annual growth rate.
- *Revenues:* Provide Exhibit 3. Note that fees charged by asset management firms are a function of the average balances held in the customers retirement account.
- *Competition:* Provide Exhibit 4. Note that this could potentially help us understand why our competitors would enter the IRA market.
 - Discount brokers are primarily in the DCP market, but could enter the IRA market. These firms are characterized by a “low price / low touch” model.
 - IRAs tend to be provided such as full-service brokers as they are more costly to manage.

Notes for Interviewer: The main purpose of this case is for the candidate to work through the numbers provided in the exhibits and estimate the market size and value for the customer segments in each market. The information on competition is to facilitate a discussion on what strategy GSC should employ when entering either the IRA or DCP market.

Case 28: Retirement Plan (Example Framework)

Main questions

- How can we estimate the potential growth for the IRA and DCP market?

Key areas to explore

- | | | |
|--|--|--|
| <p><u>Revenue</u></p> <ul style="list-style-type: none"> What is our current fee structure? How can we incentivize customers to invest with us? | <p><u>Customers</u></p> <ul style="list-style-type: none"> What are our customers financial goals? Is there an underserved portion of our market? | <p><u>Competition</u></p> <ul style="list-style-type: none"> What are the points of differentiation? Major players? |
|--|--|--|

Analysis

Age	2010 Pop.	% Growth	2014 Pop.	Est. DCP Acct. Balance	Est. IRA Acct. Balance
35-54	70M	-5%	66.5M	\$30K * 66.5M ~ \$2.0T	\$34K * 65M ~ \$2.21T
55-74	50M	+20%	60M	\$20K * 60M ~ \$1.2T	\$26K * 60M ~ \$1.56T
75+	20M	+5%	21M	\$0 * 21M = \$0	\$0 * 21M = \$0
			Est. Market	\$3.2T	\$3.77T

Recommendation

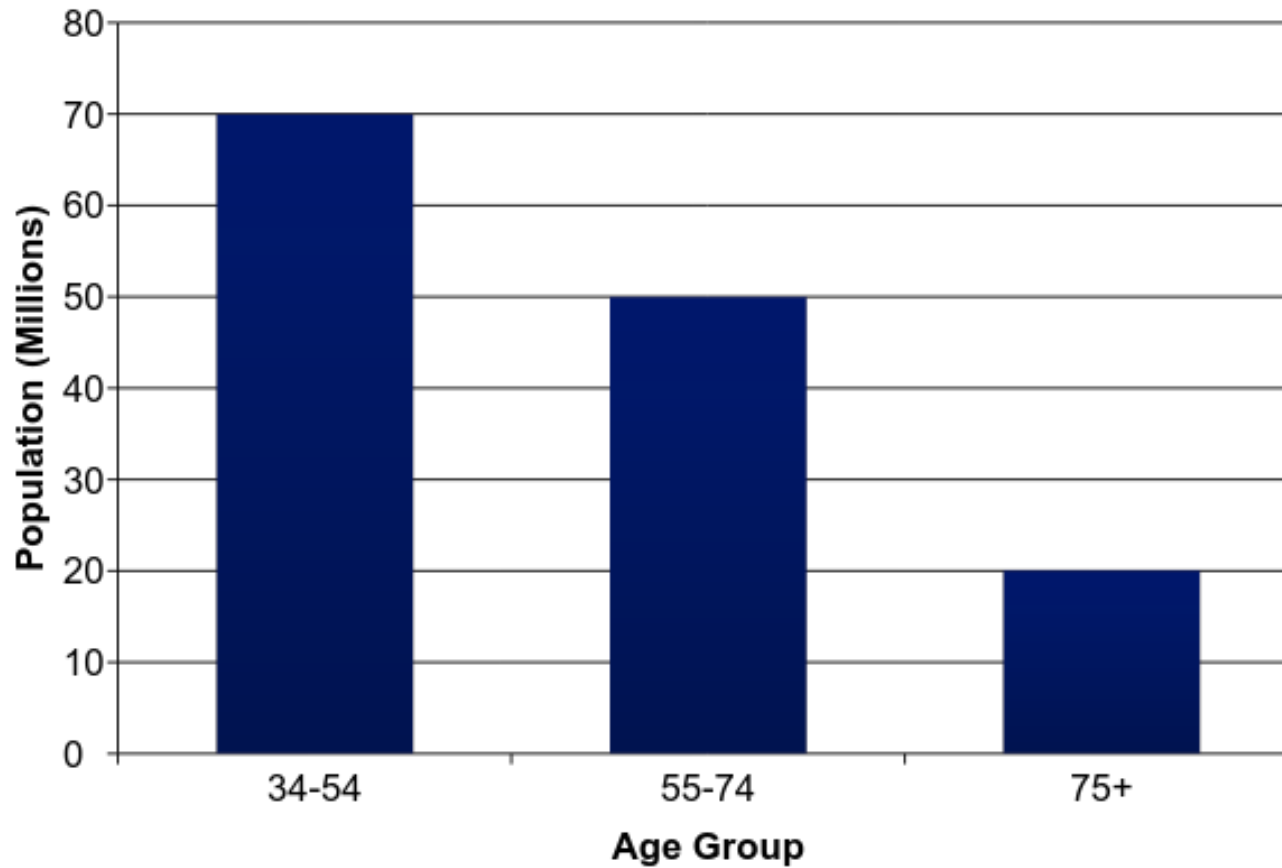
- Solution:** IRAs seem to be the best opportunity for GSC in terms of potential market. The IRA market is also good since it protects GSC from discount brokers considering the high costs of competing in the IRA market versus the DCP market. To best position GSC, it should focus on improving the professional competencies of their financial advisors to enhance their capabilities in providing personalized service.

Other factors (risks)

- Macro-economic issues?
- Enhance regulation?
- Competitive reaction?

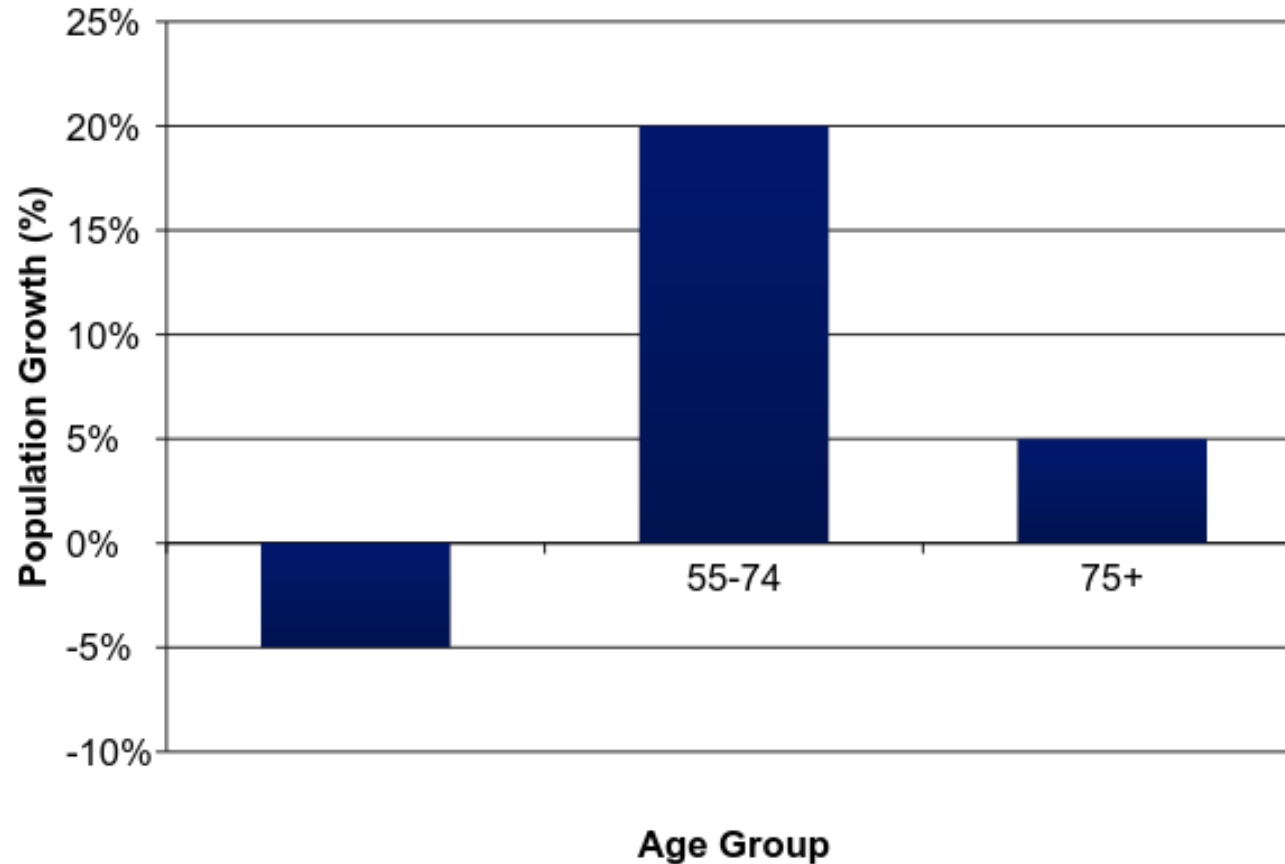
Case 28: Retirement Plan (Cont'd)

Exhibit 1: Current Population of Different Age Groups in the U.S. as of 2010



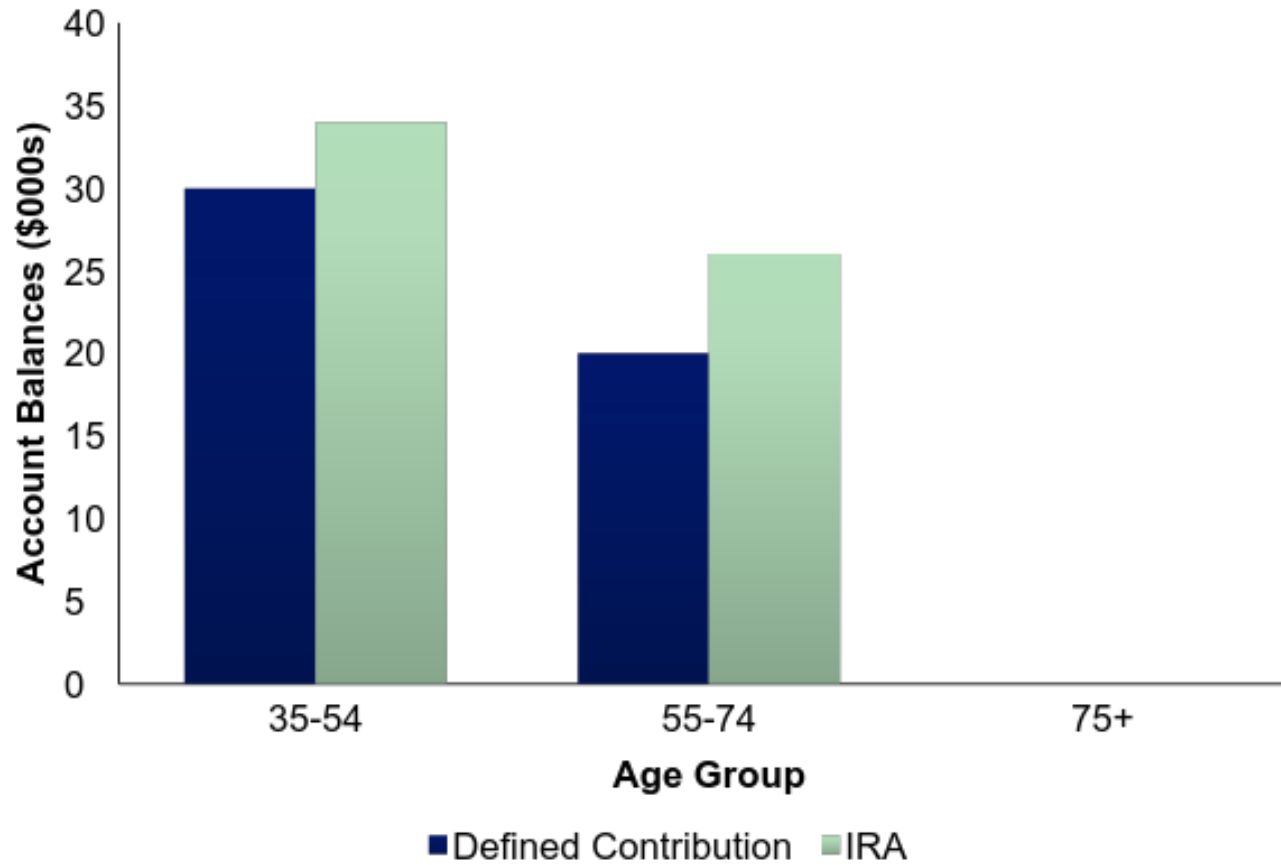
Case 28: Retirement Plan (Cont'd)

Exhibit 2: Age Group Growth, U.S. 2010-2014



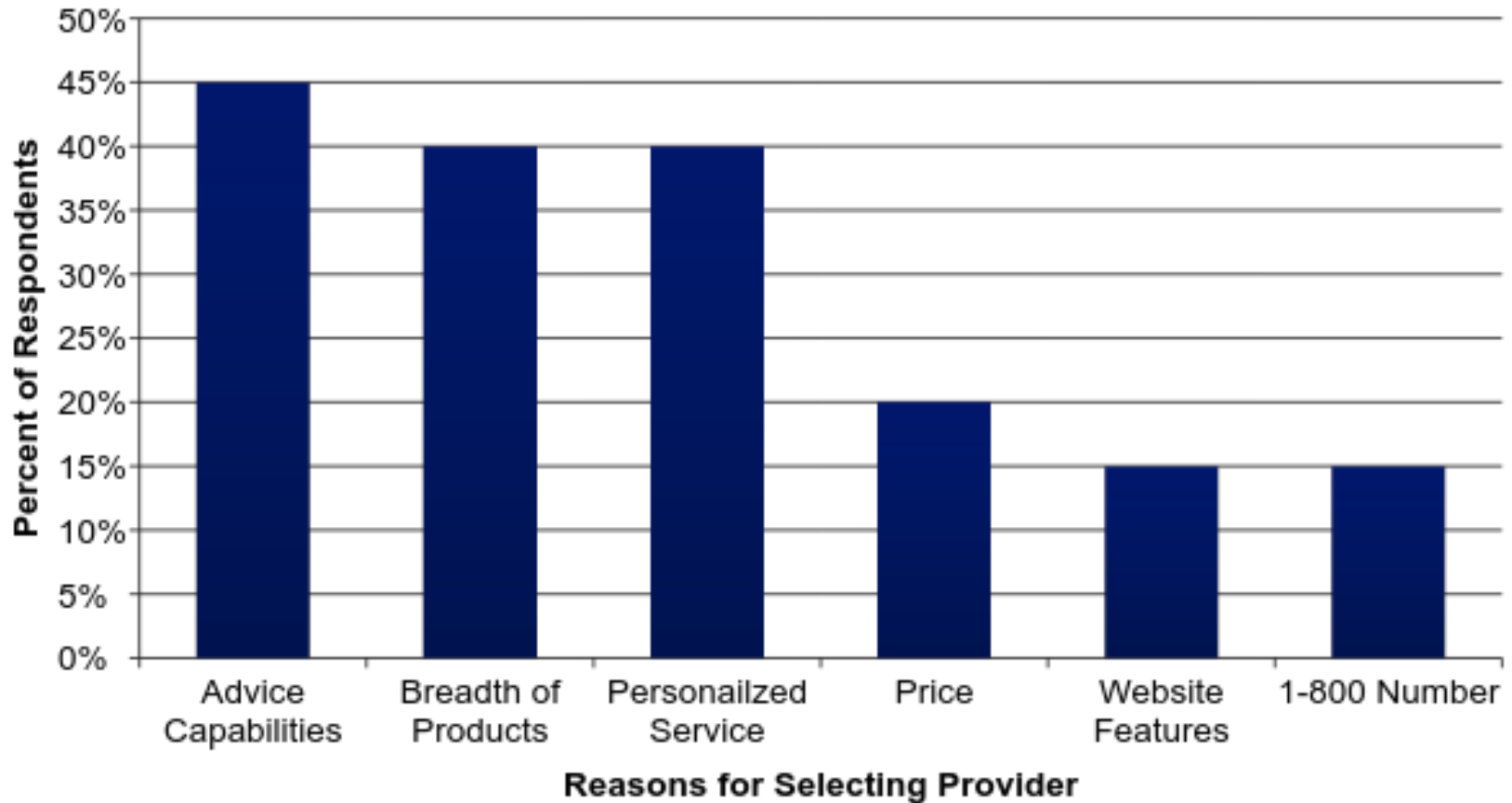
Case 28: Retirement Plan (Cont'd)

Exhibit 3: Average Balances in Retirement Accounts per Person



Case 28: Retirement Plan (Cont'd)

Exhibit 4: Reasons for Choosing an IRA Provider



Case 29: Cali Burger

Prompt: Cali Burger is a family-owned hamburger chain, with roots in Los Angeles county, and a cult following due to its commitment to quality, freshness, and dedication to hamburgers. Cali Burger currently has more than 210 locations in four states, California, Nevada, Arizona, and Utah. It is estimated that Cali generates approximately \$500 million in annual revenues. For about \$20, a family of four can each enjoy a meal of a burger, French fries, and a milkshake.

Thus far, Cali Burger's expansion has been characterized by locations within a one-day drive from Los Angeles. The majority of the locations are within a 6 hour drive from Los Angeles. Company policy dictates that locations cannot be more than 500 miles from its suburban Los Angeles distribution center so that the meat and produce do not have to be frozen. Indeed, much of the company's marketing revolves around the theme of never freezing its ingredients. The expansion has also been characterized by locations among major interstate highway routes, for example, along Interstate 5, which runs North-South through California and all the way North to the U.S.-Canada border and Interstate 15, which runs from Los Angeles to Las Vegas, Nevada - Cali Burger's first out-of-state location. Cali Burger is considering expanding to the Pacific Northwest so that it will be poised for expansion into Canada. The company has hired you to recommend whether or not to proceed.

Notes for Interviewer: Interviewee should note that the key barrier to expansion is the distribution center and the policy that the stores cannot be located more than 500 miles from a distribution center. Options to explore are building another distribution center (Interviewee should discuss various locations near Canadian border), freezing meat and produce, or any other creative recommendations interviewee can come up with. Margins are slim in the fast food / quick service restaurant industry (2-3%), market is saturated and industry is very mature. So, this is something that the company will have to consider eventually. Other things that could be mentioned—going public to finance a new distribution center, aggressive marketing campaign to increase volume, not pursuing the expansion strategy. Also ask: given the numbers provided, should Cali Burger build a 2nd distribution center? What kind of margin will Cali Burger maintain?

Case 29: Cali Burger (Cont'd)

Interviewer Information:

- Industry
 - Major competitors are McDonald's (12.7% market share), Yum! Brands (Taco Bell, Pizza Hut, KFC) (9.7%), and Wendy's/Arby's Group, Inc. (6.6%), as well as mom and pop shops
 - Industry is mature and saturated. Margins in this industry are slim (usually 2-3% of total sales)
 - The Company's current expansion strategy is to add 10-18 stores per year
- Company
 - Company is family-owned and has a very strong brand in California. The company does not want to go public (but that option should still be discussed). The company is known for high quality, fresh fast food. The menu items are priced slightly higher than its competitors, due to its higher quality
 - 2008: \$450 million in annual revenues; 2009: \$500 million
 - Profit before taxes: \$25 million
 - Claims of per store sales are \$ 2 million
- 2nd Distribution Center Option
 - Distribution center will cost \$12.5 million dollars. Cali Burger will put down 20% and take out a loan on the remainder at an interest rate of 6.5%. This includes the cost of opening 25 new stores.
 - Corporate tax rate is 30%.
 - COGS + Operating Expense constitute 90% of gross revenue.
 - Depreciation + Amortization is 5% of gross revenue

Case 29: Cali Burger (Example Framework)

Main questions	Should the company expand into Pacific NW
Key areas to explore	<u>Company</u> - Strategy? - Expansion options? - Differentiation? <u>Competition</u> - Market saturation? - Typical margins? <u>Customer</u> - Who are they? - What are their preferences? - What matters to them? - Trends/growth? <u>Product</u> - What is unique about Cali Burger? - Will this appeal to people outside California?
Analysis	Point of differentiation is freshness, so in order to expand, Company will need to open a new distribution center. Company charges a premium for fresh products so freezing is not a viable option. Debt is only method to pay for option. With the opening of 25 new stores Cali Burger will see positive profits in NW. There are no similarly priced competitors.
Recommendation	Decision: Open new distribution center and expand into the Pacific NW. Use debt financing. Other options: Consider expanding into other markets. Market is mature so growth is typically related to population growth - how does NW compare to other areas? Cali Burger could focus on trying to increase market share in current area, or could try diversifying with other complimentary food services (coffee, cupcakes, frozen yogurt, etc.). Cali Burger could also add pricier, higher-margin foods to their menus.
Other factors (risks)	What kind of marketing spend will be required? Is the Pacific NW market growing? How big is it? What kind of market share can Cali Burger hope to get in next 5 years? Market is mature so why isn't someone else occupying this space in the NW?

Case 29: Cali Burger (Interviewee Copy)

Net Income	
Revenue (from new stores)	
Total COGS + Operating Expenses	
EBITDA	
Depreciation and Amortization	
EBIT	
Interest	
EBT	
Tax (30%)	
Net Income	

Case 29: Cali Burger (Interviewer Copy)

Additional Information (if asked) (Cont.):

- Interviewee's calculations should be close to the following:

Net Income	
Revenue	\$50,000,000
Total COGS + Operating Expenses	(\$45,000,000)
EBITDA	\$5,000,000
Depreciation and Amortization	(\$2,500,000)
EBIT	\$2,500,000
Interest	(\$650,000)
EBT	\$1,850,000
Tax (30%)	(\$555,000)
Net Income	\$1,295,000

→ 25 new stores
X
\$2M revenue per
store

→ (20% down on
12.5 million loan →
\$10 million loan at
6.5%)

Case 30: Cha-Ching

Prompt: Your friend owns a gas station on I-15 right outside of Las Vegas. He is thinking about installing a slot machine in the gas station. He asked you if he should do it.

Interviewer Information:

- He is looking to install 1 slot machine, which costs \$5,000 (one time cost)
- There is annual licensing fee of \$1,000
- The average Vegas hotel casino slot machine brings in \$200/day in revenue
- Additional assumptions to be made by interviewer (provide if asked):
 - Since the slot will be located in a gas station, assume it would bring in 50% of a hotel slot machine
 - Variable costs: electricity = \$1/day
 - The gas station is open every day
 - The payout rate: 80%
 - Lifespan of a slot machine is 10 years
 - Taxes are 30%

Notes for Interviewer:

- This is a typical introductory business decision case. A good interviewee should be able to attack the numbers and quickly determine that this is a profitable decision for the gas station. A great interviewee will turn this into a discussion about the possible pro's and con's of the decision. For instance, will this increase traffic to this gas station, or will it cause cannibalization. Will the presence of a slot machine drive some customers away? Could this \$5000 be better spent? How does the structure of the \$5000 payment effect this decision? How can the gas station best promote the new slot machine? What options should the slot machine include – is it going to be a penny machine or will it cost a dollar? Even though this case appears fairly simple an interviewee should always be looking for ways to demonstrate his/her creative thinking and this case provides ample opportunity to do just that.

Case 30: Cha-Ching (Example Framework)

Main questions

- Should gas station introduce a slot machine?

Key areas to explore

- | <u>Company</u> | <u>Competition</u> | <u>Customer</u> | <u>Product</u> |
|--|---|--|--|
| <ul style="list-style-type: none"> Does this fit with company's business? What is client's goal? | <ul style="list-style-type: none"> Any other gas stations do this? Competing with local hotels? Does this give client advantage? | <ul style="list-style-type: none"> Will customer want this? Will this bring new customers? Cannibalization? | <ul style="list-style-type: none"> Expected revenue? Expected costs? |

Analysis

- Lifetime profit = \$41,545 – \$5000 = \$36,545**

Revenue	365*100 = \$36,500	Licensing Fee	\$1000
Payouts	36,500 * .8 = \$29,200	Annual Profit	\$5935
Variable Cost	365 * 1 = \$365	Annual Profit after Taxes	\$4,154.50

Recommendation

- Decision:** Install the slot machine, expected lifetime profit (ignoring time value of money) is \$36,545.
- Alternative Options:** How could the \$5000 be better spent? Maybe making the gas station nicer, or larger would provide a higher return, maybe allowing Subway or McDonalds to operate there would provide higher ROI.

Other factors (risks)

- How will other gas stations or places with slot machines react? Will this cannibalize food business (people spend spare change on slot machine instead of food), or will this increase the number of patrons?

Case 31: Bottle Shifters

Prompt: “Bottle Shifters” operates in the food/drink sector. It designs and distributes customized and standardized bottles and is the dominant player within this space. Bottle Shifters’ customers are small/medium sized liquor manufacturers. Our client, a private equity firm, has acquired Bottle Shifters and believes it should improve its profitability. Our client also believes that Bottle Shifters has already optimized their cost structure.

Interviewer information: Provide this additional information ONLY if asked by the interviewee.

- *Market Size:* Provide Exhibit 1. Note about calculating potential market size.
- *Value Chain:* Provide Exhibit 2. Note that Bottle Shifters is involved in Design and Distribution but not Manufacturing.
- *Products:* Provide Exhibit 3.
- *Customers:* Provide Exhibit 3.
- *Competition:* There is another player with half of the revenues of Bottle Shifters. The rest of the market is fragmented.
- *Price:* Steer the candidate away from exploring Price. This should send a signal to the candidate to focus more so on Quantity.
- *Store Mix/Customer Mix/Channel Mix:* No information is available.

Notes for Interviewer: The interviewee should realize the “low-hanging fruit” provided in the prompt regarding Cost and realize that this is all about improving Revenue. If the candidate tries to go down the Cost path, remind the interviewee that our client believes that Bottle Shifters’ costs are already optimized. As noted above, the point of this question is all about examining Quantity. When thinking about improving Quantity, we want to consider ideas around whether a good market exists, who are our competitors and what is the willingness of our customers to buy the product.

Case 31: Bottle Shifters (Example Framework)

Main questions	How can we improve profitability considering costs have been optimized?
Key areas to explore	<u>Market</u> - Market size? - Current trends? - Growth and future? <u>Revenues</u> - Change our price? - Improve our quantity sold? - Product Mix? <u>Customers</u> - Changes in preferences? - Purchase frequency? <u>Other Factors</u> - How is distribution handled? - Macro-economy?
Analysis	Market size calculation: 1.43B liters/0.75 liter ~ 2B bottles sold. Standardized = $2B * 0.8 = 1.6B$ bottles. Customized = $2B * 0.2 = 0.4B$ bottles. Market value: Standardized = $1.6B * \\$0.3 = \\$480M$; Customized = $.4B * \\$0.7 = \\$280M$. - Limited number of Standardized Liquor accounts. - Customized liquor is the most profitable, but comprises the smallest portion of Bottle Shifters product mix.
Recommendation	Current Position: Strengthen by focusing on standard and customized liquor. For standard liquor, consider incentivizing sales managers to better relationships with their clients by enhancing outreach efforts. For customized liquor, leverage Bottle Shifters' competitive advantage in Design and Manufacturing as a means of differentiation. Adjacencies: Stronger marketing materials. Contract negotiations with clients.
Other factors (risks)	- Financial investment of expanding customized product line? - Alienating current customer base? - Competitive reaction? How do we best position Bottle Shifters in this regard?

Case 31: Bottle Shifters (Cont'd)

Exhibit 1: Market Data

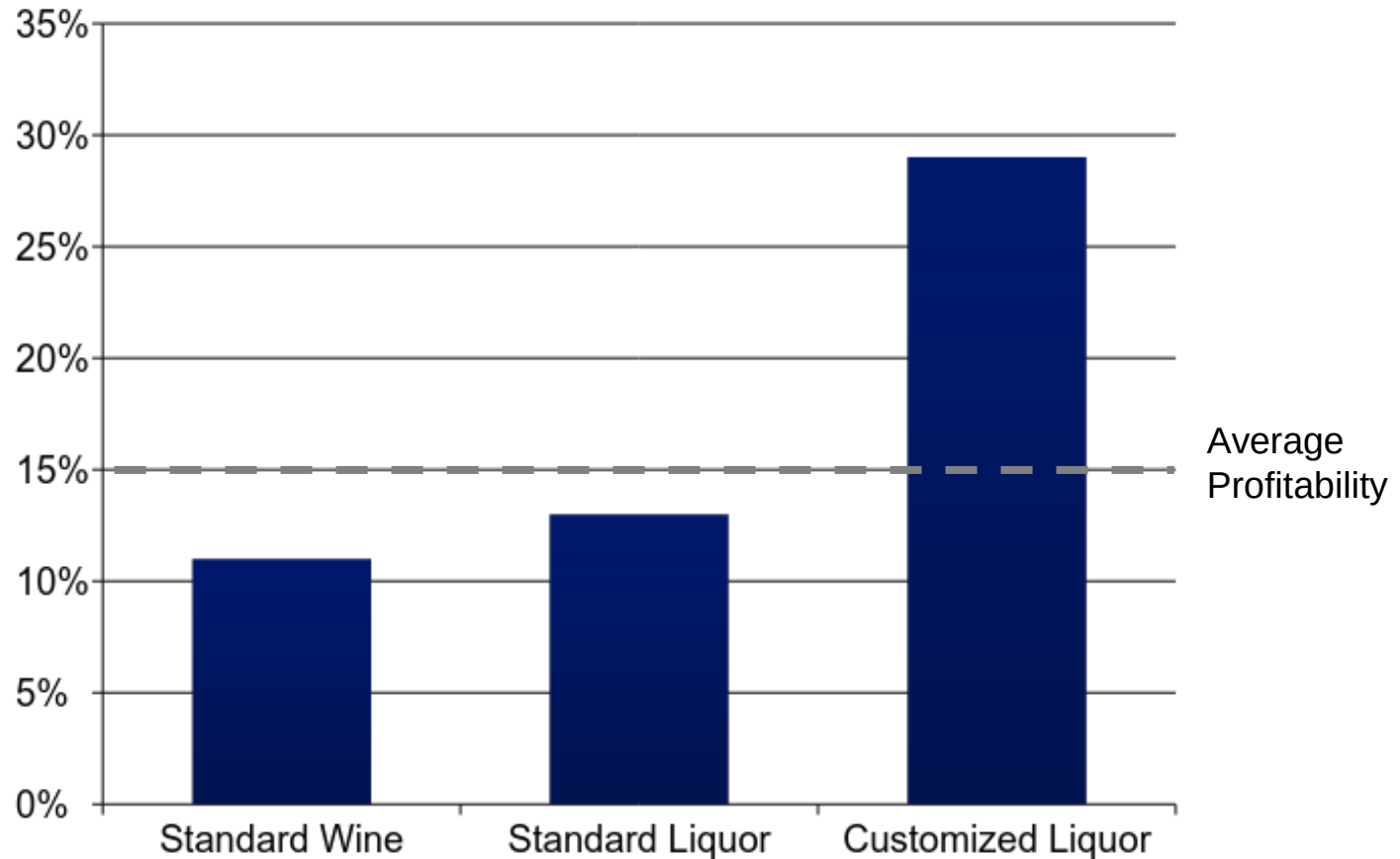
Total volume of liquor sold in the U.S. annually: 1.43B liters
Average size of a bottle: 0.75 liters
Standardized: 80% (price: \$0.30)
Customized: 20% (price: \$0.70)

Exhibit 2: Value Chain



Case 31: Bottle Shifters (Cont'd)

Exhibit 3: Product Mix and Customers



	Standard Wine	Standard Liquor	Customized Liquor
# of Accounts	698	2	2
Revenues	1.4M	8.6M	0.7M

Case 32: Death by Math

Prompt: Your client is the Minister of Public Health and Safety of a developing nation. He has asked you to determine the best way to allocate the Ministry's funds. The Health Ministry has sufficient funds to reduce death from any one of the three following causes by 5% (each costs the same):

1. Infant mortality due to malaria (malaria kills 10% of children under the age of 5)
2. Adult mortality due to injuries (half of all citizens who make it to age 5 die before reaching their 60th birthday at an average of 30 years; of those, 10% die from injuries)
3. Senior mortality due to infections (of those who survive to age 60, 40% die from infections and 60% die of natural causes)

For the sake of this case, assume that malaria affects only infants, injuries affect only adults between 5 and 60, and infections affect only seniors. What are a citizen's chances of dying from each of these three causes? What are the "life years" lost from each cause? What cause of death do you recommend the Minister address?

Notes for Interviewer: Ask the candidate to setup a framework of how to address this problem. **Don't let the candidate begin doing math during their framework.** The point of their framework is to see how they'd answer the question if all the math is given. Once candidates have discussed their framework, begin by giving candidates their sheets one at a time to sequentially determine: 1) death rates, 2) average life expectancy, 3) effect of malaria on life expectancy, 4) effect of injuries on life expectancy, 5) effect of infections on life expectancy. Have the candidates fill out the blank-underlined portions.

These sheets should test the candidate's "math in public" abilities. Before giving candidates the subsequent sheets try to get the candidate to discuss the "so what" of what they've discovered, if they're not doing so unprompted. Once candidates have completed all of their sheets, encourage them to return to their framework and make a recommendation.

Case 32: Death by Math (Example Framework)

Main questions	Which option should the government invest in?
Key areas to explore	<u>Quantitative</u> - Which option saves the most lives? - Which option has the greatest effect on life expectancy <u>Qualitative</u> - What are the moral implications of each option? - Is it better to increase the odds of an infant living, to increase the odds of someone who may have a family making it to a later life, or increase the life expectancy of someone who has paid their debt to society? - Which option would have the greatest economic effect on society?
Analysis	- Malaria has the greatest impact on life expectancy, so a 5% change to that will have the greatest effect. Treating infections can have the greatest number of lives saved. - Fighting injuries has a low effect on both lives saved and life expectancy.
Recommendation	- Invest in fighting malaria, has greatest effect on average life expectancy and “Saved population” has the most productivity left to offer the country. Potential Alternatives: Possibly allocate less than \$5M to multiple options. Look at other health measures beyond 3 mentioned which will have greater than 5% impact.
Other factors (risks)	- Political ramifications: in America senior citizens are one of the most significant voting blocks, the political influence of that voting demographic could play a role. - What are the costs of not doing an option? How much does it cost the country when a 30 year old parent dies from injury vs. other options? - What are the “hidden costs” of an option? Does country have capacity to support increased life expectancy?

Case 32: Death by Math (Cont'd)

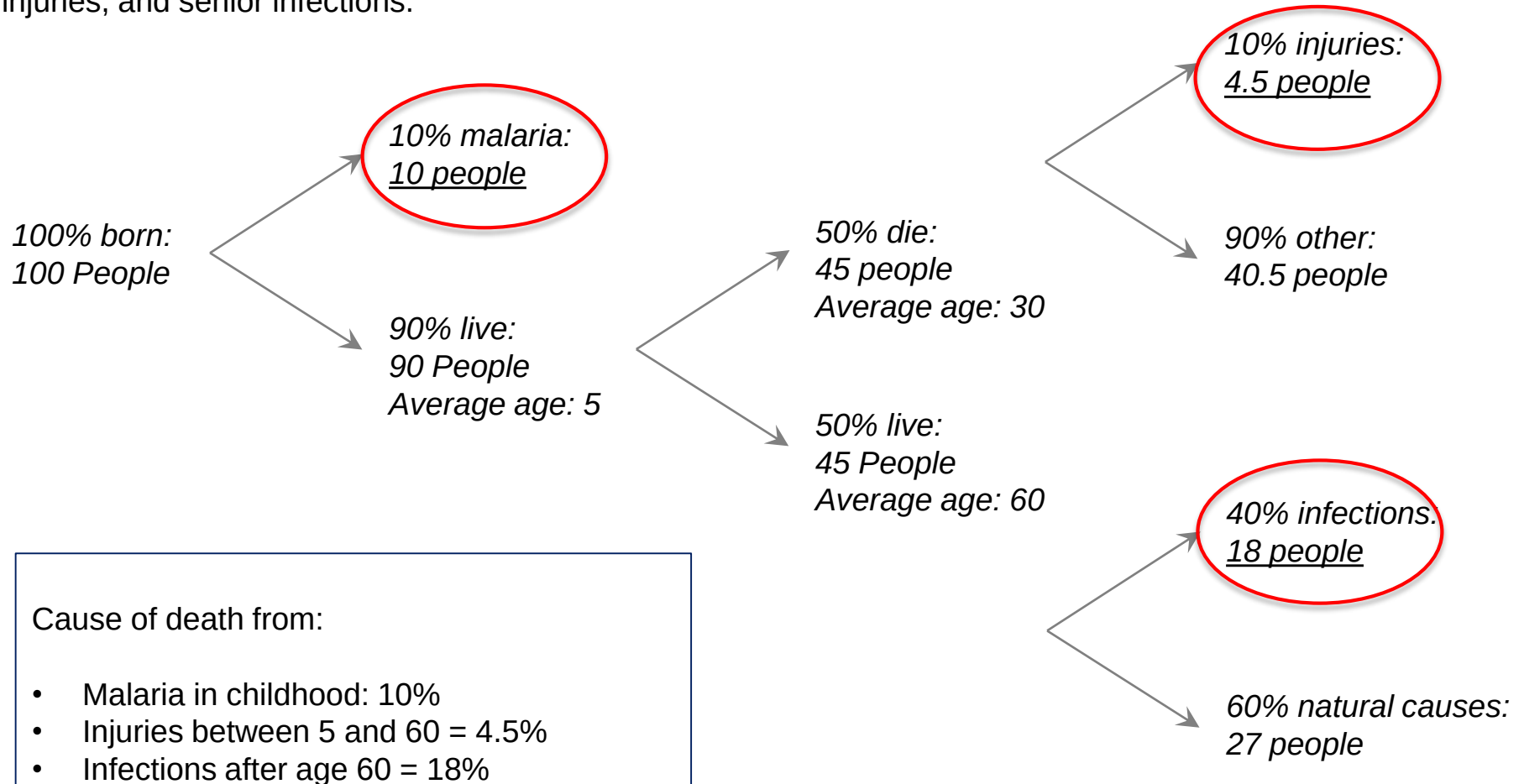
Exhibit 1: Life Expectancies and Ages

- Life expectancy at birth is 47.9 years
- Malaria:
 - The average victim of malaria is 2 years old
 - 10% younger than 5 die
- Injuries:
 - The average victim of injuries is 30 years old
 - 10% of those who die between ages 5-60, die from injuries
- Infections:
 - The average victim of infections is 70 years old
 - 40% of those who live past 60 die of infections
- Natural Causes:
 - The average victim of natural causes is 80 years old

Case 32: Death by Math (For Interviewer)

Step 1: Diagram Death Rates

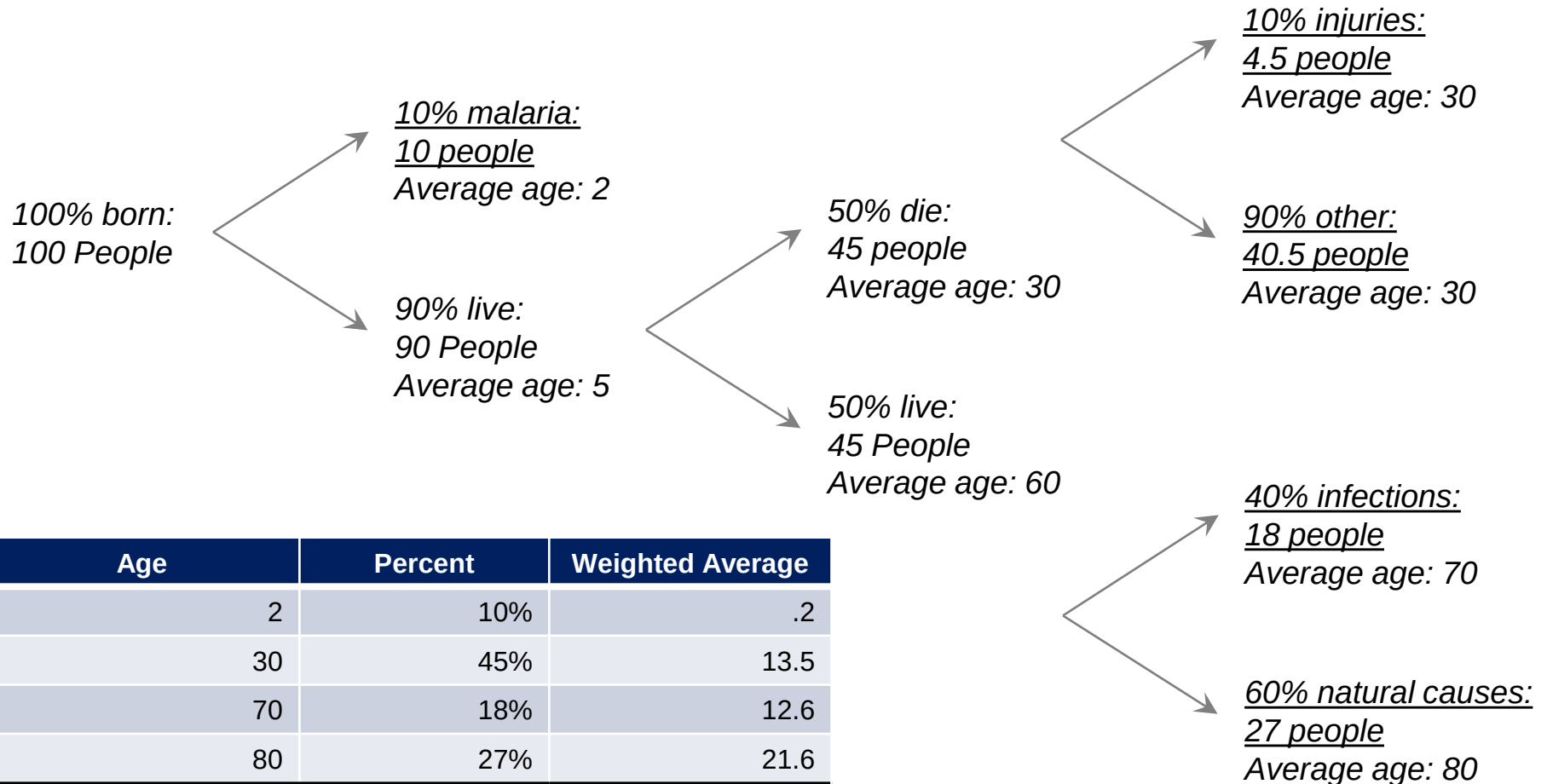
This diagram shows the death rates from the three causes of death in the prompt: childhood malaria, adult injuries, and senior infections.



Case 32: Death by Math (For Interviewer)

Step 2: Calculate Life Expectancy

Now the interviewee can calculate the total life expectancy. But, ask the interviewee which cause of death leads to the greatest loss of “life years”

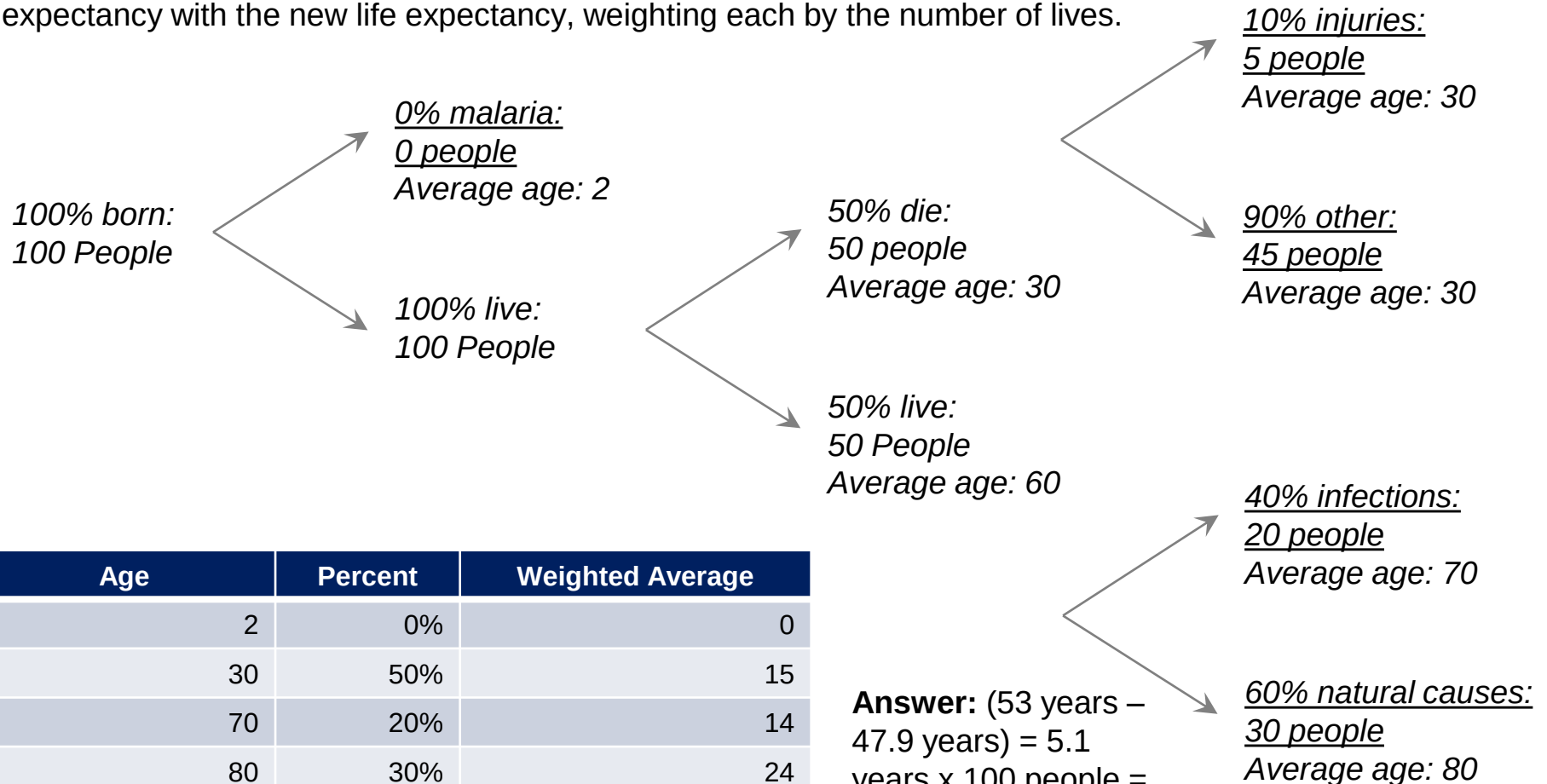


Age	Percent	Weighted Average
2	10%	.2
30	45%	13.5
70	18%	12.6
80	27%	21.6
Weighted average: 48	100%	47.9

Case 32: Death by Math (For Interviewer)

Step 3: Calculate “Life Years” Lost from Malaria

Calculate the number of “life years” lost by each cause of death. To do this, calculate the life expectancy by removing one cause of death to see its effect on the life expectancy. Then, combine the original life expectancy with the new life expectancy, weighting each by the number of lives.



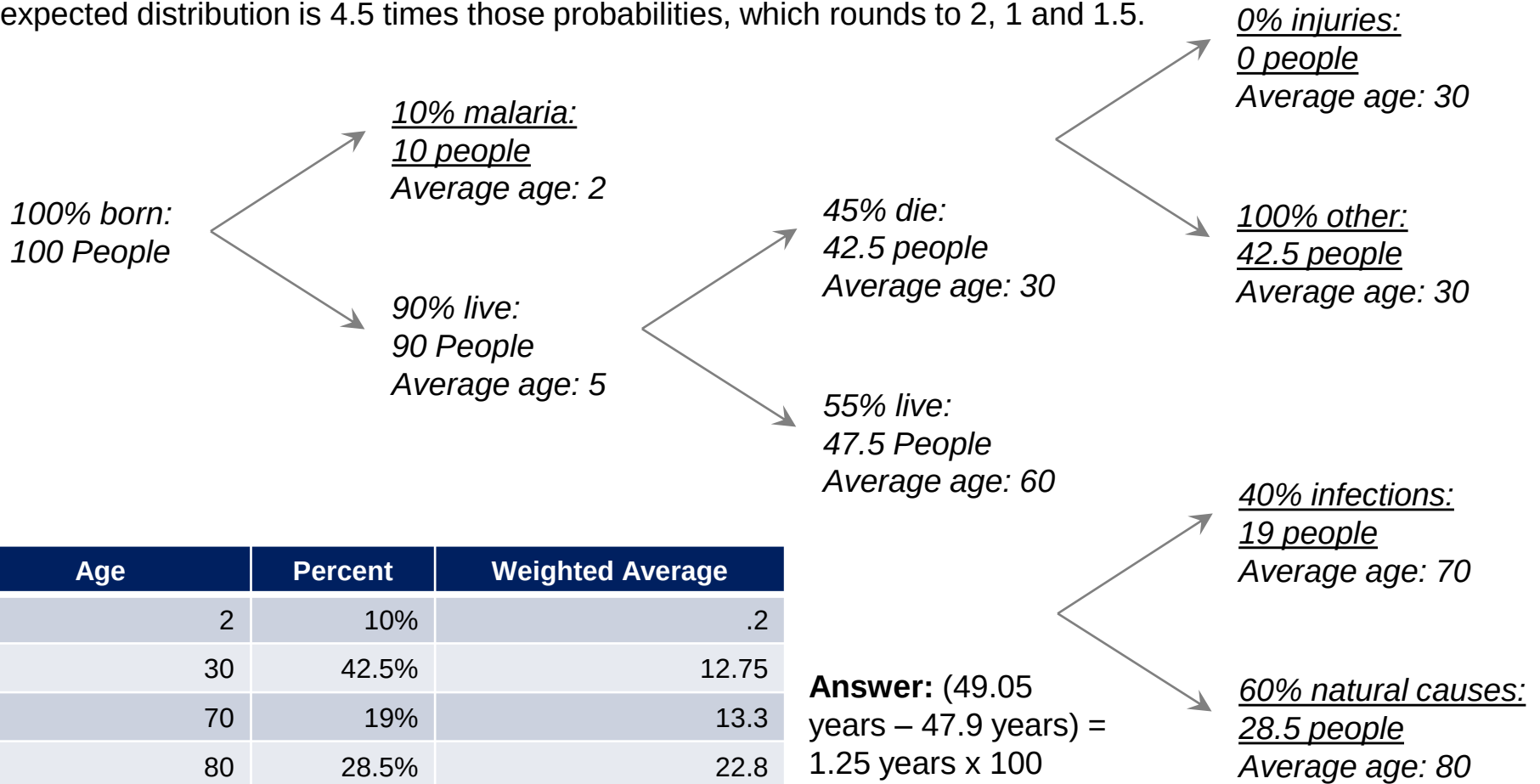
Age	Percent	Weighted Average
2	0%	0
30	50%	15
70	20%	14
80	30%	24
Weighted average:	100%	53 years

Answer: (53 years – 47.9 years) = 5.1 years x 100 people = **510 years**

Case 32: Death by Math (For Interviewer)

Step 4: Calculate “Life Years” Lost from Injuries

Do the same step as was done for malaria, but now for injuries. Previously, 4.5 people died from injuries, but now they live, and must be distributed among remaining options (40.5 people die of other, 18 people die of infections, 27 die from natural causes, which results in probabilities of 47.3%, 21%, and 31.7%). Thus the expected distribution is 4.5 times those probabilities, which rounds to 2, 1 and 1.5.



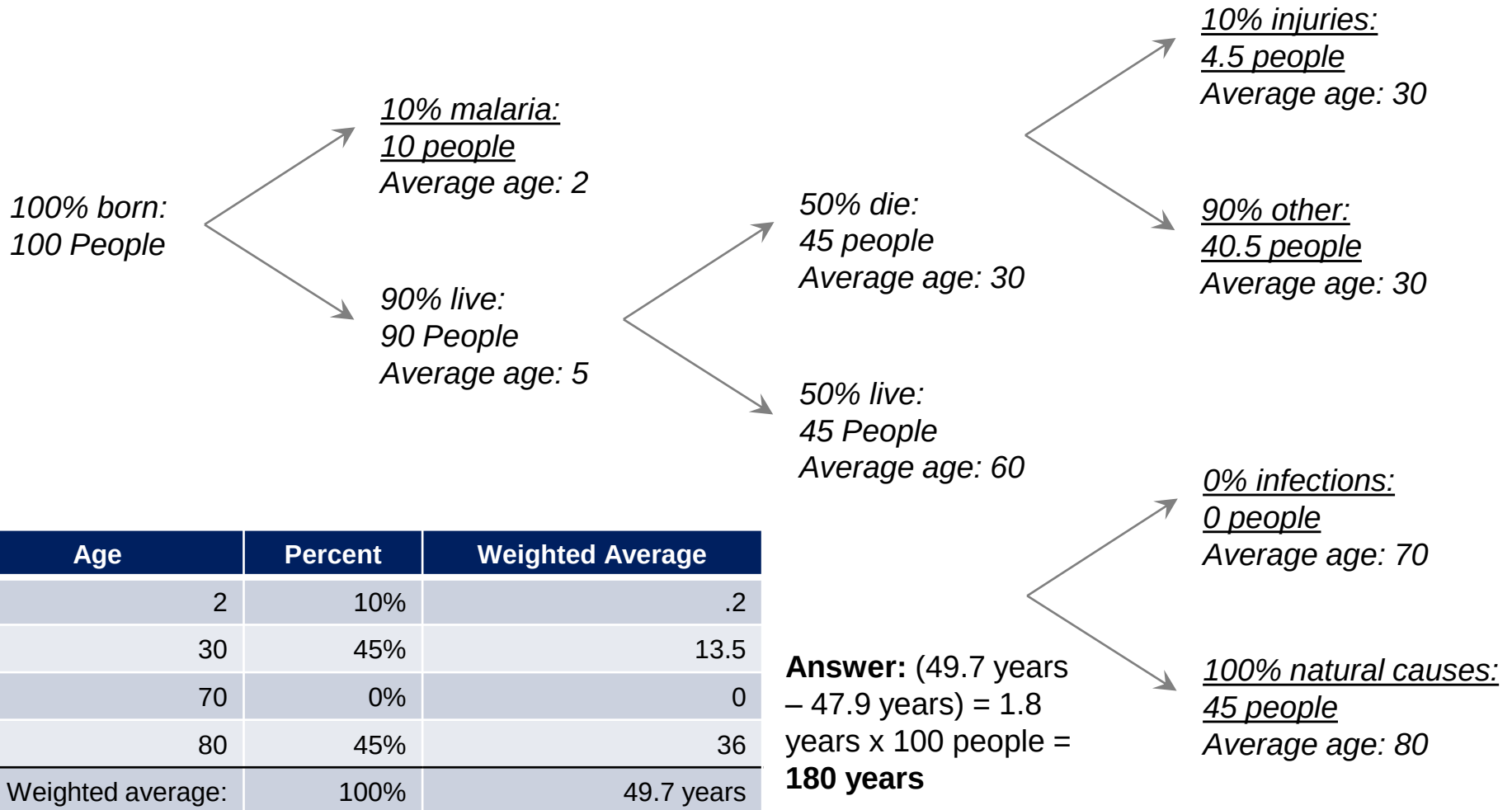
Age	Percent	Weighted Average
2	10%	.2
30	42.5%	12.75
70	19%	13.3
80	28.5%	22.8
Weighted average:	100%	49.05 years

Answer: (49.05
years – 47.9 years) =
1.25 years x 100
people = **115 years**

Case 32: Death by Math (For Interviewer)

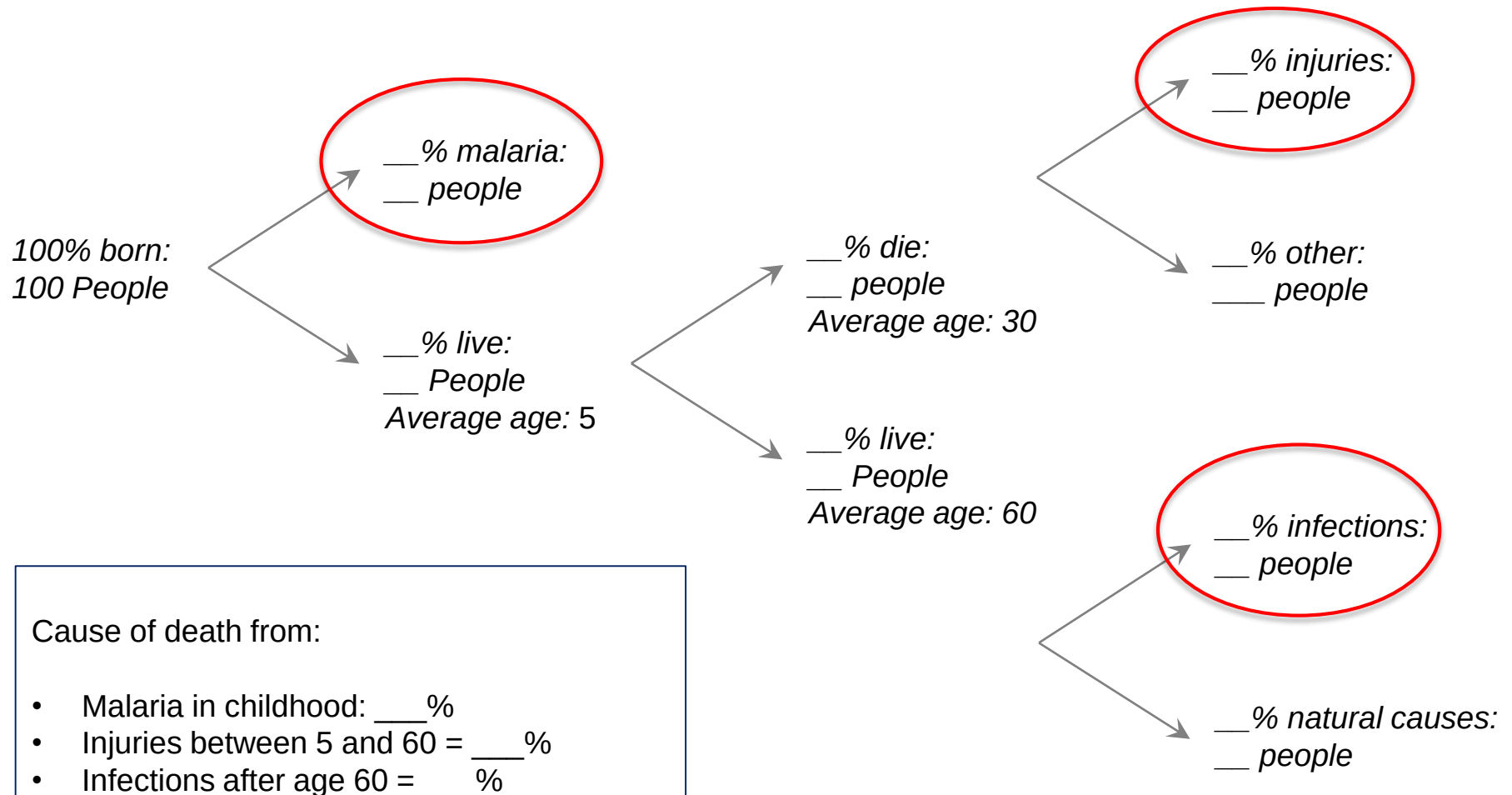
Step 5: Calculate “Life Years” Lost from Infections

This is the easiest step; do the same logic as before. The 17 People who died of infections now die of natural causes.



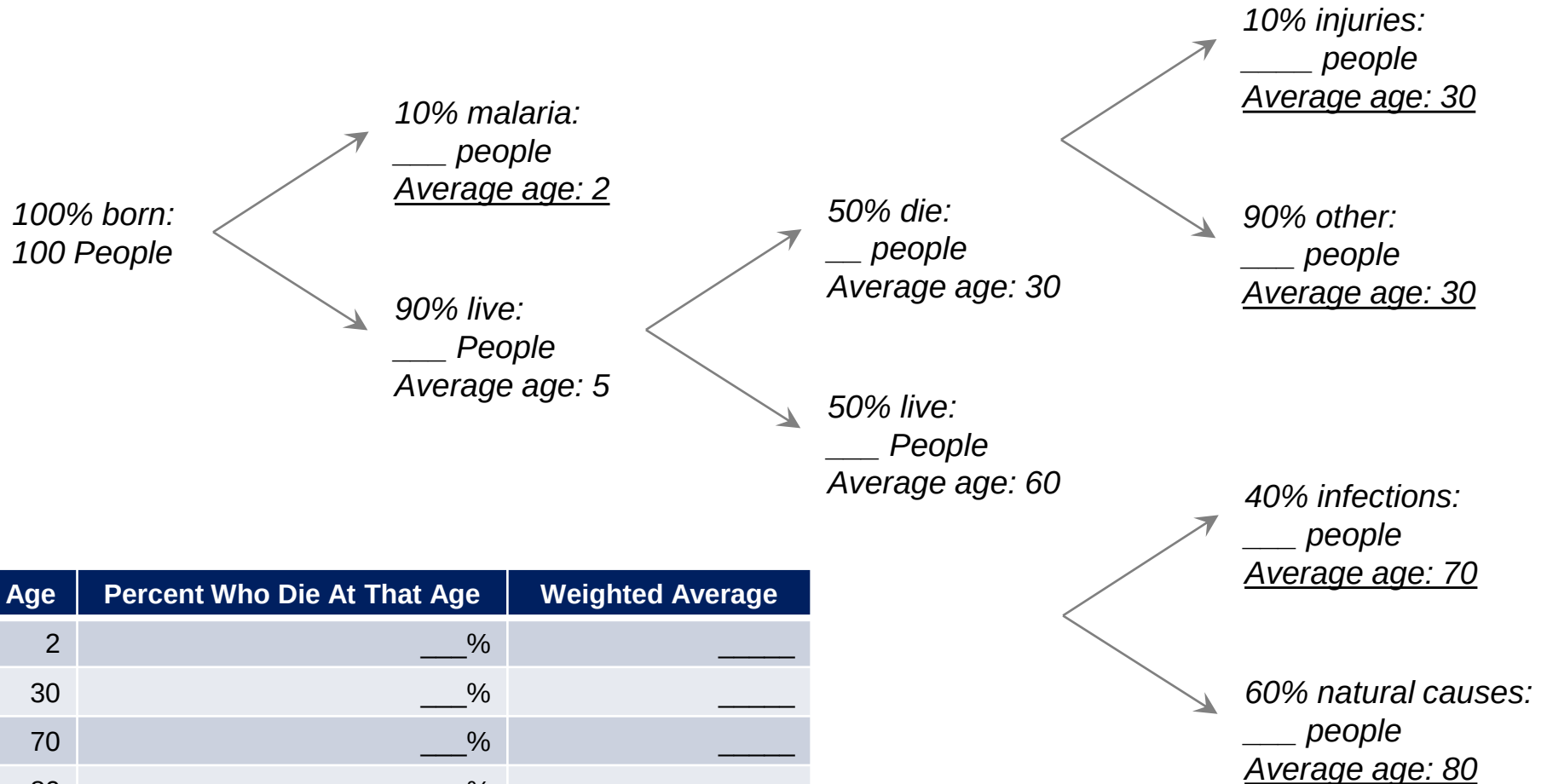
Case 32: Death by Math (For Interviewee)

Step 1: Diagram Death Rates



Case 32: Death by Math (For Interviewee)

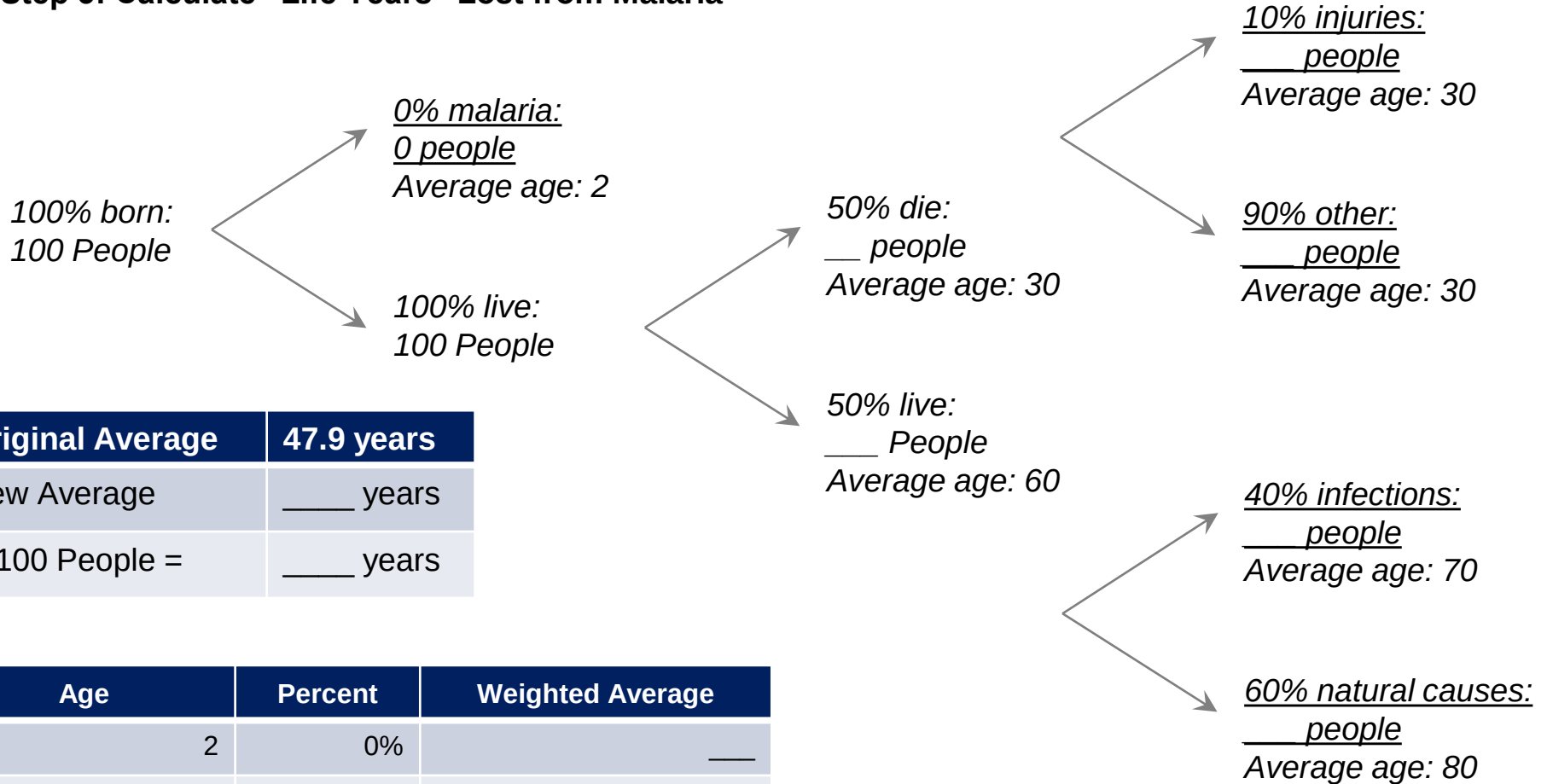
Step 2: Calculate Life Expectancy



Age	Percent Who Die At That Age	Weighted Average
2	___%	_____
30	___%	_____
70	___%	_____
80	___%	_____
	100%	_____ yrs

Case 32: Death by Math (For Interviewee)

Step 3: Calculate “Life Years” Lost from Malaria

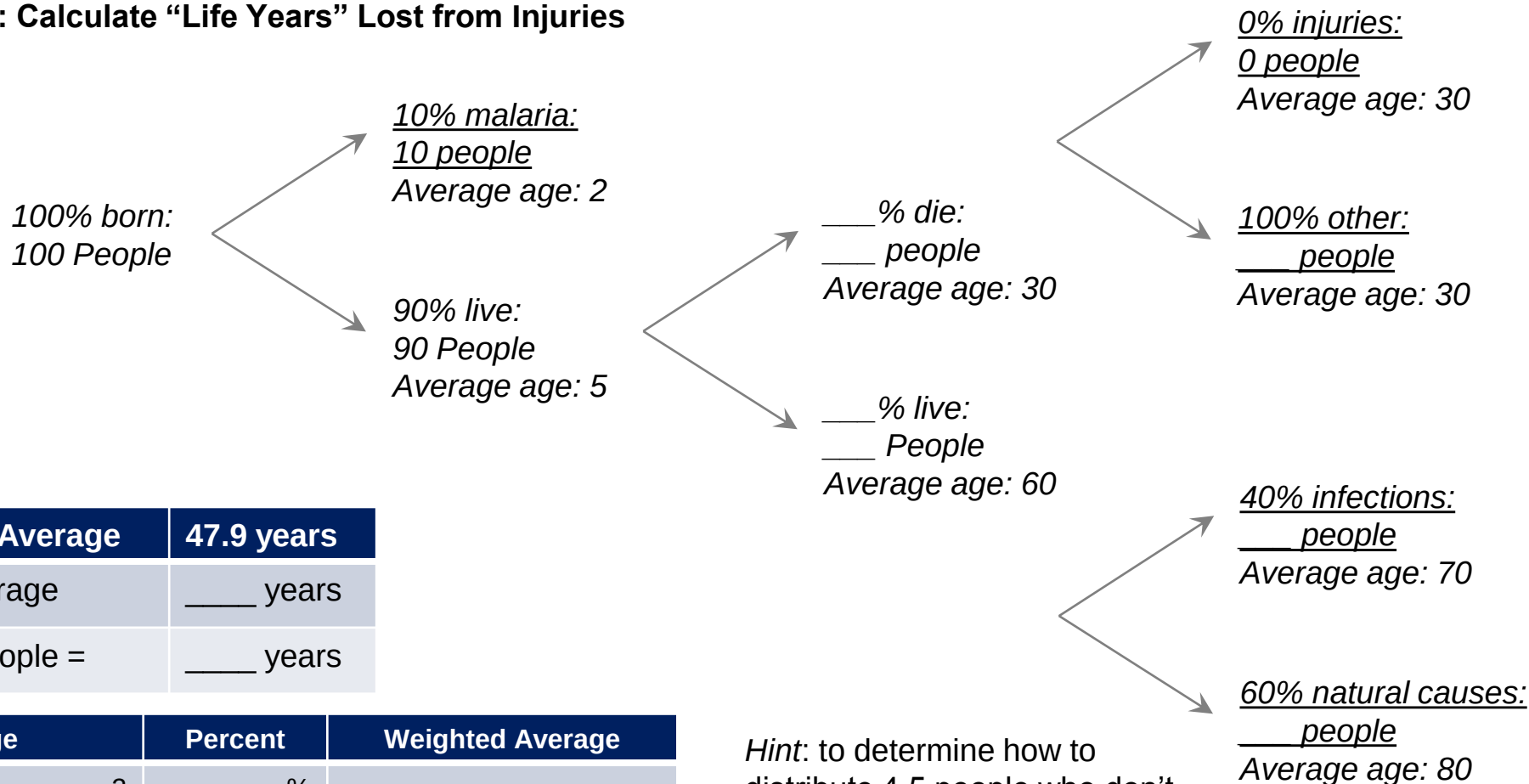


Original Average	47.9 years
New Average	___ years
X 100 People =	___ years

Age	Percent	Weighted Average
2	0%	___
30	___%	___
70	___%	___
80	___%	___
Weighted average:	100%	___ years

Case 32: Death by Math (For Interviewee)

Step 4: Calculate “Life Years” Lost from Injuries



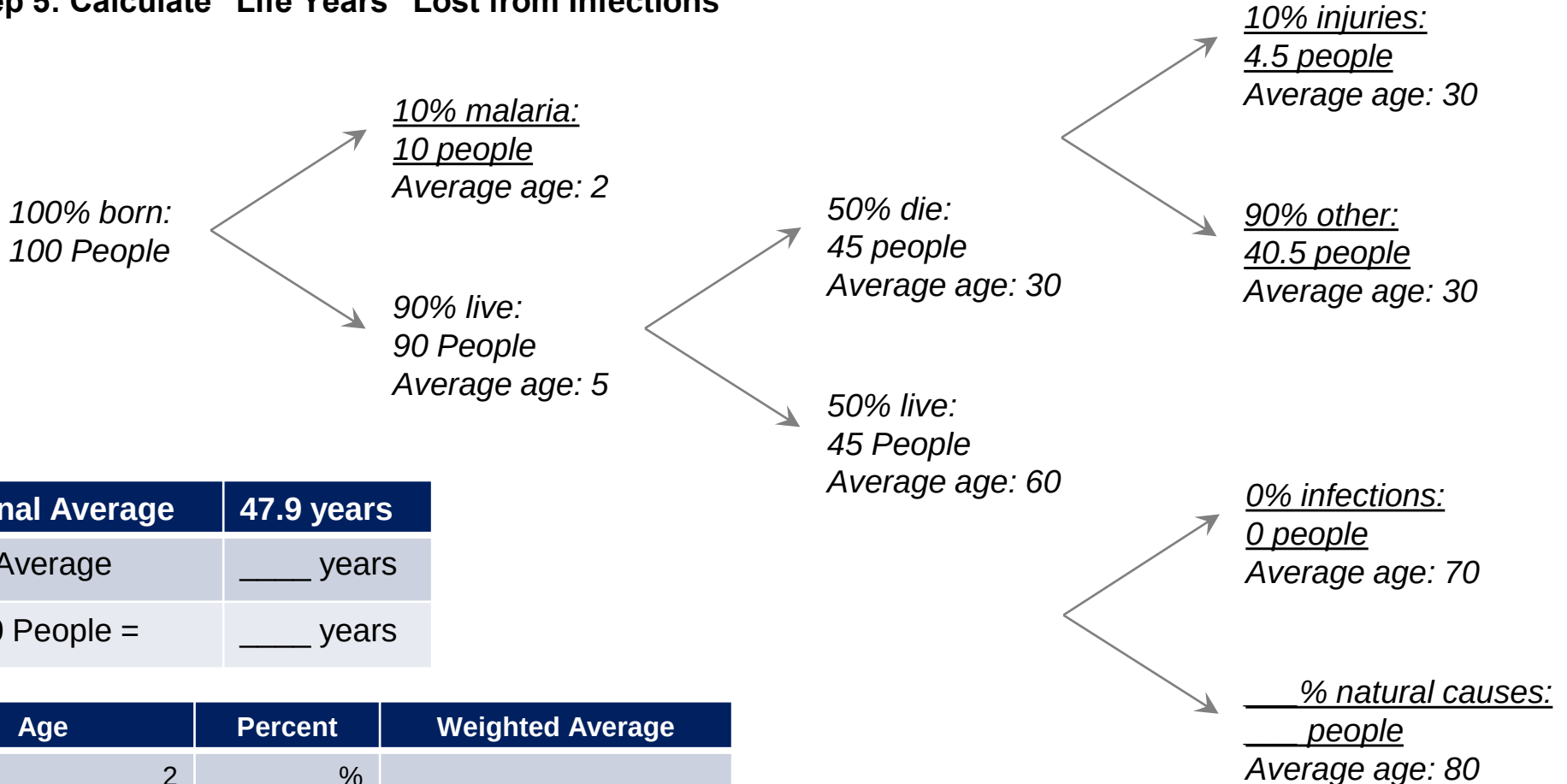
Original Average	47.9 years
New Average	___ years
X 100 People =	___ years

Age	Percent	Weighted Average
2	___%	___
30	___%	___
70	___%	___
80	___%	___
Weighted average:	100%	___ years

Hint: to determine how to distribute 4.5 people who don't die from injuries, look at the probability of dying from “other”, “infections” and “natural causes” for people who don't die from malaria or injuries.

Case 32: Death by Math (For Interviewee)

Step 5: Calculate “Life Years” Lost from Infections



Original Average	47.9 years
New Average	____ years
X 100 People =	____ years

Age	Percent	Weighted Average
2	____%	____
30	____%	____
70	0%	____
80	____%	____
Weighted average:	100%	____ years

Case 33: Olympic Trials

Prompt: Los Angeles is considering placing a bid to host the upcoming Summer Olympic Games. Our client, the L.A. bidding committee, approached us to ask for our opinion as to what the important considerations should be. Of major concern is how to convince the state and federal governments that this would be beneficial for the city. How would you recommend they go about doing this?

Interviewer Information:

- The Olympics are to be held over 15 days
- Revenues
 - Sponsorship: \$75 per seat (occupied)
 - IOC will give \$1.1B
 - TV Rights will give \$4B
 - Merchandise: \$2B
- Costs (give all at once)
 - \$2.6 billion for the operation of the games
 - \$2.6 billion for freeway upgrades
 - \$6.0 billion for new and upgraded venues
 - \$2.5 billion for new airport terminal
 - \$1.1 billion for athlete facilities

Notes for Interviewer: This case is designed to test the interviewee's ability to generate a framework outside of what they might be accustomed to; there is no market, no competition, a very unusual product. The case is designed to initially revolve around profitability, which will show a great deficit. At this point, the interviewer should ask if hosting the games is a bad idea. If the interviewee says that it is, then the case can turn into a brainstorming session on how the committee might convince the government that it is a good idea after all. Let the interviewee float around with some questions, before hinting that we should be looking at this from a profitability angle.

Once the interviewee starts looking at this from a profitability angle they will quickly discover that this is an unprofitable venture. However, Olympic cities rarely make money, so why do so many cities wish to host the Olympics? The answer is that it helps in a lot of other manners: it allows for large infrastructure investments, it creates a tourism boom both before and after the games, and it increases the city's reputation. The question then becomes: are these indirect impacts worth the investment for the city of LA?

Case 33: Olympic Trials (Example Framework)

Main questions	Should LA host the Olympics?
Key areas to explore	<u>Costs</u> - Construction: - Olympic village? - Infrastructure? - Airport? - Temporary venues? <u>Revenues</u> - Ticket sales? - Merchandise? - TV? <u>Indirect Costs</u> - Cost of increased traffic/construction? - Increased security? <u>Indirect Benefits</u> - Benefits of infrastructure upgrade? - Tourism benefits? - Reputation benefits?
Analysis	Explicit Costs: \$14.8B. Explicit Revenues: \$9B → Explicit Profit: -\$5.8B. Unknown benefits include the long term benefits from improving LA's infrastructure, and any increases in tourism during/after the games. Unknown costs include the downside to productivity in LA from construction before the games and any increased costs for security upgrades.
Recommendation	Decision: Do not host the games. The games are expected to cost the city of LA \$5.8B, and unlike lesser known cities in underdeveloped countries, LA will not likely see enough of an increase in tourism to justify the almost \$6B loss. Additionally, LA already suffers from significant traffic issues and adding pre-Olympic construction to the mix could have a detrimental effect on traffic/productivity in LA.
Other factors (risks)	What is the political situation in LA? Do the local people and politicians support LA hosting the games? What is the security situation? Has security in LA improved or is LA an easy potential target?

Case 33: Olympic Trials (Interviewer Exhibit)

Interviewee Sheet				Interviewer Answers	
Seating Type	Price per Ticket	Total Number of Seats per Day	Expected Capacity	Total Occupied Seating (15 Days)	Total Revenue (15 Days @ \$75 per seat)
A	\$200	420,000	95%	5,985,000	\$1.197B
B	\$120	700,000	60%	6,300,000	\$0.756B
			Exact Totals	12,285,000	\$1.953B

Case 33: Olympic Trials (Interviewee Exhibit)

Exhibit 1: Seating

Seating Type	Price per Ticket	Total Number of Seats per Day	Expected Capacity
A	\$200	420,000	95%
B	\$120	700,00	60%

Case 34: Nabisco's Market Share

Prompt: The salted food division at Nabisco has been steadily losing market share over the past two years, from a high of 20% to the current level of 18%. Profits as a percentage of sales, however, have been growing. What could be causing this?

Interviewer Information:

- *Market* is growing at a steady rate.
- *Competition:* mainly two multinational consumer product companies that have a similar product mix to Nabisco. The two companies have ~50% combined market share.
- Differentiation from Competitors: Nabisco's sales reps are regarded as the best in the industry
- Change in Product Line: Nabisco has not introduced any new product lines.

Information about exhibit (to be provided only if asked):

- Reasons for Sales Cut: Sales force cut to reduce costs, but number of outlets unchanged
- Cause of Change in the Marketing Budget: The changes come from reduced trade promotions
- Sales Channels: Products primarily sold in large grocery store chains and convenience stores
- Sales Force/Customer Interaction: Sales force visits each customer at least once per quarter
- Timing of Promotions: Promotions usually occur at the end of each quarter
- Impact of Promotions: Promotions required for end of aisle displays and advertising space

Notes for Interviewer: The data show a large decrease in sales force and marketing expenditure. Most of the marketing reduction was in trade promotions. Product is sold through grocery chains and convenience stores, which are traditionally driven by periodic trade promotions. The reduction in trade promotions brought about a loss of shelf space, which led to a decrease in market share. Also, the product line did not change in a product category where new products and line extensions are routine. The market has been growing, indicating a missed opportunity for new products in the market. Lastly, profitability increased due to lower costs, but it may not be sustainable.

Case 34: Nabisco's Market Share (Example Framework)

Main questions	Why is market share declining while profitability is increasing? Is this sustainable?
Key areas to explore	<u>Internal</u> - Market share changes - Where is profit increase coming from? - Any new products or changes to company's product offering? <u>External</u> <u>Competition</u> - Changes in market share for competitors - Possible new entrants <u>Market</u> - Steady growth?
Analysis	- Market is growing and largest competitors have seen market share growth - Profit increase has come from a reduction in spending on marketing and sales personnel, specifically a lack of promotional spending → less shelf space = less sales - No new product lines, despite frequency of new products in packaged food market
Recommendation	Operational: Increase promotional spending. At a minimum return sales force and marketing budgets to earlier levels. Investigate changing cycle of sales force visits and promotions to allow Nabisco products to stand out more easily. Potential Adjacencies: Offer a new snack products, sell in new geographies, explore new distribution channels, explore others ways to reach customers beyond shelf-space.
Other factors (risks)	- What gain in market share would be required for this to make sense? - What is the cost of doing nothing? - How will competitors respond?

Case 34: Nabisco's Market Share (Cont'd)

Exhibit 1: Company Cost Structure

Cost	Current	Two Years Ago
Raw Ingredients	28%	26%
Conversion Costs	24%	24%
Distribution	8%	9%
Marketing	16%	18%
Sales Force	7%	9%
Pre-Tax Profit	17%	14%

Contents

Section	Page #
☐ Introduction	3
☐ Consulting Industry Guide Industry Overview Firm Overviews (by type)	5
☐ Interview Preparation Interview Overview Theoretical Frameworks / Helpful Accounting Information Sample Case Frameworks	9
☐ Practice Cases Case Certification Schedule 34 Practice Cases	27
☐ Additional Resources	149

Additional Resources

Below are some of the best resources you can use to help prepare for your case interviews:

Resume / Cover Letter Guidance:

- Career Center Orgsync Page
- GSCG Orgsync Page
- Career Coaches
- Peer Advisors

Case Interview References:

- David Ohrvall's website - <http://www.mbacase.com/>
- Marc Cosentino's website – <http://casequestions.com/>
- Victor Cheng's website - <http://www.caseinterview.com/>
- GSCG Orgsync Page
- <http://www.consultingcase101.com/>
- <https://www.caseinterview.com/math/home.php>
- <http://www.mbacase.com/books-courses/crack-the-case-fastmath/>

Case Interview Practice:

- Peers
- Career Coaches – schedule through MCV
- Peer Advisors – schedule through MCV

Corrections?

We are constantly in the process of revising and improving our casebook. If you spot any changes that you think should be made, please use this google form to let us know:

https://docs.google.com/a/georgetown.edu/forms/d/1ixm0bQv8Mnq0_biMwCDPQzCabaitr94PZRhvJjFFCT4/viewform